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Integrated Geospatial and Long-Term Electricity Planning for Universal Access and Decarbonization in Burkina Faso

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**Integrerad geospatial och långsiktig elplanering
för universell tillgång och avkarbonisering
i Burkina Faso**

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Sammanfattning

Denna avhandling utvecklar omfattande, geospatiala, långsiktiga elplaneringsvägar för Burkina Faso, ett land som strävar mot hållbar tillväxt och inkluderar universell tillgång till elektricitet i denna tillväxt. Genom att integrera Open Source Energy Modeling System (OSeMOSYS) och Open Source Spatial Electrification Tool (OnSSET), för att skapa Open Source Geo-spatial Energy Modeller (OSGEM), analyserar studien tre elektrifieringsscenarier: Business-as-Usual (BAU), Balanced (BAL) och Ambitious (AMB). Resultaten visar att även om BAU-scenariot leder till stegvisa framsteg, så är det kortare än universella åtkomst- och hållbarhetsmål. BAL-scenariot uppnår en mer optimal balans mellan att utöka tillgången, hantera kostnader och minska utsläppen, medan AMB-scenariot, även om det kräver de högsta investeringarna, ger snabb elektrifiering och de största utsläppsminskningarna. Känslighetsanalyser visar att tidigare nettonollmål och lägre solceller (PV) kapitalkostnader kan påskynda användningen av förnybar energi och sänka de totala systemkostnaderna. Studien belyser vikten av integrerad, fysiskt explicit planering och starkt politiskt stöd för utbyggnad av förnybar energi. Begränsningar inkluderar dataosäkerheter och behovet av mer detaljerad modellering av efterfrågan och systemtillförlitlighet. Resultaten ger handlingskraftiga insikter för beslutsfattare och ger en grund för framtida forskning om hållbar elplanering i Burkina Faso.

Nyckelord

Geospatial elektrifiering, långsiktig energiplanering, Burkina Faso, förnybar energi, Tillgång till el

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Abstract

This thesis develops comprehensive, geospatial, long-term electricity planning pathways for Burkina Faso, a country striving toward sustainable growth and including universal electricity access in that growth. By integrating the Open Source Energy Modelling System (OSeMOSYS) and the Open Source Spatial Electrification Tool (OnSSET), to create the Open Source Geo-spatial Energy Modeller (OSGEM), the study analyzes three electrification scenarios: Business-as-Usual (BAU), Balanced (BAL), and Ambitious (AMB). Results show that while the BAU scenario leads to incremental progress, it falls short of universal access and sustainability goals. The BAL scenario achieves a more optimal balance between expanding access, managing costs, and reducing emissions, whereas the AMB scenario, though requiring the highest investments, delivers rapid electrification and the greatest emission reductions. Sensitivity analyses reveal that earlier net-zero targets and lower photovoltaic (PV) capital costs can accelerate renewable adoption and lower overall system costs. The study highlights the importance of integrated, spatially explicit planning and strong policy support for renewable energy deployment. Limitations include data uncertainties and the need for more granular modeling of demand and system reliability. The findings offer actionable insights for policymakers and provide a foundation for future research on sustainable electricity planning in Burkina Faso.

Keywords:

Geospatial Electrification, Long-Term Energy Planning, Burkina Faso, Renewable Energy, Electricity Access

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Nomenclature

Abbreviations

ECOWAS	Economic Community of West African States
GDP	Gross Domestic Product
GHG	Greenhouse Gas
GWh	Gigawatt-hour
HDI	Human Development Index
HFO	Heavy Fuel Oil
LCOE	Levelized Cost of Electricity
LPSE	Energy Sector Policy Letter 2016-2020
MtCO ₂ e	Metric tons of Carbon Dioxide equivalent
MUSD	Million US Dollars
MW	Megawatt
ND-GAIN	Notre Dame Global Adaptation Index
NDC	Nationally Determined Contribution
OnSSET	Open Source Energy Modelling System
OSeMOSYS	Open Source Spatial Electrification Tool
OSGEM	Open Source Geo-spatial Energy Modeller
PANEE	Plan d'Action National d'Efficacité Energétique 2015-2030
PANER	Plan d'Action National des Energies Renouvelables
PNDES-II	National Plan for Economic and Social Development 2021-2025
PPP	Purchasing Power Parity
PV	Photovoltaic
SDGs	Sustainable Development Goals
SSA	Sub-Saharan Africa
TJ	Terajoules
WAPP	West African Power Pool

Chapter 1

Introduction

In this chapter, an introduction to the central themes of this thesis are presented, giving context to the study. Burkina Faso is introduced as the primary focus, highlighting its energy challenges and the importance of geospatial electricity planning in the country

1.1 Electricity Access

Access to electricity is described as a household that has an electricity connection, either to the grid or an off-grid system, that provides some basic level of energy service, with the definition of "basic" varying geospatially and temporally [1]. The rate at which a region or country has or attains a certain level of access is given as the percentage ratio of the total number of people with access to electricity in their residences to the total number of people living in that region or country. Access to electricity is a critical component of improving the quality of life for people all over the globe [2]; so much so that it is encapsulated in the Sustainable Development Goals (SDGs) as Goal 7, proclaiming that universal access to modern energy services (electricity included) should be achieved by 2030, where the energy future here is more renewable and efficient, built on international cooperation, financing and, technical and human capacity building [3]. In 2022, approximately 700 million global citizens (about 9% of Earth's population) did not have access to electricity. Of those around 700 million, 86% of them reside in Sub-Saharan Africa (SSA), highlighting SSA as the region of focus to achieve universal access to electricity. Different degrees of unmet demand exist due to affordability and availability issues with the service and/or appliances to make use of that service [4]. Progress has been made over the years as about 20% of the world's population was without electricity in 2000, but at the current rate, it is likely universal access will not be achieved by 2030 [5]. Therefore, additional measures should be investigated to accelerate electricity access.

The SDGs, as a whole, capture the importance of energy, as 143 of the 169 SDG targets have synergies in achieving SDG7 due to this achievement resulting in increased welfare, infrastructure development and environmental management [6]. The widening access to electricity highlights these intersections in many ways, one being that an affordable electricity service can decrease the amount spent monthly on liquid or solid fuels used to provide light

to homes. An electric service provides light for longer periods of time. This allows these funds to be reinvested in other household expenses or invested in income-generating work done by household residents [7]. Moreover, local pollution is negated by switching energy carriers, improving health outcomes for those who are otherwise exposed and preventing many respiratory diseases. Electricity access also improves community and regional health outcomes by providing energy to community health centers that need to use medical, information and communication equipment to provide health services. Increasing electricity access can induce a positive environmental impact, as studied in SSA where when looking at how access and public borrowing affect environmental sustainability, it was shown that electricity access provides the setting to alleviate environmental damage, and the actual alleviation comes through making sustainable energy choices like using renewable energy [8]. Emphasis on access was given to rural communities where the need is more severe in terms of the tailoring of policy to provide affordable electricity and appliances to make use of that electricity. An increase in access can also produce economic growth. Looking at a case study of the Indonesian electricity and industrial sectors, how grid expansion affects firm turnover was studied. It was revealed that the increased electrification increases the number of industrial firms and workers, along with the industrial output, due to lowering costs and encouraging a more competitive environment that redistributes resources in a market towards an optimal state [9].

1.2 Energy Systems for Sustainable Development

For communities to receive electricity, they must engage in the use of electrical energy systems, a connection of components responsible for generating, transmitting, distributing, and consuming electricity. Households can gain electricity access by employing three types of technologies: stand-alone systems, mini-grid systems and the grid. Stand-alone and mini-grid systems can operate as off-grid systems, independent of the grid, or in a grid-tied fashion, running in sync with the grid where the grid essentially acts as a battery [1]. All of these energy systems can be powered by a plethora of energy sources. Stand-alone systems tend to be mainly powered by solar photovoltaic (PV) (with battery storage) sources, diesel, or PV-diesel hybrid configuration. In addition to these fuels, mini-grids can be powered by wind, biogas, and hydro resources, with the option of being connected to the grid at a later time. The grid is often powered by large-capacity utility solar PV, wind, natural gas, coal, or even nuclear power plants dedicated to producing power in large quantities to be delivered over a transmission and distribution network.

The increase in electricity access then occurs by the proliferation of off-grid systems, which has been less common, or by the build-out or extension of the national grid, where high and medium voltage distribution lines are introduced to non-grid electrified areas to be connected to households, which is the more common approach to electrification by governments and utilities

[10]. Nearly 100% of all investments in the electricity sector go toward extending the grid, as it often is more cost-effective at providing higher quality electricity. However, off-grid systems are appealing when the connection cost per energy consumed is less than that for a grid connection, the grid is not as reliable as would be necessary (like for a community clinic) or if it will take an inordinate amount of time for the grid to arrive in a community. Because situations arise where one type of system is preferable to another, depending on where *and when* service is needed, investment and thus, electricity access requires planning.

1.3 Electricity Planning

Governments are interested in optimizing and managing the systems that provide their populations with electricity, especially in developing countries where increasing electricity access has an immense impact on development, as previously described. The challenge for countries in the 21st century is to achieve this while simultaneously decarbonizing the energy systems used to provide energy services. Given advances in technology, it is prudent to consider the degree of decentralization that can be introduced, which can take advantage of local energy sources but also give consumers the power to evolve into prosumers and generate for their communities, adding a level of resilience, but also an added dimension of choice and empowerment to their lives [11]. Given these dimensions, actions must be coordinated at the national, corporate, and local level among many stakeholders such as government, regulatory agencies, and the community.

Electricity planning can also form a subset of a country's broader climate strategy regarding national and international commitments. Countries are interested in reducing their impact on the natural environment to reduce the frequency and severity of climate-related events such as hurricanes, drought, flash floods, and rising sea levels. Actions to address the negative effects of climate change often appear in a country's Nationally Determined Contribution (NDC), where they explain how they will adapt to and mitigate the effects they experience. In the same document, one can find how they plan to achieve the SDGs most relevant to their circumstances. Additionally, national policies might seek to take advantage of market conditions and trends, whether that is negotiating cross-border electricity trade or securing more cost-effective PV panels to distribute in a decentralized fashion. The policies are essential for countries to receive international funding to achieve the policy objectives as it relates to SDGs. It is crucial to be able to demonstrate how funds would be allocated over time and describe their intended effect related to improving economic outcomes and quality of life assurances [12]. For these reasons, it is interesting to governments and national utilities to create strategies to plan and monitor the development of the electricity sector in their countries.

Electricity sector modeling is one tool that governments and utilities can use to manage and plan investments. It is helpful because governments are bound by the technical capacity,

resources, and budget they possess and the energy disruptions and security concerns that must be considered and addressed. Modelling allows the user to, by using mathematics and computation, assess the electricity demand in the medium and long term, and determine the best arrangement of infrastructure additions and retirements in time and space to achieve cost, environmental and operational goals [4]. To not optimize for these goals would result in a sub-optimal system configuration. Governments use results from models and scenarios to glean insights to inform their decision-making processes. This can help them to create policies with respect to renewable energy integration, increasing energy security and efficiency, and reducing greenhouse gas (GHG) emissions.

1.4 Why Electricity Planning for Burkina Faso

This thesis aims to demonstrate the use of models applied to the country case of Burkina Faso, a Least Developed Country in Western Africa with very poor access to electricity, much lower than the continental average, highlighting the urgent need for strategic electricity planning. The country has goals of very high electrification penetration, specifically targeting rural areas that are disproportionately affected. Simultaneously, Burkina Faso depends greatly on neighbouring countries for electricity despite having resources it can use to its advantage. Its resources are even sustainable, but just as pockets of demand are spread over the whole country, so are these resources. Use of these sources (solar and wind) can increase the energy sovereignty of Burkina Faso, cementing its upward economic and social mobilization. The government already has policies that promote sustainability and electricity access but it should still determine how to best make use of its resources that vary by location and strength, while achieving other objectives. Acceptable outcomes of these policies can occur as a result of planning for the optimal mix of grid expansion, mini-grids, and stand-alone systems, requiring careful geospatial analysis. Using geospatial tools will allow policymakers to determine cost-effective solutions tailored to given locales, maximizing investment efforts. It is hoped that this thesis can provide insight as to how Burkina Faso can chart its electrification pathway for its development in this regard.

1.5 Outline

Chapter 2 will speak to the energy Background of Burkina Faso, the country case in question, Chapter 3 will describe the Objective of this Master's thesis, Chapter 4 will explain the Methodology used to perform this work, Chapter 5 present the results of this work, Chapter 6 will those discuss those results of this work, and Chapter 7 will provide concluding remarks on the way forward given what this thesis has accomplished.

Chapter 2

Background

This chapter presents a review of literature relevant to Burkina Faso's energy context, highlighting past work done in electrification and energy planning, and identifying research gaps in the fields.

2.1 Country Context

Burkina Faso is a landlocked West African country of 270,000 sq km, bordered by Ghana, Togo and Côte d'Ivoire to the south, Benin to the southeast, Mali to the north and west, and Niger to the northeast (see Figure 2.1.1) [13, 14]. Burkina's landscape experiences a gradient from dry conditions in the north (the Sahelian region close to the Sahara Desert) to Savannah plains in the center of the country, to a more luscious and fertile woody forest in the south, due to more rainfall occurring in this area [15]. Agriculture and fishing are more prominent in this area, not only due to rainfall, but also to the concentration of river and delta lands in the south of Burkina Faso. The country has three rivers that form the Volta River system: Mouhoun (Black), Nazinon (Red) and Nakambé (White) Volta, Black Volta being the major one. This river system has a transnational relationship with Ghana as these rivers feed into Volta Lake in Ghana. Despite the presence of these rivers, the country is generally dry, as other smaller rivers fill seasonally, and has only one small lake that is used as a reservoir.



Figure 2.1.1: Geographical Map of Burkina Faso

Burkina Faso can in part attribute its geography to its climate. The climate can be described as dry and tropical, with long and short rainy seasons [16]. The northern part of the country receives less than 600 mm of rain, the central part 600-900 mm, and the southern part over 900 mm. The country experiences temperatures between 27-30°C (80-86 °F). The dry and wet seasons are expected to become more extreme with some uncertainty in future rainfall patterns. For temperatures, they have annually increased by 0.6°C since 1975, and could rise by 1.4 - 1.6°C by 2050 and 3 - 4°C by 2080-2099.

Burkina Faso's population stood at 23.03 million in 2023, which grew on average 2.53% per year between 2018 and 2023 and 4.05% per year between 2000 and 2023 [17]. By 2050, the population is expected to be 43.2 million due to an estimated average annual growth rate of 3.05% between 2023 and 2040 and 2.36% between 2040 and 2050. In 2020, Burkina had a projected urban population share of roughly 30%, up from approximately 18% in 2000, which is expected to grow to 50% by 2050 [18]. In addition to being mostly rural at present, the population of Burkina Faso is also young. 45% of Burkinabè are under 15 years of age, and approximately 70% are under 30 [14]. This age distribution has been the case for at least the last decade. One contributing factor is that Burkinabè of working age (making up a third of the population) typically migrate to surrounding countries for work, although in more recent times, this has become more challenging.

Burkina Faso has agriculture as its main industry, with as many as 90% of the population being involved [14]. Farming is carried out mainly on the subsistence level, with the excess being sold for cash. Foods such as maize, rice, cassava, beans, and sorghum feed the people locally, and excess cotton, sugarcane, and shea nuts are exported for profit. The rearing of livestock such as cattle, camels, chickens, horses, and sheep is also a major source of revenue. Mining exports also support the economy, but there is more that can be taken advantage of. Burkina

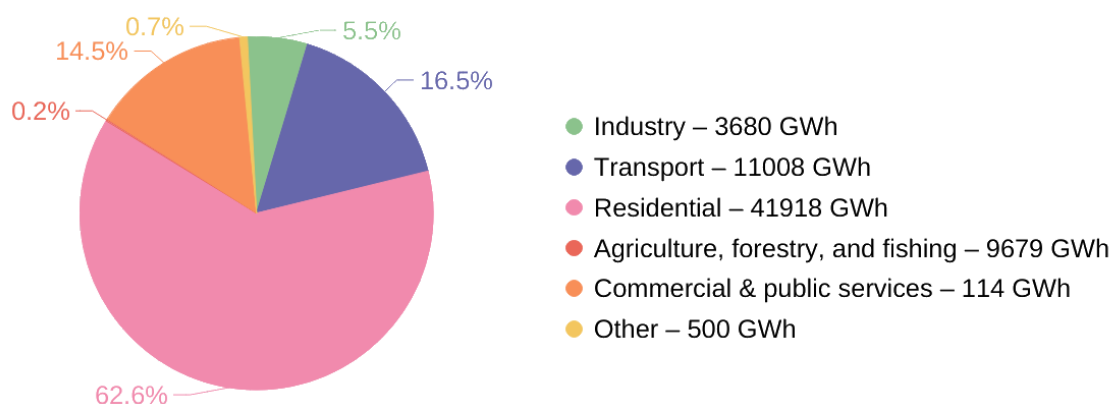
Faso has mines for various minerals such as gold, manganese, nickel, bauxite, zinc, and others, the mining of which is hampered by poor transportation facilities. Industrial plants also play a role in the economy, where value is added to raw materials in the creation of food, drinks, and textiles.

Burkina Faso had a GDP per capita (PPP, 2017 international dollars) of roughly 2930 in 2023, which rose from 953 in 2000 at an average annual rate of 4.67%. The GDP per capita is estimated to grow on average by 3.56% per year between 2023 and 2030 to roughly 3660 [19]. The country also featured a Human Development Index (HDI) of 0.449 in 2021 [20]. This index, in conjunction with its GDP per capita, is indicative of Burkina Faso's poor prosperity, education, and longevity of its people relative to the globe, with an expected lifespan of 60 years and an average schooling period of 2.3 years [21].

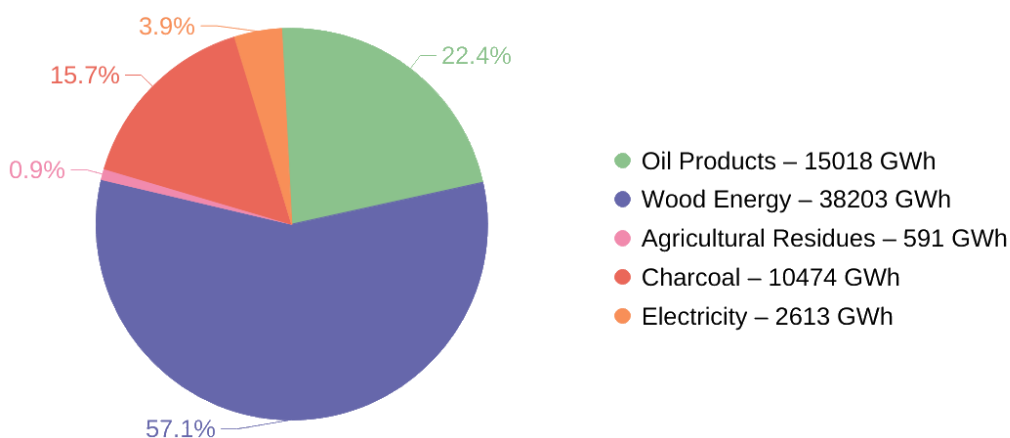
In 2020, Burkina Faso emitted 56.36 MtCO₂e, the largest portion (43.1%) coming from the agriculture sector and 8.2% coming from the energy sector [22]. This has grown from an annual emissions tally of 29.78 MtCO₂e in 2000, and from 53.89 MtCO₂e in 2017 at an average annual rate of 0.82%. Despite Burkina Faso's relative contribution to global GHG emissions of 0.08%, it has a Global Climate Risk Index rank of 99.33 and ND-GAIN Index rank of 159th, indicating the country as one of the most vulnerable countries to climate change [16]. Burkina's population and industries can expect challenges like erratic weather, deforestation and desertification, floods, droughts, heat waves, locusts, and more to affect quality of life and agricultural yield.

2.2 Energy Access and Energy Balances

Burkina Faso, across all sectors, consumed 246.6 TJ of energy in 2020, the lion's share going to the residential sector (Figure 2.2.1a) [23]. Almost all wood energy and charcoal use (which make up more than 72% of energy use by fuel in Figure 2.2.1b) in Burkina Faso can be attributed to the residential sector for cooking and other residential heating purposes. The transport and commercial sectors are the next largest demand sectors and nearly use the same amount of energy. The entire Burkina Faso's transport sector is powered by oil and other petroleum products, which is the most significant demand for petroleum products themselves, helping the oil products category to be the second-most fuel of final use (Figure 2.2.1b). The commercial sector is largely powered by wood energy, which uses the remainder of the wood energy that is not used by the residential sector.



(a) By Sector



(b) By Fuel

Figure 2.2.1: Total Final Energy Consumption in 2020 (data from [23])

Only about 4% of the final energy used is sourced from electricity, half of which is used by the industrial sector and a quarter of which is used by the residential and commercial sectors together (Figure 2.2.2). About half of the electricity supplied to the grid is imported, and the other half is generated in the country from biomass, solar, hydro, and oil energy sources. Oil is the primary energy source for electricity, as renewable primary energy stood at 25% in 2020. Only 6% of the energy is lost in transmission.

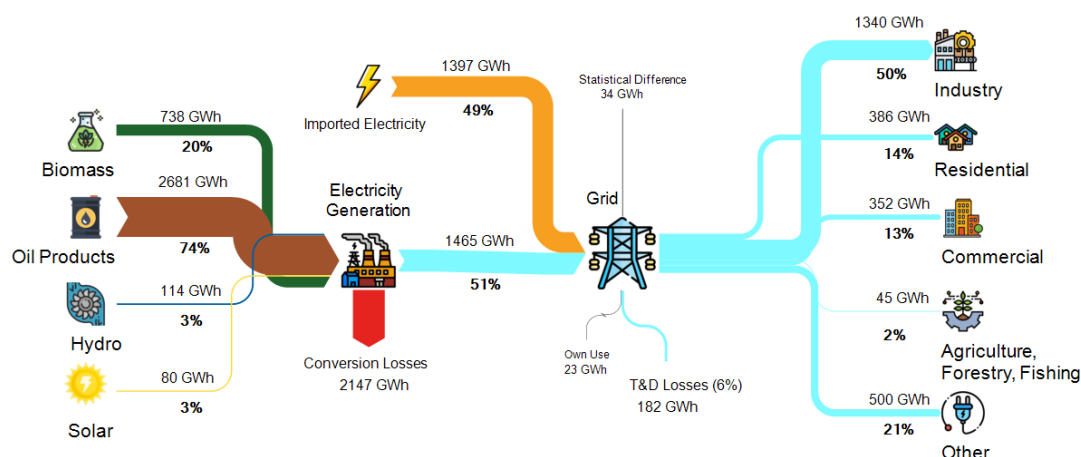


Figure 2.2.2: Electricity Sector Sankey Diagram for 2020 (data from [23])

Burkina Faso is challenged with poor electricity access, one of the lowest in the world in fact, as its electrification rate is much lower than the continental electrification rate; 19% in 2021 as opposed to 57% in 2023 [24, 25]. The lack of access is exasperated in rural communities where only 3% of rural people in Burkina Faso (and 28% in sub-Saharan Africa) have access, while 60% of those in Burkinabè urban communities (and 78% in sub-Saharan Africa) have access [26]. Among Burkina's neighbours, it is notable for its low access rate, but all its neighbours are not equal in this regard [27]. Niger has similar electricity access to Burkina and its rate of growth (roughly 0.5% per annum) resembles that of Burkina's as well, probably due to similar economic challenges (Figure 2.2.4). Mali, Togo, and Benin form a class of medium-electricity access achievements of around 55% in 2022, which also have similar development rates over time. Côte d'Ivoire and Ghana have the highest electricity access rates in this group (approximately 70% and 85%, respectively) and have for a long time, probably due to economic standing.

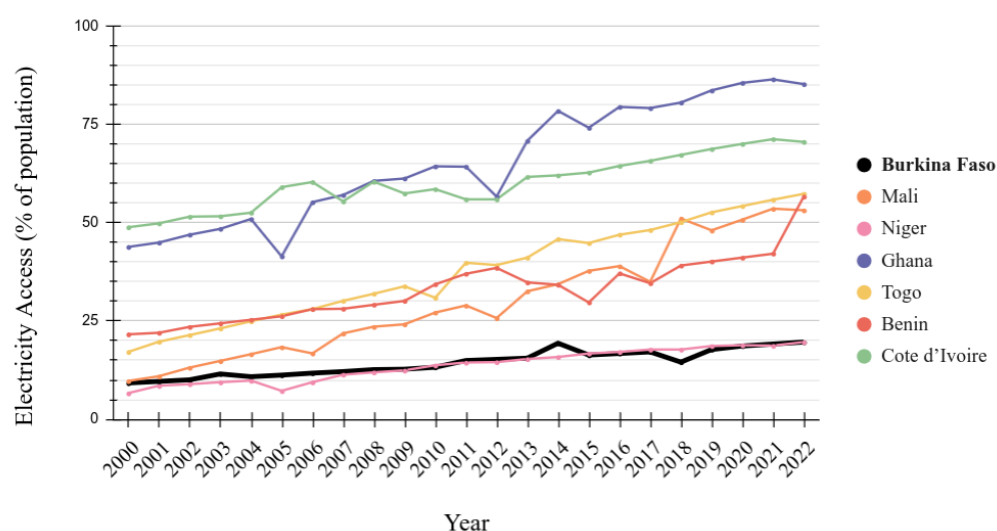


Figure 2.2.3: Electricity Access of Burkina Faso and its Neighbours [26]

Burkina Faso is also challenged with poor access to clean cooking fuels, relative to the world, but is keeping up with the SSA region, 17% in 2022 compared to the globe's 74% and Sub-Saharan Africa's 21% [28]. A similar disparity between urban and rural communities exists in clean cooking access as it did in electricity access where only 3% of rural people in Burkina Faso (and 6.6% in SSA) have access, while 48% of those in Burkina's urban communities (and 40% in SSA) have access. For clean cooking, Burkina is in the middle of the pack with respect to its neighbours, probably due to its experience with biogas and biofuels, despite all except one of its neighbours having comparable clean cooking statistics in the year 2000 [29]. Côte d'Ivoire and Ghana once again lead this set of countries, even exceeding the regional average, in a way that reflects their economic status.

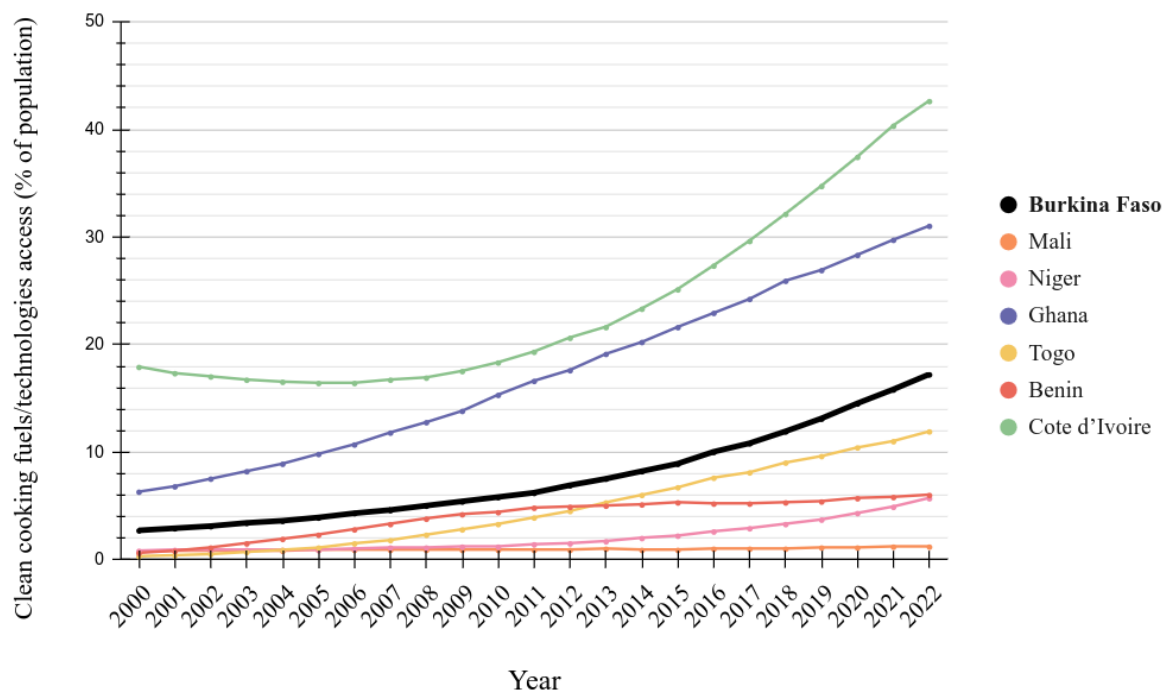


Figure 2.2.4: Clean Cooking Access of Burkina Faso and its Neighbours [29]

2.3 Current and Future Operating Infrastructure in the Electricity Sector

Burkina Faso's energy use is supported by the country's existing electricity generation infrastructure of roughly 340 MW of HFO and Diesel power plants, comprising 78% of Burkina's total capacity. This is followed by a 56 MW penetration of Solar PV assets (13%) and 35 MW of Hydro assets (8%). Since 2000, the oil assets have grown sevenfold, with solar investments starting in earnest in 2017.

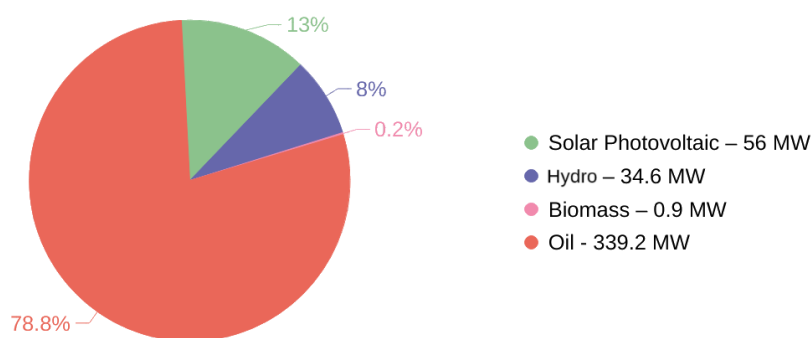


Figure 2.3.1: Power Plant Capacities by Type in 2020

In 2020, Burkina Faso imported electricity from three of its neighbours, Ghana, Côte d’Ivoire, and Togo, with almost all of the electricity coming from Ghana and Côte d’Ivoire at a ratio of 2:1 [30]. Both of the interconnecting transmission lines are rated at 327 MW with the Ghana-Burkina line being built in 2018 and the Côte d’Ivoire-Burkina Faso line existing since at least 2011 [31]. Burkina Faso is a country that currently only imports electricity.

Burkina Faso is a member of the West African Power Pool (WAPP) and as a part of overall system development in the region, there are planned capacity additions and interconnectors that concern Burkina Faso. The general guiding plan for this network is to take advantage of the massive solar energy potential in northern Sahelian countries such as Burkina and Niger, but also the lower marginal cost of electricity (due to natural gas power plants) in the larger countries in the southern part of the region like Ghana and Nigeria and to transmit power from north to south in the day and in the reverse direction at night. This arrangement benefits Burkina Faso as it will be able to lower its electricity costs by locally generating cheap solar, generating revenue through exports, and having energy security at night through imports, which would be affordable at times when power is needed most, without having to build all that generation infrastructure itself.

In terms of generation projects, it is planned that 600 MW of solar PV should be built between 2022 and 2030 and 75 MW of Wind in 2029. It was expected that by 2022, two more transmission lines would be built connecting Burkina Faso to Niger and Mali, with the line to Niger rated at 250 MW and the line to Mali rated at 760 MW. According to WAPP’s plan, power would flow both ways except for the Côte d’Ivoire line which would only see imports into Burkina Faso. This behaviour would continue until 2033 where exports to Côte d’Ivoire would start to occur. It is noted that not all imported and exported power is generated/consumed by Burkina Faso as Burkina could act as a transit country to send power across the region.

2.4 Energy and Climate Policies

Burkina Faso has several strategies that shape its energy sector ambitions. The Energy Sector Policy Letter 2016-2020 (LPSE) sought to stabilize the energy sector by making it more efficient, but also by incentivizing both fossil fuel electricity generation and renewable electricity [32]. Other goals included doubling the number of persons using on-grid electricity to 1 million, increasing installed capacity from 300 MW to 1000 MW (400 MW of which are fossil fuel power plants, and 300 MW of which are renewable electricity generation plants) increasing the electricity coverage rate to 80%.

Burkina Faso's Plan d'Action National des Energies Renouvelables (PANER), Renewable Energy Action Plan in English, contains targets that are interested in increasing energy access and increasing renewable energy use [32]. By 2030, according to the plan, the installed renewable electricity capacity should be 318 MW, generating 36% of electricity, nearly 27% of the rural population should consume electricity from renewable off-grid systems, and 79% of Burkina Faso residents should use efficient cookstoves and 32% should use butane as cooking fuel. There are also targets for solar water heater and biofuels use.

The nation's Plan d'Action National d'Efficacité Energétique 2015-2030 (PANEE), National Energy Efficiency Action Plan in English, also addresses energy efficiency in multiple sectors through standards and labelling [32]. PANEE by 2030 would see that grid losses be reduced to 10%, lighting savings amount to roughly 340 GWh/year, and that 90% of public and private buildings are designed or renovated with energy efficiency measures among other measures.

The National Plan for Economic and Social Development 2021-2025 (PNDES-II) the successor to the original PNDES, by 2025, aims to increase the electricity coverage rate to 56%, the total electrification rate to 35%, the rural electrification rate to 15%, the percentage of households on the grid to 35%, the percentage of people with electricity access to 60%, and the available power on the grid to 1500 MW [33].

Burkina Faso wishes to achieve, by 2030, 95% electricity access for urban areas and 50% for rural areas [34]. The nation has also made a Paris Agreement commitment (NDC) of reducing its GHG emissions by 29.42% (19.6% unconditionally, 9.8% conditionally) by 2030 compared to its BAU scenario, through mitigation [35]. In the NDC, the country has promised the construction of nearly 490 MW of solar PV and 10 MW of biomass-waste to energy plants before 2030. Burkina also seeks to achieve GHG emissions reductions through adaptation and resilience efforts, by another 30.8% (20.7% unconditionally, 10.1% conditionally) by 2030 compared to its BAU scenario. Additionally, Burkina Faso is party to Agenda 2063 whose mission is to bring about an African continent that is prosperous and grown sustainably, with its citizens living lives of high quality characterized by, inter alia, sustainable access to clean

and affordable energy (in line with SDG 7), water, transportation, and modern agriculture and infrastructure [36].

Burkina Faso's National Climate Change Adaptation Program aims to, among other things, improve public health, reduce the impact of climate change on the energy sector, protect persons and goods from climate events and natural disasters, and maintain food and water sustainability and accessibility, from now until 2050. Some actions/measures under this include: cultivating early varieties of drought-resistant crops, introducing community and participative management of forestry and fauna resources, diversifying energy sources, introducing water planning and management plans where climate forecasts predict a slight increase in rainfall, and increasing resilience of vulnerable communities and households to food and nutrition insecurity by improving their livelihood [37]. Other related programs to climate include the National Partnership Program for Sustainable Land Management, the National Sustainable Development Policy, and the National Multi-Risk Plan for Disaster Risk Preparedness and Response, which informed Burkina's NDC [16].

2.5 Literature Review and Research Gap

Hermann et al. performed a CLEWs study on Burkina Faso where it demonstrated that assessing the resource nexus between climate, land, energy and water is a useful endeavour at the national level when dealing with a country that is restricted in its use of resources, such as Burkina Faso [38]. The effect of agricultural policy was studied and it was found that increasing gains from the agricultural sector is strongly correlated with energy use, and energy use with water restrictions. Therefore, it was found that a cohesive strategy to address energy, water, and food was needed to increase productivity in all sectors.

Falchetta et al., in an effort to better estimate electricity demand projections for rural electrification, created the Multi-sectoral Latent Electricity Demand (M-LED) platform which processes geo-spatial information and applies it to energy impoverished areas as an application to combine big data with energy planning [4]. Kenya was used as a case for this application to investigate the local water-energy-agricultural-development nexus. Falchetta et al. found that this application can help policymakers prioritize development funds so that they have the greatest impact, such as accruing benefits in the agricultural sector from electrification.

Salhberg et al. used the Open Source Spatial Electrification Tool (OnSSET), which determines how to electrify unserved areas with a least-cost methodology, to examine the effect of myopia on modelling [39]. Pathways to universal electricity access in Somalia were studied using different timelines. The study found that longer timelines favour grid infrastructure development at least-cost (and higher demand) and shorter timelines favour mini-grid and stand-alone technology build-outs. It was also found that increasing the myopia of the model increased

the costs of the technology mixes and that planning for longer time-spans results in overall cheaper costs. This then suggest policies should be implemented to confirm the success of long-term electrification strategies. The results are also useful for other Sub-Saharan African nations.

Another paper by Salhberg et al. used Burkina Faso as a case study, applying scenario discovery to geospatial electrification modelling [40]. By combining seven variables, 1944 electrification simulations were examined. When examining these possible scenarios, it was found that a high electricity cost in-country would be due to a high grid cost and an intense (forced) grid-extension policy. It was then concluded that lowering the grid price and extending the grid in a less aggressive way would cheapen the cost of electricity in Burkina Faso. The discount rate was also found to be a source of higher electricity prices, when it was on the higher side. The study showed that this structured approach is capable of analyzing the effect of more parameters than a normal study. Although this method was good for reducing bias and handling more uncertainty, it requires additional computational resources.

Pappis et al. explores electrification strategies available to Ethiopia to achieve universal electricity access and its electricity requirements between 2015 and 2065 [41]. The authors combined the geospatial electrification prowess of OnSSET and the long-term cost optimization of OSeMOSYS to determine how policy and power technological advancement would affect Ethiopia's energy development through the analysis of several scenarios. Results include increasing electricity supply by increasing CO₂ emissions (through increased natural gas usage) and hydropower, wind, solar and geothermal power production. In terms of the prevalence of power systems used, nearly two-thirds of the population with use the grid to access electricity, with stand-alone systems supplying the majority of the remaining fraction, supplying remote areas. The results also demonstrate the role of policy, economic growth, and investment timing in affecting electricity demand, CO₂ emissions, and export revenues. The findings inform energy policy and strategic planning, emphasizing sustainable energy transition and regional electricity trade.

Pea Balderrama et al. expanded the estimation method for LCOEs of decentralized energy systems for large-scale geospatial electrification modeling, with OnSSET in particular [42]. The refinement comes from generating hourly demand profiles for different settlements using the open-source RAMP model, techno-economically optimizing the microgrid factors that affect the LCOE calculation like reliability targets and renewable generation variability, and creating a surrogate model based on multivariate regression which feeds into OnSSET for large-scale scenario analysis. This captures the real-world reliability and cost dynamics in electrification planning. The authors applied this framework to Bolivia's electrification strategy, which demonstrated that the results are more accurate with the refined framework than with the simpler approaches.

Parzen et al. feature the creation of an open-source global energy system model that performs analyses of large-scale power system planning with high spatial and temporal resolution called PyPSA-Earth [43]. It automatically integrates and cleans global geo-spatial datasets from OpenStreetMap, satellite data, and other open sources to create detailed network topologies, capacity potentials, and demand profiles. Through a modular workflow, the temporal and spatial scope can be adjusted to tailor-make customized energy system models with transmission grids and a range of generation and storage technologies, which can be used to perform transition, technology evaluation, diversification, and electricity market studies. The model was demonstrated and validated for Africa as a continent and for Nigeria, specifically for a 2020 power dispatch optimization and evaluating the country's target of net-zero emissions by 2060 which showed that PyPSA-Earth is suitable for data-driven electricity planning studies and providing insights to decision-makers.

Ouedraogo et al. investigates the feasibility of hybrid solar PV–diesel energy systems for the village of Pissila, Burkina Faso to provide cost-effective off-grid electricity in the country's rural areas [44]. After estimating local electricity demand and resource potential in the village of Pissila, the study compares three system options: pure diesel, pure PV, and a hybrid PV–diesel setup and compares them using the levelized cost of electricity (LCOE) technique, with cost determinations performed by the energy design software, HOMER. The optimum configuration was shown to be the hybrid PV-diesel configuration which yields the lowest LCOE, which is 54% cheaper than diesel-only power. Sensitivity analyses show that higher diesel fuel costs and discount rates increase electricity prices, but stronger solar radiation and declining PV module costs lower overall generation costs. Takeaways include well-designed policies like lower interest on loans, exemptions on value-added tax (VAT), and local financial and industrial investments can enhance the uptake of hybrid systems, make solar-based electrification more affordable, and develop rural areas in Burkina Faso.

Azoumah et al. explored how hybrid solar PV–diesel systems can sustainably meet electricity needs for rural and peri-urban populations in sub-Saharan Africa, and in Burkina Faso in particular [45]. The authors analyze hybrid systems that substitute diesel for locally produced biofuels and reduce reliance on batteries. Different hybrid configurations (such as series, switched, and parallel) are examined, and critical technical elements are discussed, including how to control and manage generation and loads. The concept is implemented at a school in Burkina Faso and shows how solar energy can lower diesel consumption, with added flexibility for biofuel integration. Hybrid systems were also shown to cut operating expenses and reduce subsidies tied to rising fossil fuel costs; simultaneously, the approach lowers environmental impact by decreasing carbon emissions and avoiding large-scale battery disposal. This overall results in affordable, locally sourced off-grid power for off-grid communities in Burkina Faso, supporting development and energy independence.

Abid et al. looked at how electricity access can be improved by merging solar-PV systems with energy storage [46]. Techno-economic feasibility analyses using HOMER software were performed to examine two scenarios: rural off-grid system and urban grid-connected electrified areas. Pumped hydro (PHS) and lithium-ion batteries were the energy storage options investigated. Results show that solar PV with PHS is cost-effective if suitable reservoir sites exist, whereas battery integration remains comparatively expensive due to higher capital costs and limited lifetimes. For remote rural areas, an off-grid solar-PHS system competes favorably with grid extension only if villages lie beyond about 100 km from existing transmission lines. Meanwhile, in urban settings, large-scale solar with PHS significantly lowers overall power costs. The cost of solar PV modules are a prevailing factor in the cost of electrification and are key to its reduction in Burkina Faso.

Studies and assessments have been commissioned by various bodies related to electricity access in Burkina Faso. Artelia Group drafted the Burkina Faso Electricity Master Plan (2020-2040) for use by Burkina Faso's Ministry of Energy [47]. It was a detailed plan that outlines strategic investments in generation, transmission, and distribution so that Burkina Faso could meet its growing electricity demand, featuring several demand scenarios. Renewable energy integration, rural electrification, and network stability were focal points. IRENA, in conjunction with the Ministry of Energy, Mines, and Quarries of Burkina Faso produced a Renewables Readiness Assessment report that identifies key drivers to accelerate Burkina Faso's energy transition (like affordable sustainable energy, rural energy access, economy diversification, renewable energy deployment and financing strategies) [32]. Recommendations from this report include updating the country's integrated resource plans with roadmaps for the implementation of energy storage, developing business models to electrify rural communities, and assessing the country's bioenergy potential. Burkina Faso's Ministry of Energy and SONABEL, with support from IFC, developed a roadmap that shows how Burkina Faso can increase its renewable energy penetration by using battery storage [48]. It suggests that deploying 60-70 MW of independent battery energy storage could significantly reduce reliance on fossil fuels and energy imports, leading to annual savings and reduced carbon emissions.

The literature review presented has some research gaps. There are a limited number of studies that use a hybrid OSeMOSYS-OnSSET model for electricity planning, not to speak of for Burkina Faso specifically. OSeMOSYS and OnSSET have been used separately for many countries on the African continent, but not many for both in tandem. In general, there is a lack of geo-spatial electrification studies that focus on Burkina Faso. Geo-spatial models have instead been applied for neighbouring countries. More should be explored in this regard, so as to investigate the interplay of being a landlocked country that is reliant on energy imports, and possesses a dispersed population, on electrification in Burkina Faso. Additionally, there is a lack of long-term electricity planning for Burkina Faso. Many reports focus on the medium-term of until 2040 but not projecting further (beyond 2040) restricts policy-makers' insights into

how projected population growth, evolving demand, and climate adaptation can affect energy adequacy into the future.

Also, the number of studies that look at hybrid electrification strategies using grid, mini-grid, and stand-alone systems are few. Most studies prefer to examine or optimize one or two, but not all three at once. Moreover, as a country, Burkina Faso currently finds itself importing a large fraction of its electricity from neighbouring countries, but many models neglect the importation aspect. More research is needed to ascertain the implications of energy trade with respect to: changes in regional energy trade prices, over-reliance on imports, and feasibility of increasing local generation. Current studies are also deficient when it comes to modelling the impact of different Burkinabè energy policies (e.g., PANER, PANEE, WAPP) and energy technologies like battery storage, green hydrogen energy storage and waste-to-energy solutions.

Chapter 3

Research Objective

3.1 Objective

The objective of this thesis is to assess least-cost electrification pathways for Burkina Faso developed with a geospatial, long term energy model. The analysis of these pathways will lead to policy recommendations that stakeholders can use to enact sustainable development strategies.

This thesis seeks to answer the following questions:

- *In each least-cost electrification pathway, what is the relative prevalence of the grid, mini-grid, and stand-alone technology in the electricity system, in electrified population share?*
- *By what percentage does the capacity and electricity mix change when changes in PV costs are modelled?*

3.2 Scope

The scope of this thesis is spatially limited to the national boundaries of Burkina Faso, with a temporal examination of the period 2021 to 2070. Within the national boundaries, the only energy system that will be investigated is the electricity system. However, the method developed in this thesis can be used for other national or regional studies. The expected results from this inquiry include:

- a soft-linking modelling framework connecting OSeMOSYS with OnSSET,
- an OSeMOSYS energy system model of Burkina Faso integrated with an OnSSET geo-spatial model,
- electricity capacity and generation mixes representing different fuels
- the prevalence of different electricity types to urban and rural communities,
- the effect of these generation mixes on CO₂ emissions, and

-
- key insights on system costs and their strategic value.

Chapter 4

Methodology

This chapter covers the methodologies and tools utilized in developing the country model in this degree project.

4.1 Open Source Energy Modelling System (OSeMOSYS)

OSeMOSYS, used for medium- and long-term energy planning, uses a bottom-up and least-cost methodology based on exogenously defined demand to find the optimum development of the energy infrastructure to generate an energy mix that meets that demand [49]. It also has the ability to analyze large spatial areas from countries to regions to continents. Its free and open-source nature lends transparency and ease of understanding to increase energy systems communications, insights, and capacity building. It is useful for investigating scenarios for varying combinations of costs, capacity, and other technical factors, and emission trends/expectations. The results of these models can prove useful to policy-makers and other agencies and companies in aiding in their recommendations and strategies. This thesis will make use of OSeMOSYS-PuLP, a Python implementation of OSeMOSYS, to facilitate linking to OnSSET, also written in Python [50].

4.2 Open Source Spatial Electrification Tool (OnSSET)

OnSSET takes advantage of location-specific data using GIS to determine the optimum residential electrification strategy to meet electrification goals for electrified and unelectrified urban and rural populations using grid, mini-grid (PV, Wind, Hydro, Diesel), and stand-alone (PV, Diesel) technologies [51]. Using the levelized cost of electricity (LCOE) of these delivery options, the least-cost option is chosen for every populated area in the study. OnSSET is also free and open-source allowing for flexibility in the construction of scenarios to model different policies.

4.3 Open Source Geo-spatial Energy Modeller (OSGEM)

OSGEM, the name given to the modeller created for this thesis, is a hybrid modelling tool that combines the functionality of OSeMOSYS and OnSSET and covers their independent weaknesses. OSeMOSYS within OSGEM has the objective of finding the least-cost grid development pathway, but OSeMOSYS lacks geo-spatial factors that might help it determine the best energy mix for smaller non-electrified settlements in a region. OnSSET within OSGEM is skilled in determining, given an electrification goal, what types of energy systems should be deployed in residential settings using a GIS-based approach. However, it is weaker when extending its sight into the far future as one would need more information about the future, inclusive of what the cost of electricity from the grid would be as OnSSET does not contemplate technologies powering the grid. Combining them, therefore, allows one to compensate for the oversight of the other. The soft-linking modelling approach is depicted in Figure 4.3.1.

The code for OSGEM and the documentation to use it is available on GitHub and the approach is as follows [52]:-

1. The long-term electricity model in OSeMOSYS is ran to find the cost-optimal electricity supply system for Burkina Faso, driven by demands representing every sector (ie. residential, commercial, industrial, etc.). At this stage, the contribution of demand from newly electrified settlements is nil as OnSSET has not yet performed its analysis.
2. The LCOE of the grid is calculated for each period (decade in this thesis) using the investment costs and calculated energy generation from the OSeMOSYS optimization.
3. A check is then made. If the newly calculated grid LCOEs are the same as (or close enough to, as defined by the user) the grid LCOEs used in the last iteration of the OnSSET analysis, then the optimization is considered complete and the loop will be exited.
4. Otherwise, the new LCOEs are added as inputs to OnSSET which will calculate the least-cost electrification technology mix to meet urban and rural residential on-grid and off-grid electricity demand.
5. The OnSSET model is ran for every period defined by the user (which is the same as the number of LCOEs calculated) in the study covering the same length of time that the OSeMOSYS results cover.
6. After the OnSSET optimization is performed, the demand from newly electrified settlements is extracted.
7. Another check is then performed. If the newly calculated demand from OnSSET is the same as (or close enough to, as defined by the user) the most previously calculated

OnSSET demand (from the last iteration of the analysis), then the optimization is considered complete and the loop will be exited.

8. If the loop continues, the additional demand from the newly electrified settlements is incorporated into the OSeMOSYS model for the next iterative run, which would change the set of grid LCOEs. This process will be repeated until exited.

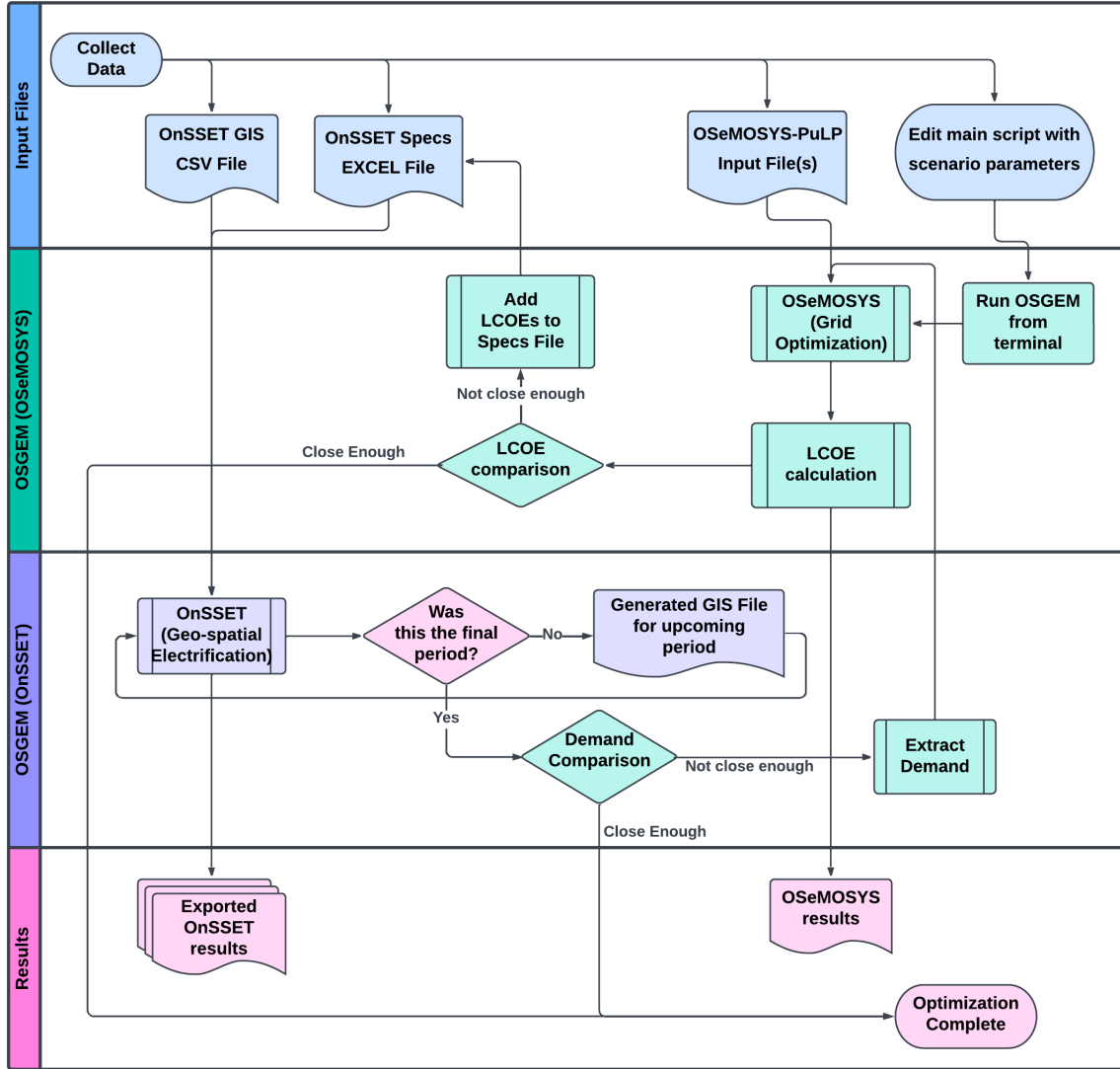


Figure 4.3.1: OSGEM Soft-Linking Modelling Approach

The LCOE will be calculated using the following for each technology:

$$LCOE = \frac{\sum_{t=1}^n \frac{I_t + O\&M_t + F_t}{(1+r)^t}}{\sum_{t=1}^n \frac{E_t}{(1+r)^t}} \quad (4.1)$$

Where in year t , I_t is the annualized investment cost, $O\&M_t$ is the operation and maintenance

cost in year t , F_t represents the fuel cost in year t , E_t is the electricity generated in that year t , r is the discount rate and n is the length of the period you want the LCOE to be calculated.

While OSeMOSYS and OnSSET have been soft-linked before (outputs manually inserted as inputs into each tool), the programmatic nature of OSGEM makes the hybrid modelling more convenient, automatically routing results as inputs into the respective tools. Parameters were added to the OSeMOSYS input files for identifying which technologies should contribute to the LCOE calculation, the annualized costs of existing assets relevant for the LCOE calculation, and the energy produced by the technologies relevant for the LCOE calculation. These input parameters allow for the results to be parsed for processing by the LCOE function in OSGEM. OSGEM also allows the user to predetermine the period over which the LCOE is calculated, for the associated OnSSET geo-spatial distributed energy system LCOE comparison. The user, additionally, is able to set which parameter and fuel newly electrified demand (from OnSSET) is applied to in OSeMOSYS. Further flexibility is introduced to easily convert OSeMOSYS units to OnSSET units and vice versa. To be able to achieve multiple OnSSET runs in succession, an update function was created to reproduce the original GIS and specification files but with update population data relevant to the (myopic) period of analysis. Convergence of OSGEM is determined by calculating the percent difference between the LCOEs and newly electrified demand of the previous iteration and the current iteration. The user also has the ability to tighten the threshold for convergence or introduce another method of convergence that fits the user's needs. Because these improvements are encapsulated in an open source framework, those in the open science community can further improve this work to satisfy their own needs as they see fit and add to the body of work. Validation of this work can be done by running the individual tools independently in their current form and performing the manual calculations for LCOE, GIS file updating and convergence. Also, validation can be done by performing previous OSeMOSYS-OnSSET studies with the OSGEM framework.

4.4 Reference Energy System

The reference energy system (RES) for this thesis is divided into four parts (Figure 4.4.1). Burkina Faso has a selection of sources, *primary energy sources*, that are currently available and that could be utilized in the future for electricity generation. Electricity from wind and biomass sources could be important to the energy mix as they represent a feasible renewable energy source when solar PV plants do not produce. Nuclear- and natural gas-sourced electricity could also be a significant player given offers from nations with nuclear energy experience and the prevalence of natural gas use in neighbouring countries. The RES also reflects the energy conversion of primary fuels to electricity by thermal, photovoltaic, and mechanical energy conversion plants, as a *secondary energy source*. This electricity is destined for the national transmission grid or localized micro-grids. The electricity delivered to the transmission grid is

met by electricity imported from neighbouring countries in the WAPP. *Tertiary energy* carriers represent the electricity that is delivered to the end-use sectors after losses to the grid. That may be for export to the WAPP network or to the main sectoral consumption areas: residential, commercial, industrial, and agriculture. The *final demand* in these sectors is what drives the optimization.

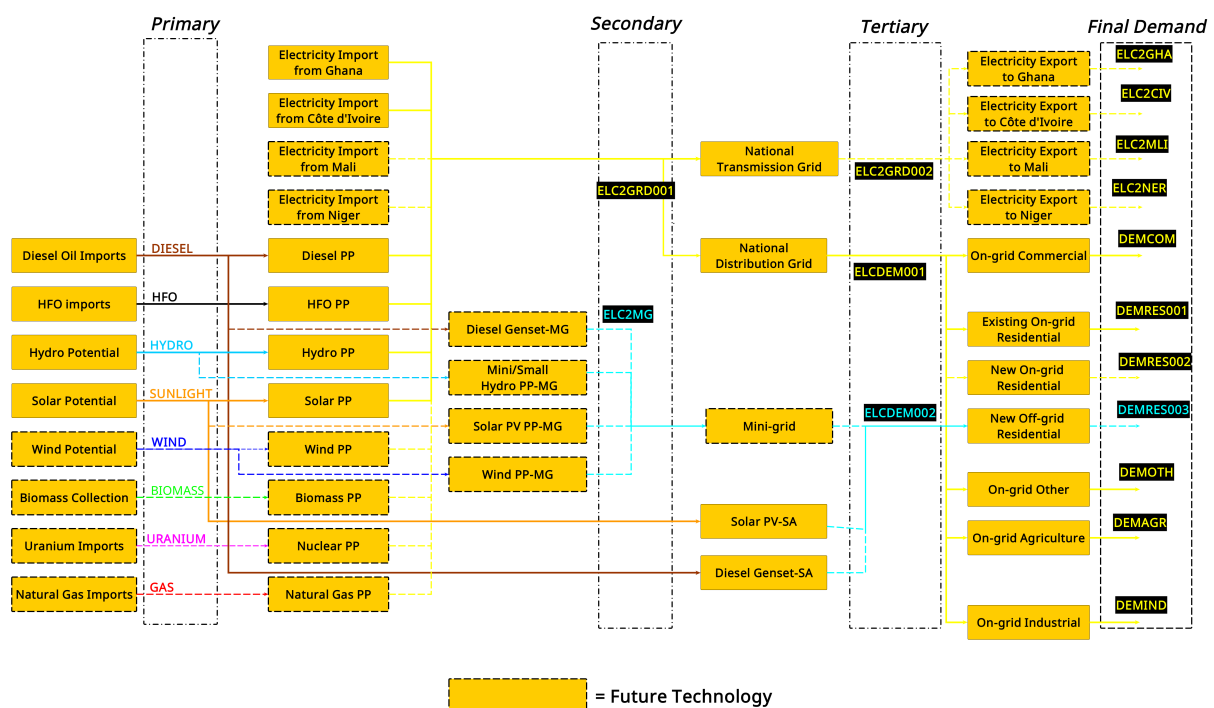


Figure 4.4.1: Reference Energy System

Table 4.4.1: Reference Energy System Fuels

Fuel	Description	Fuel	Description
HFO	Heavy Fuel Oil	DEMRES001	Existing On-grid Residential Demand
DSL	Diesel Oil	DEMRES002	New On-grid Residential Demand
BIO	Biomass	DEMRES003	New Off-grid Residential Demand
HYD	Hydro Energy	DEMOTH	Other Demand
SOL	Solar Energy/Sunlight	DEMAGR	Agriculture Demand
WND	Wind Energy	DEMIND	Industry Demand
GAS	Natural Gas	ELC2MG	Electricity to Minigrid
URN	Uranium	ELCDEM002	Mini-grid Electricity to Demand
ELC2GRD001	Generated Electricity sent to Centralized Transmission Grid	ELC2GHA	Electricity exported to Ghana
ELC2GRD002	Electricity to Centralized Distribution Network	ELC2CIV	Electricity exported to Cote d'Ivoire
ELCDEM001	Consumed Grid Electricity by All sectors	ELC2MLI	Electricity exported to Mali
DEMCOM	Commercial Demand	ELC2NER	Electricity exported to Niger

4.5 Model Structure

As OSGEM is a hybrid modeller, two model structures were created to facilitate OSeMOSYS and OnSSET analyses. The details are summarized in Tables 4.5.1 and 4.5.2. The model start year is 2021 in light of the availability of data in that year and the desire to remove the COVID-19 effect from the modelling process. The model ends in 2070 due to the availability of projected data. A time resolution of 48 timeslices for OSeMOSYS modelling was determined by partitioning the year into 4 seasons (namely January - March, April - June, July - September, and October - December) and 12 sections of a day representing 2 hour blocks in a 24 hour day. The discount rate is taken from a previous project completed in Burkina Faso by the African Development Bank [53]. Diesel and HFO resources potentials are treated as unlimited in their context of imports. Renewable resource potentials are taken from an IRENA renewables readiness report for Burkina Faso [32]. The OnSSET portion of the code is run myopically over the full model period to simulate the effect of making short-sighted decisions in a similar way as politicians do. Each myopic period lasts 10 years with midpoints of 5 years. The average household size, and population projections are taken from GlobalDataLab and the UN, respectively [18, 54]. The residential electricity consumption tiers are inherited from the GIS data file used in this thesis from previous work done at KTH division of Energy Systems [55].

Table 4.5.1: Model Structure in OSeMOSYS [32, 53]

Modelling Period	2021 to 2070
Time Resolution	48 timeslices
Discount Rate	12%
Reserve Margin	15%
Resource Potentials	
Diesel	99.9 million GWh
HFO	99.9 million GWh
Hydro	0.13 GW
Solar	95 GW
Wind	1.96 GW
Biomass	1 GW
Nuclear	Scenario Dependent
Natural Gas	Scenario Dependent

Table 4.5.2: Model Structure in OnSSET [18, 54, 55]

Modelling Period	2021 to 2070
Number of Scenarios/Periods	5
Myopia (Length of Period)	10 years
Discount Rate	12%
Average Household Size	<i>persons/household</i>
Rural	6.88
Urban	5.1
Base Year Electrification Rate	
Total	19%
Rural	3%
Urban	60%
Residential Electricity Consumption Tiers	<i>kWh/capita/year</i>
1	8.15
2	46.1
3	169
4	446
5	630
6 (custom)	260
Population (Urban Population Ratio)	<i>million people (%)</i>
2021	20.90 (30.31)
2030	27.38 (37.12)
2040	34.95 (43.72)
2050	43.21 (50.17)
2060	52.50 (57.02)
2070	62.63 (63.62)

4.6 Energy Supply Technologies and Fuels

Tables 4.6.1 and 4.6.2 present a summary of the techno-economic characteristics of the technologies used in this study contributing to the resulting capacity, energy mix, and technology configurations in Burkina Faso. At present, most of the capital costs of existing

and possible technologies are similar, with biomass and hydropower plants having elevated costs and nuclear power plants having much higher costs. Over time, solar and wind costs are expected to decrease. Of the fossil fuels, HFO and natural gas fuel costs are comparable while diesel is 66% more expensive. Fossil fuel prices are expected to increase in the future. The technologies are available for use for most of the year, with solar and wind being virtually always available, given their minimal maintenance. Concerning the technology lifetime, most technologies are operable for roughly 25-30 years with hydro and nuclear power plants having lifetimes of 50 years, extending their ability to provide electricity far into the future. Data is sourced from CCG Starter Data Kit for Burkina Faso, and an updated TEMBA model [56, 57]. Techno-economic data for the OnSSET portion of the analysis comes from OnSSET defaults and the Global Electrification Platform collection for Burkina Faso [58].

Table 4.6.1: Technologies and their Techno-economic Characteristics in Burkina Faso (OSeMOSYS) [56, 57]

Name of Technology	Existing Capacity	Capacity Factor	Efficiency	Availability	Technical Lifetime	Capital Cost	Fixed O&M	Variable O&M	Fuel Cost
	<i>MW</i>	<i>%</i>	<i>%</i>	<i>%</i>	<i>Years</i>	<i>USD/MW</i>	<i>USD/GWh,e</i>	<i>USD/GWh,e</i>	<i>USD/GWh,f</i>
Diesel PP	34	85	35	90	25	1,200,000	35,000	0.00001	53,590
HFO PP	355	85	35	90	25	1,200,000	45,000	0.00001	32,230
Hydro PP	35	35	–	95	50	3,000,000	90,000	0.00001	–
Solar PP	57	18	–	100	24	1,299,200	16,890	0.00001	–
Wind PP	–	19	–	100	25	1,429,400	57,176	0.00001	–
Biomass PP	–	49	35	93	30	2,500,000	75,000	0.00001	–
Nuclear PP	–	85	33	90	50	6,137,000	184,110	0.00001	510
Natural Gas PP	–	85	48	90	30	1,200,000	35,000	0.00001	31,570

Table 4.6.2: Technologies and their Techno-economic Characteristics in Burkina Faso (2030, OnSSET) [58]

Name of Technology	Capital Cost	O&M costs	O&M of Transmission and Distribution Lines	Connection Cost per Household	Capacity Factor	Technical Lifetime
	<i>USD/kW</i>	<i>%</i>	<i>%</i>	<i>USD/household</i>	<i>%</i>	<i>years</i>
Grid	–	–	0.0001	152	95	50
Mini-grid Hydro	3000	3	0.0001	100	35	50
Mini-grid Wind	2800	4	0.0001	100	–	25
Mini-grid PV	1033.44	1.3	0.0001	100	–	24
Stand-alone PV >1kW	6950	1.3	–	–	–	24
Stand-alone PV >0.1kW and <1kW	4470	1.3	–	–	–	24
Stand-alone PV >0.05kW and <0.1kW	6380	1.3	–	–	–	24
Stand-alone PV >0.02kW and <0.05kW	8780	1.3	–	–	–	24
Stand-alone PV <0.02kW	9620	1.3	–	–	–	24
Mini-grid Diesel	261	3	0.0001	100	70	25
Stand-alone Diesel	261	3	–	–	90	25

4.7 Scenarios

The analysis of Burkina Faso electricity system occurs in three scenarios: BAU (Business-as-Usual), BAL (Balanced), and AMB (Ambitious). These scenarios vary in load expectations, emission limits, renewable energy targets, technological availabilities, and trade, which are explained below.

4.7.1 BAU (Business-as-Usual)

BAU represents the evolution of Burkina Faso's electricity sector according to historical trends, existing plans for the WAPP as a whole, and Burkina's place in them. All technologies except natural gas and nuclear technologies are utilized. There are no emissions limitations, and there are no prescribed renewable energy targets or limits.

To model a reasonable rate of technology uptake:-

- wind turbine additions to the grid were limited to 100 MW per year from 2024-2034, and
- biomass power plant additions to the grid were limited to 35 MW per year from 2024-2034.

BAU Demand Projections and Considerations

The demands in the electricity consumption sectors, namely industrial, commercial, agriculture, residential, and other sectors, drive the optimization in OSGEM as summarized in Table 4.7.1. The demands for all sectors except the residential sector were projected using data from IRENA that estimated the growth in electricity demand between 2021 and 2030 and that growth was continued at the same pace until 2070 (320 GWh/year) [32]. The distribution of that 320 GWh was split according to the base year share of electricity between those sectors, which was maintained until 2070. Those demands and inputted into the OSeMOSYS portion of OSGEM. In OSGEM, the residential demand for the OSeMOSYS portion of code is determined by the output given by OnSSET, but in the initial run, the demand is the same for all years as it is in the base year. Industrial and Other demand sectors have the strongest influence over the optimization.

For demands in OnSSET, only residential is considered, where demand tiers were set for the urban and rural populations. Given a determined specific demand for Burkina Faso of 160 kWh/year/capita in the base year, a combination of tiers were selected to main that specific demand until 2070 in this BAU scenario. The historical electrification growth rate (roughly 5% per decade) was also carried through to 2070. For BAU, a modest increase in the maximum grid extension distance was chosen that doesn't expect much in extension capacity in the near future but increased a bit towards the end of the modeling period to reflect sluggish growth.

Table 4.7.1: BAU Demand Projections and Considerations

(a) OSeMOSYS

Sector	2021	2030	2040	2050	2060	2070
	<i>GWh</i>	<i>GWh</i>	<i>GWh</i>	<i>GWh</i>	<i>GWh</i>	<i>GWh</i>
Industrial	1334	2774	4374	5974	7574	9174
Commercial	347	776	1137	1553	1969	2385
Agriculture	53	111	175	238	302	366
Other (excl. Residential)	560	1165	1837	2509	3181	3853
Residential	374	–	–	–	–	–

(b) OnSSET

Period Ending	2030	2040	2050	2060	2070
Electrification Tier Target (Urban)	3	3	3	3	3
Electrification Tier Target (Rural)	1	1	1	1	1
	%	%	%	%	%
Electrification Rate Target (mid-period)	20.88	25.58	30.28	34.98	39.68
Electrification Rate Target (end year)	23.23	27.93	32.63	37.33	42.03
	<i>km</i>	<i>km</i>	<i>km</i>	<i>km</i>	<i>km</i>
Maximum Grid Extension Distance	50	75	100	100	100

BAU Energy Trade

Energy trade plans relating to Burkina Faso are taken from the WAPP Master Plan and the implications on the modelling procedure is summarized in Tables 4.7.2 and 4.7.3. In the initial year of the model period, Burkina Faso imports electricity from two neighbouring countries apart of WAPP, Ghana and Côte d'Ivoire, using 327 MW transmission lines each, with no exports to them. The WAPP Master Plan is split into three milestone years, 2022, 2025, and 2033. By 2022, new imports are taken from Mali and Niger with newly built lines of 250 MW and 760 MW capacities, respectively. Imports are expected to fall from the year from Ghana and Côte d'Ivoire as that load is shared with Mali and Niger. Imports are expected to increase generally over time up to 2033 as the general plan for WAPP evolves. The strategy is that cheap hydro and solar energy are generated in the more northern countries in the day

with the intention of exporting to the more southern countries, then when night comes, the more southern countries with natural gas power plants can export that lower-cost electricity to the more northern countries. Exports are higher in 2033 to Mali and Niger as Burkina Faso acts as a medium to transmit power from countries like Nigeria to the Sahelian nations. Due to a lack of information after 2033, the import limit and export demand remain the same from 2033 to 2070. For the years in between the main years given by the tables, the data is linearly interpolated.

Table 4.7.2: BAU Electricity Import Maximum Limit [31]

Exporting Country	2021	2022	2025	2033	2070
	<i>GWh</i>	<i>GWh</i>	<i>GWh</i>	<i>GWh</i>	<i>GWh</i>
Ghana	707	241	362	994	994
Côte d'Ivoire	381	153	310	300	300
Mali	0	514	236	692	692
Niger	0	369	681	423	423

Table 4.7.3: BAU Electricity Export Demand [31]

Importing Country	2021	2022	2025	2033	2070
	<i>GWh</i>	<i>GWh</i>	<i>GWh</i>	<i>GWh</i>	<i>GWh</i>
Ghana	0	266	266	167	167
Côte d'Ivoire	0	0	0	9	9
Mali	0	71	71	247	247
Niger	0	26	26	337	337

4.7.2 BAL (Balanced)

BAL builds upon BAU but possesses a different demand projection as explained below, in general portraying a higher rate of development for Burkina Faso. In addition to the changes in demand, the following changes are made:

- Minimum of 27% of electricity generated by renewable energy sources by 2030 (according to PANER [32])
- Natural Gas and Nuclear generation technologies are available from 2040, and
- Implementation of emission limits according to Burkina Faso's NDC

To model a reasonable rate of technology for the new technologies in this scenario uptake:-

- natural gas additions to the grid were limited to 600 MW per year from 2040 - 2049, and
- nuclear power plants were limited to 500 MW for the whole modelling period after 2040.

The REMinProductionTarget was supplied with a linear interpolation between an 18% RE penetration in 2021 and 27% RE target in 2030 and 27% thereafter until the end of the modelling period. The rate of technology uptake for wind and biomass is the same as they were in BAU. Natural gas is included in this scenario as Burkina Faso's regional neighbours have access to natural gas and conceivably Burkina Faso could contemplate this technology. Nuclear energy is also modelled as an option for Burkina Faso as nuclear is present in West Africa already and as the country is strengthening ties with Russia, a nuclear power, it is possible that closer cooperation could lead to the introduction of that technology.

BAL Demand Projections and Considerations

The demands that are implemented (in the OSeMOSYS portion of OSGEM) for the consumption sectors (save residential) are the mean values of those from the BAU and AMB scenarios (see more about the demand in AMB in the below sections) and are summarized in Table 4.7.3. Residential demand in the OSeMOSYS portion is still given due to the optimization from the OnSSET portion of the code. For demands in OnSSET, electrification tier targets are increased to the highest tier over the modelling period, with urban populations reaching tier 5 by 2050 and rural populations by 2060, targeting high electricity usage rates for everyone (100% electrification) by 2065. The ability of the government/utility to extend the grid is also increased over the BAU reflecting a balanced approach to strong electricity consumption in time for Agenda 2065.

Table 4.7.3: BAL Demand Projections and Considerations

(a) OSeMOSYS

Sector	2021	2030	2040	2050	2060	2070
	<i>GWh</i>	<i>GWh</i>	<i>GWh</i>	<i>GWh</i>	<i>GWh</i>	<i>GWh</i>
Industrial	1334	2858	5877	8890	11683	14471
Commercial	347	743	1528	2313	3038	3762
Agriculture	53	114	235	356	467	578
Other (excl. Residential)	560	1200	2468	3736	4907	6078
Residential	374	–	–	–	–	–

(b) OnSSET

Period Ending	2030	2040	2050	2060	2070
Electrification Tier Target (Urban)	3	4	5	5	5
Electrification Tier Target (Rural)	2	3	4	5	5
	<i>%</i>	<i>%</i>	<i>%</i>	<i>%</i>	<i>%</i>
Electrification Rate Target (mid-period)	26.36	44.77	63.18	81.59	100.00
Electrification Rate Target (end year)	35.57	53.98	72.39	90.80	100.00
	<i>km</i>	<i>km</i>	<i>km</i>	<i>km</i>	<i>km</i>
Maximum Grid Extension Distance	50	100	150	150	150

BAL Energy Trade

The energy trade situation in the BAL scenario is the same as it is in the BAU scenario with Tables 4.7.4 and 4.7.5 summarizing the same.

Table 4.7.4: BAL Electricity Import Maximum Limit [31]

Exporting Country	2021	2022	2025	2033	2070
	<i>GWh</i>	<i>GWh</i>	<i>GWh</i>	<i>GWh</i>	<i>GWh</i>
Ghana	707	241	362	994	994
Côte d’Ivoire	381	153	310	300	300
Mali	0	514	236	692	692
Niger	0	369	681	423	423

Table 4.7.5: BAL Electricity Export Demand [31]

Importing Country	2021	2022	2025	2033	2070
	<i>GWh</i>	<i>GWh</i>	<i>GWh</i>	<i>GWh</i>	<i>GWh</i>
Ghana	0	266	266	167	167
Côte d’Ivoire	0	0	0	9	9
Mali	0	71	71	247	247
Niger	0	26	26	337	337

4.7.3 AMB (Ambitious)

AMB builds upon BAU but possesses a different demand from BAU or BAL, as explained below, and in general, portrays a Burkina Faso very interested in strong self-development, using all the resources at its disposal. In addition to the changes in demand, the following changes are made:

- Minimum of 48% of electricity generated by renewable energy sources by 2030 (according to ECREEE [32])
- Natural Gas and Nuclear generation technologies are available from 2035,
- Implementation of emission limits that reflect net-zero emissions by 2050 according to Burkina Faso’s Vision 2050 [59], and
- No electricity exports and unlimited imports from 2035

The changes in the rate of technology uptake are as follows:-

- natural gas additions to the grid were limited to 600 MW per year from 2035-2040, and
- nuclear power plants were limited to 500 MW from 2035-2039 and are unlimited for the remainder of the modelling period.

The REMinProductionTarget was supplied with a linear interpolation between an 18% RE penetration in 2021 and 48% RE target in 2030 and 48% thereafter until the end of the modelling period. The rate of technology uptake for wind and biomass are the same as they were in BAU.

AMB Demand Projections and Considerations

The demands that are implemented (in the OSeMOSYS portion of OSGEM) for the consumption sectors (save residential) are values generated from a JRC report concerning energy projections for African countries, and are summarized in Table 4.7.6 [60]. Electricity consumption grows by 357 GWh/year on average until 2030, when the growth rate increased to 887 GWh/year on average until 2050, when it decreased slightly to 794 GWh/year until 2070. The growths were distributed to the sectors according to their share in total consumption in the base year. The output given by OnSSET once again determines the residential demand for the OSeMOSYS portion of code, but in the initial run, the demand is the same for all years as it is in the base year. For demands in OnSSET, given the ambitious nature of the scenario, the top electrification tier targets are targeted to be reached more quickly than in the BAL scenario, with both urban and rural populations reaching tier 5 by 2040 and universal electricity access targeted to be reached by 2035, due to the 95% access rate being a goal by the country by 2030. The ability of the government/utility to extend the grid is also further increased to the point of being able to reach large pieces of the country more quickly than in the BAL, overall capturing fast invested growth.

Table 4.7.6: AMB Demand Projections and Considerations

(a) OSeMOSYS

Sector	2021	2030	2040	2050	2060	2070
	<i>GWh</i>	<i>GWh</i>	<i>GWh</i>	<i>GWh</i>	<i>GWh</i>	<i>GWh</i>
Industrial	1334	2942	7381	11820	15794	19768
Commercial	347	764	1919	3073	4106	5139
Agriculture	53	117	295	472	631	790
Other (excl. Residential)	560	1236	3100	4964	6633	8303
Residential	374	–	–	–	–	–

(b) OnSSET

Period Ending	2030	2040	2050	2060	2070
Electrification Tier Target (Urban)	4	5	5	5	5
Electrification Tier Target (Rural)	3	5	5	5	5
	<i>%</i>	<i>%</i>	<i>%</i>	<i>%</i>	<i>%</i>
Electrification Rate Target (mid-period)	52.78	100.00	100.00	100.00	100.00
Electrification Rate Target (end year)	95.00	100.00	100.00	100.00	100.00
	<i>km</i>	<i>km</i>	<i>km</i>	<i>km</i>	<i>km</i>
Maximum Grid Extension Distance	50	100	200	200	200

AMB Energy Trade

The energy trade situation is the same until 2035 as the situations in the BAU and BAL scenarios. From 2035 onward, each transmission line is freed up to import as much electricity as is economically best in the determination of OSGEM and the lines are only used for import. This is summarized in Tables 4.7.7 and 4.7.8

Table 4.7.7: AMB Electricity Import Maximum Limit [31]

Exporting Country	2021	2022	2025	2033	2070
	<i>GWh</i>	<i>GWh</i>	<i>GWh</i>	<i>GWh</i>	<i>GWh</i>
Ghana	707	241	362	994	99,999,999
Côte d'Ivoire	381	153	310	300	99,999,999
Mali	0	514	236	692	99,999,999
Niger	0	369	681	423	99,999,999

Table 4.7.8: AMB Electricity Export Demand [31]

Importing Country	2021	2022	2025	2033	2070
	<i>GWh</i>	<i>GWh</i>	<i>GWh</i>	<i>GWh</i>	<i>GWh</i>
Ghana	0	266	266	167	0
Côte d'Ivoire	0	0	0	9	0
Mali	0	71	71	247	0
Niger	0	26	26	337	0

4.8 Sensitivity Analysis

Sensitivity analyses were performed by varying the *net zero year* and the *capital cost of solar PV panels* in the AMB scenario.

The net zero year puts pressure on the electricity system to cease its use of fossil fuels. Fossil fuels have been prevalent in this scenario, and it would be interesting to see what the electricity sector looks like under an increasingly relaxing net zero regime. The years were chosen in 5 year increments between 2050 and 2070.

Given Burkina Faso's massive potential to generate electricity from solar resource, solar PV technology is expected to be featured heavily in the electrification pathway determined for Burkina Faso. Given this prevalence, it is interesting to see how changes in the capital cost on the grid supply side and on the local off-grid side simultaneously affect the technology distribution and energy mix.

Chapter 5

Results

This chapter presents the results from the Business-as-Usual (BAU), Balanced (BAL), and Ambitious (AMB) scenarios that were optimized using OSGEM which examine the effects of changing demand, levelized cost of electricity (LCOE), costs, emission limits and renewable energy penetration targets on the energy and capacity mix in the electrification pathway for Burkina Faso. Discussion and analysis will follow in the subsequent chapter.

5.1 Business-as-Usual (BAU)

As a result of the OSGEM optimization for the BAU scenario explained in the methodology, Figure 5.1.1 presents the annual electricity production according to the generation sources. The first 5 years of the modelling period sees a domination by HFO. HFO's influence, however, is decreasing over time as electricity from imports, solar and biomass begins to ramp up. Burkina Faso, in this scenario, is reliant upon HFO, imports and solar until 2035, which is when wind substantially enters the electricity mix. From here on, electricity sourced from wind is used at full national capacity until the end of the modelling period and the model sources electricity from biomass at faster rates to continue to meet demand. This quick uptake in biomass occurs until about 2046 when it plateaus while HFO's contribution becomes almost nil. After this inflection point, HFO's prominence appears again, increasing from near nil to a sizable portion of the electricity mix in 2070 (but definitely not the same share as in 2021), with an increased rate analogous to the rate of increase in demand. Electricity imports remain steady at their maximum use, while biomass remains steady as well. Electricity from hydro which was not so relevant prior to 2046, is used at maximum capacity until the end of the modelling period. Solar ever increases in prominence, but its rate of growth is less than that of HFO while always maintaining a greater contribution to the electricity mix than HFO until 2070 in this scenario.

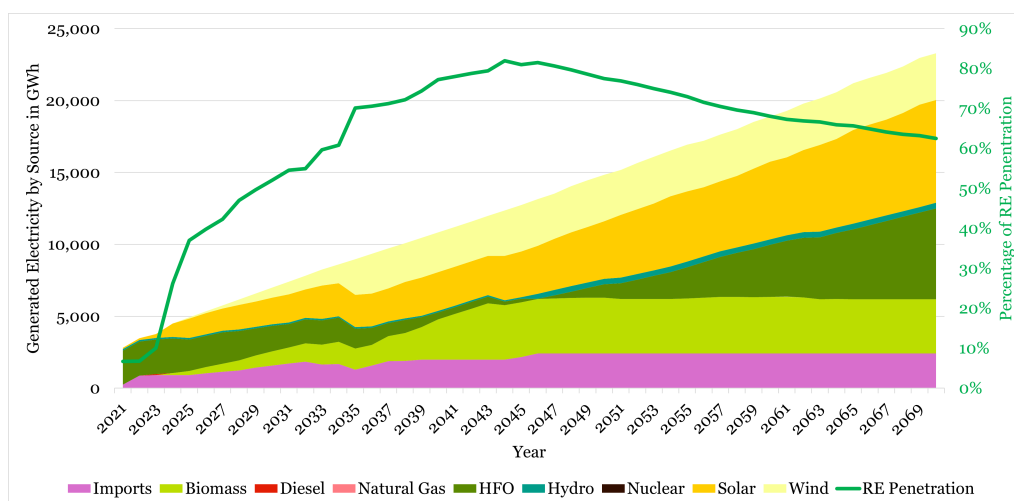


Figure 5.1.1: BAU Annual Production by Technology

The OSGEM optimization for the BAU scenario also yielded the installed capacity that exists annually according to generation source as displayed in Figure 5.1.2. At the beginning of the modelling period, nearly all installed capacity are fueled by HFO and diesel as any shortfall is made up by imports. Coinciding with the energy contribution, the solar capacity shoots up and makes up the majority of installed capacity by 2025. By this time, wind starts to come on scene and gradually increases until experiencing a jump in 2035. Meanwhile, solar and biomass capacity also grows gradually, while HFO and diesel assets gradually decrease to near nil in 2046. The inflection point is once again seen in 2046 where biomass capacity stabilizes and HFO capacity grows, although at a slower rate than the overall growth rate in total capacity in this scenario. Wind capacity is also relatively stable while capacity growth in solar power plants more closely match the over rate of growth in capacity totally. Post-inflection point also see the growth and maxing out of hydro power plants.

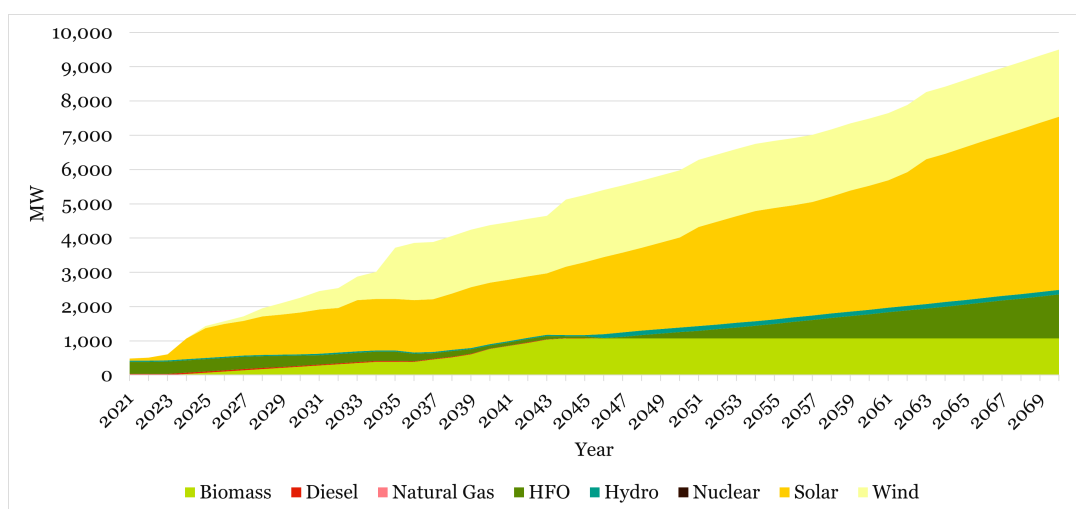


Figure 5.1.2: BAU Total Annual Capacity by Technology

A zoomed-in view of electricity imports in BAU portrayed in Figure 5.1.3 shows that electricity is imported from Côte d'Ivoire, Ghana and Mali throughout the whole modelling period. Most electricity comes from Ghana, which grows from the model's inception until 2033 and maintains that level until 2070. Fluctuations in imports are seen in electricity from Mali which dips to 0 for one year (2035) before maxing out and stabilizing thereafter. Imports from Niger are not utilized until 2046 which plateau two years later and the plateau remains until 2070.

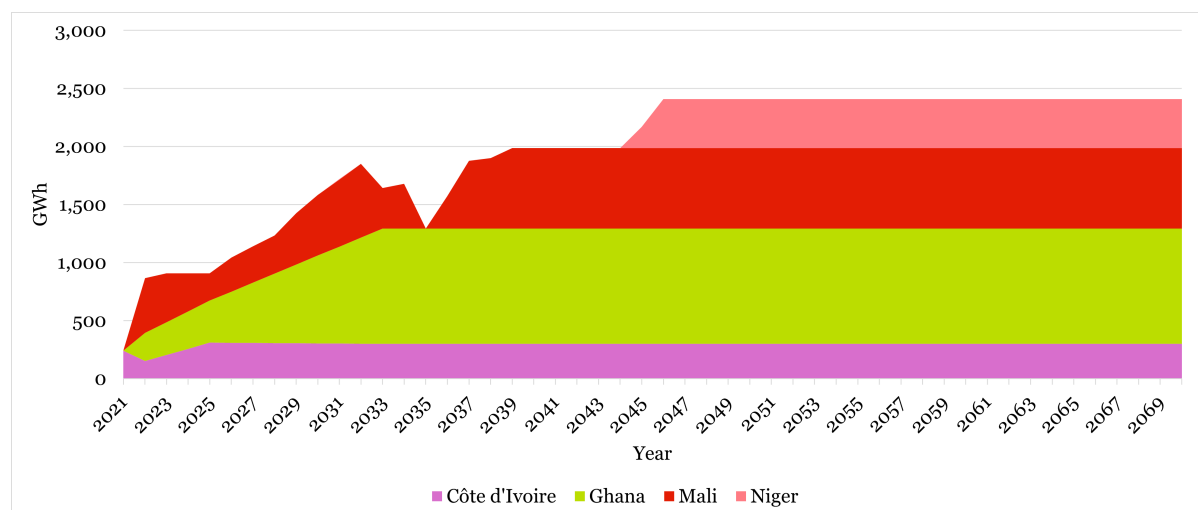


Figure 5.1.3: BAU Electricity Imports by Country

Figure 5.1.4 displays the annual emissions from electricity generated for Burkina Faso in the BAU scenario. The emissions hover around 2000 kTon CO₂eq until 2025. Thereafter, it decreases gradually in a linear fashion with small plateaus around 2026 and 2033, until 2040 to 449 kTon CO₂eq. In the early 2040s, the emissions increase to roughly 800 kTon CO₂eq before dropping to just 46 kTon CO₂eq in 2046, the lowest of the modelling period. After 2046, the emissions trend up linearly to around 5000 kTon CO₂eq by 2070.

The carbon intensity of the electricity generated in Burkina Faso follows a similar trend. In 2021, the carbon intensity amounted to 0.68 g CO₂eq per kWh. From then on, the carbon intensity sharply decreases to 0.36 g CO₂eq per kWh by 2025 and 0.18 g CO₂eq per kWh by 2031. After 2031, the changes in carbon intensity closely follow the trend in the emissions, where 2040 sees a carbon intensity of 0.04 g CO₂eq per kWh but increases in the years soon after to around 0.07 g CO₂eq per kWh before decreasing to 0.005 g CO₂eq per kWh in 2046. After 2046, the carbon intensity grows again linearly to around 0.22 g CO₂eq per kWh by 2070.

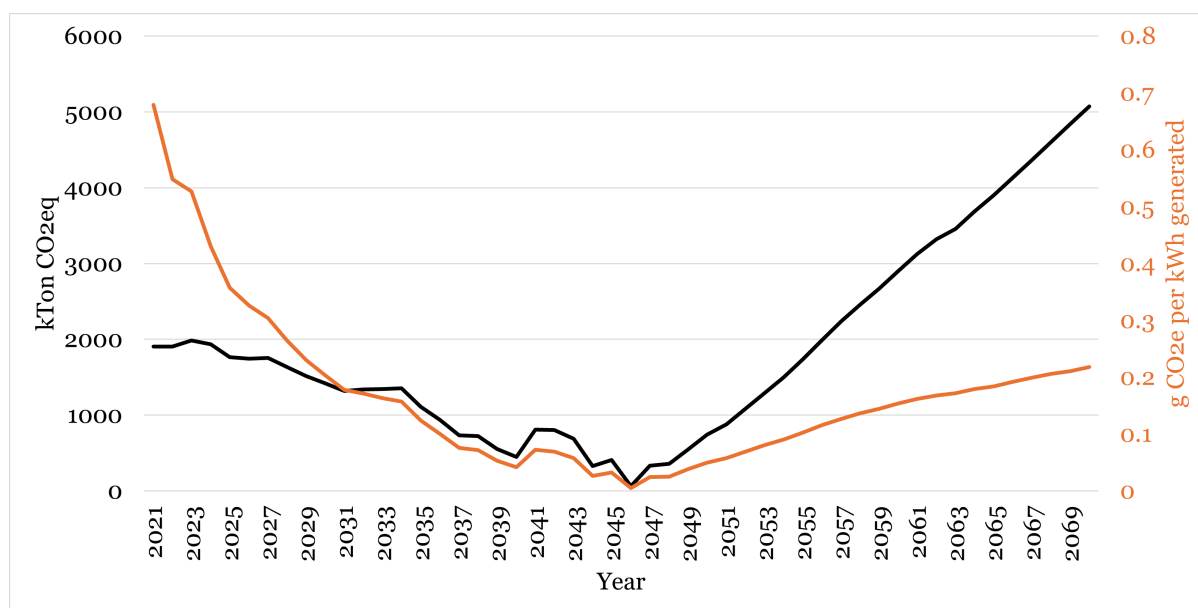


Figure 5.1.4: BAU Electricity Emissions

The annual capital investments into the grid for the BAU scenario is shown in Figure 5.1.5. In the first decade of investments, there is steady investment in biomass technology at nearly 100 million USD per annum, while solar technologies see a high peak of investment around 400 million USD in 2024. Wind investments see its major spikes in 2035 and 2060 at roughly 600 million USD each time. Most investments between 2035 and 2046 are made in biomass, coinciding with the uptake in biomass use seen in the graphs prior. In the latter half of the modelling period, we see the same trends that are seen in annual production and total capacity; hydro investments occur between 2046 and 2050, HFO power plant investments are steady on the level of 70 - 80 million USD per year and constant investments into solar to meet demand.

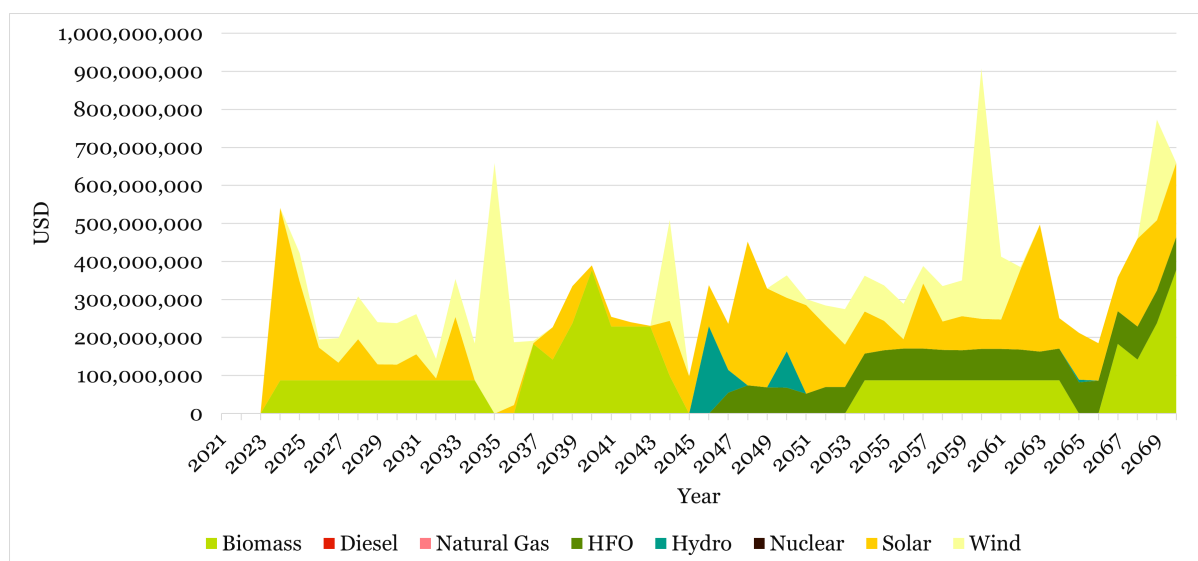


Figure 5.1.5: BAU Grid Capital Investment

Given the production from generation sources and their associated costs, OSGEM has calculated the LCOEs for each myopic period of the optimization, the values of which are presented in Table 5.1.1. The LCOE of the grid decreases in until 2050 where it remains virtually the same until 2060 and increases again slightly to 2070, but it still remains lower than the LCOE for the period ending in 2040.

Table 5.1.1: BAU LCOEs (Grid Cost)

Year	LCOE (USD/kWh)
2021-2030	0.138
2031-2040	0.093
2041-2050	0.056
2051-2060	0.055
2061-2070	0.065

In the BAU scenario, OSGEM yielded the annual amount of demand that newly electrified residences/settlements have added to the grid, as shown in Figure 5.1.6. Demand increases almost linearly, accelerating over time, going from 560 GWh in 2025 to 2564 GWh in 2050 and 5193 GWh in 2070. This newly electrified load made up an increasing share of the demand throughout the modelling years, from 12% in 2025 to 18% in 2050 and 23% in 2070. The number of loops that the optimization needed to converge was two before the change in LCOE between runs was less than the threshold for change.

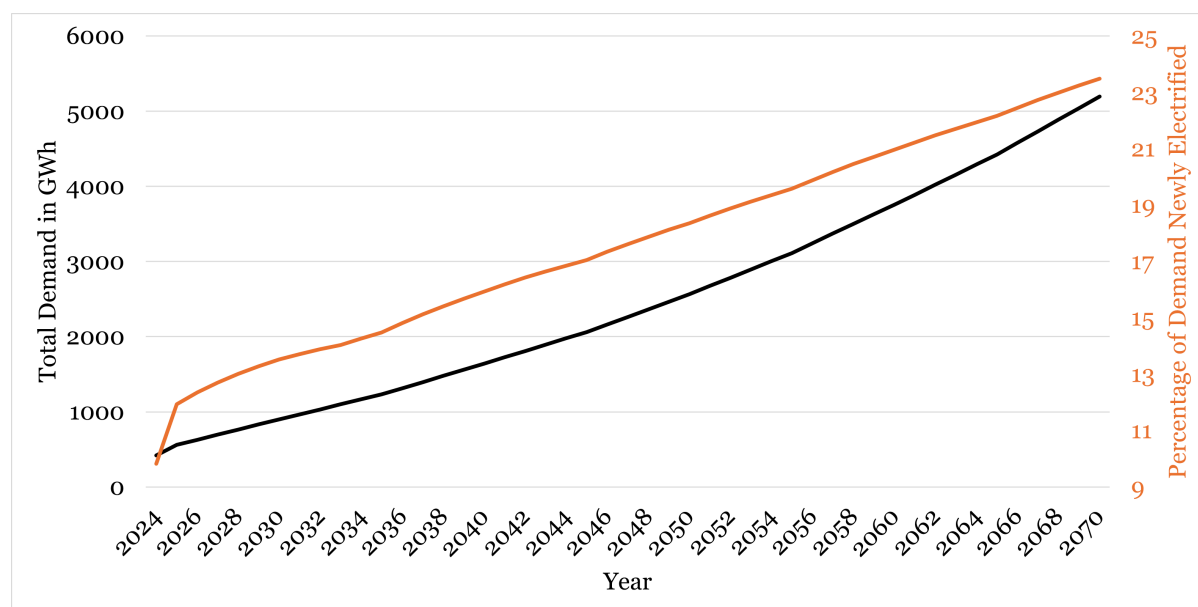
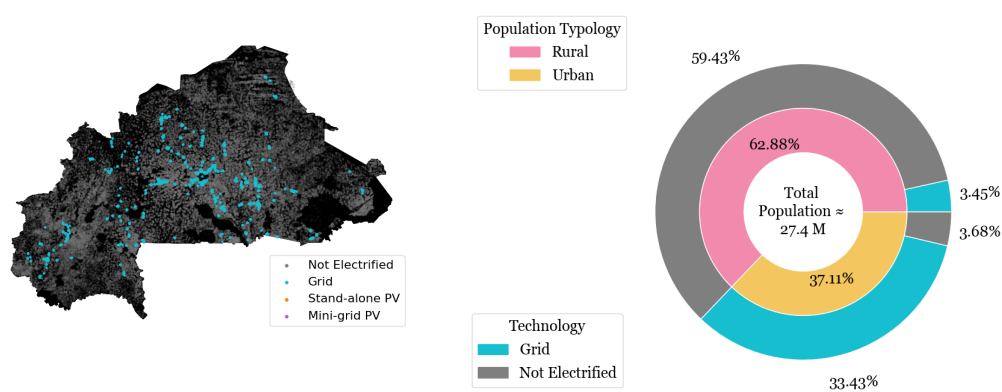


Figure 5.1.6: BAU Electricity Demand for Newly Electrified Settlements

Given the inputs of OSGEM, the electrification of Burkina Faso in BAU is plotted over time by population and technology is displayed in Figures 5.1.7 - 5.1.11. In 2030, most of those who

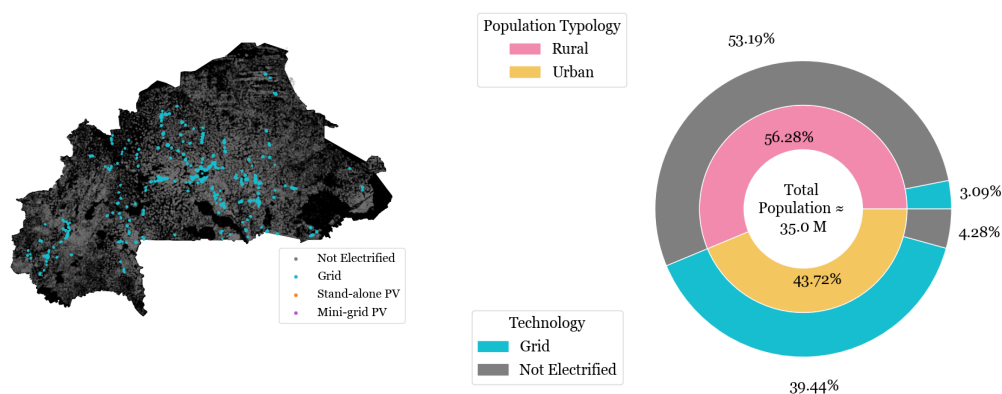
have access to electricity are living in urban areas, with the geo-spatial distribution of electrified areas resembling as they did at the beginning of the modelling period. Over time, as the urban areas become more populous, the share of electrified people increases, with the relative share of the electrified in rural communities remaining similar throughout the modelling period, as rural communities get smaller overall. Notwithstanding the increase in total electrified population over time up to 2070, the geographical areas that are electrified do not appreciably change over the whole period, and neither does the technology mix, with only the grid providing power.



(a) BAU 2030 Geo-spatial Technology Distribution

(b) BAU 2030 Electrification by Population Type

Figure 5.1.7: BAU 2030 Electrification



(a) BAU 2040 Geo-spatial Technology Distribution

(b) BAU 2040 Electrification by Population Type

Figure 5.1.8: BAU 2040 Electrification

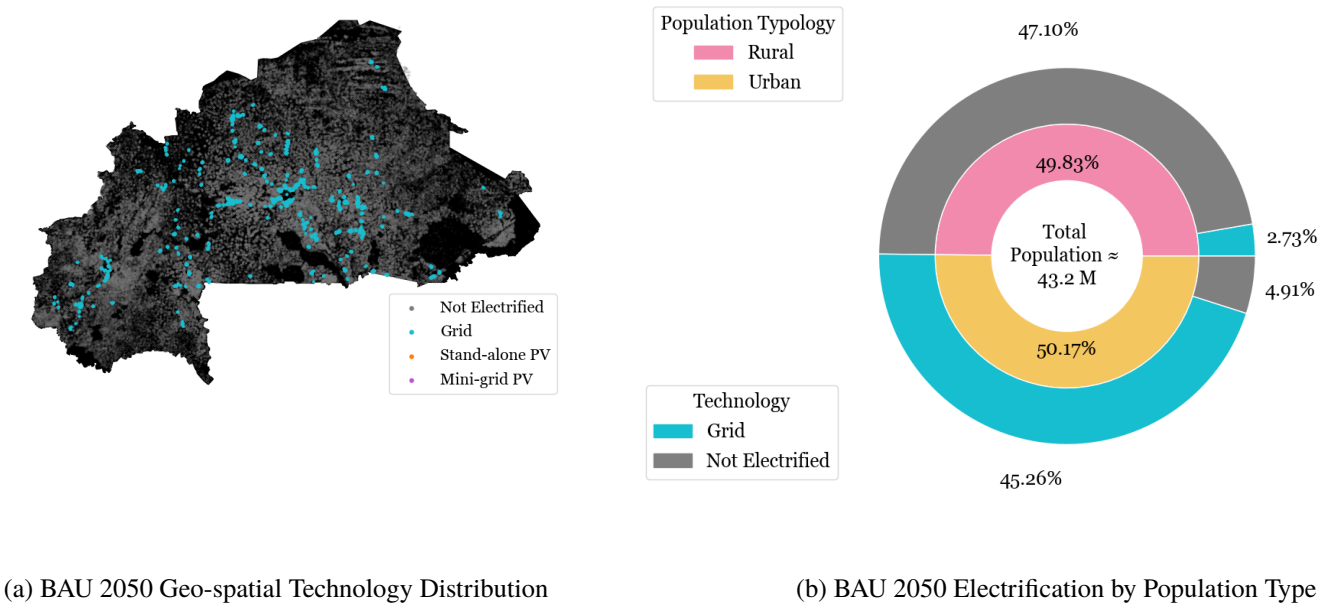


Figure 5.1.9: BAU 2050 Electrification

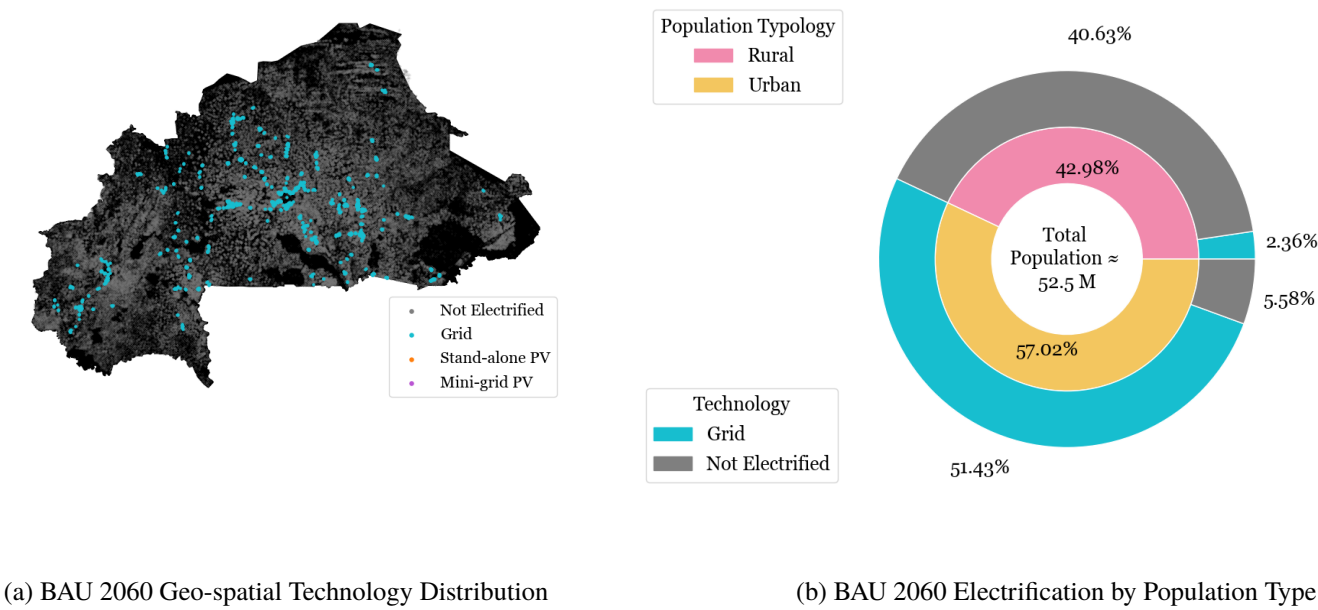


Figure 5.1.10: BAU 2060 Electrification

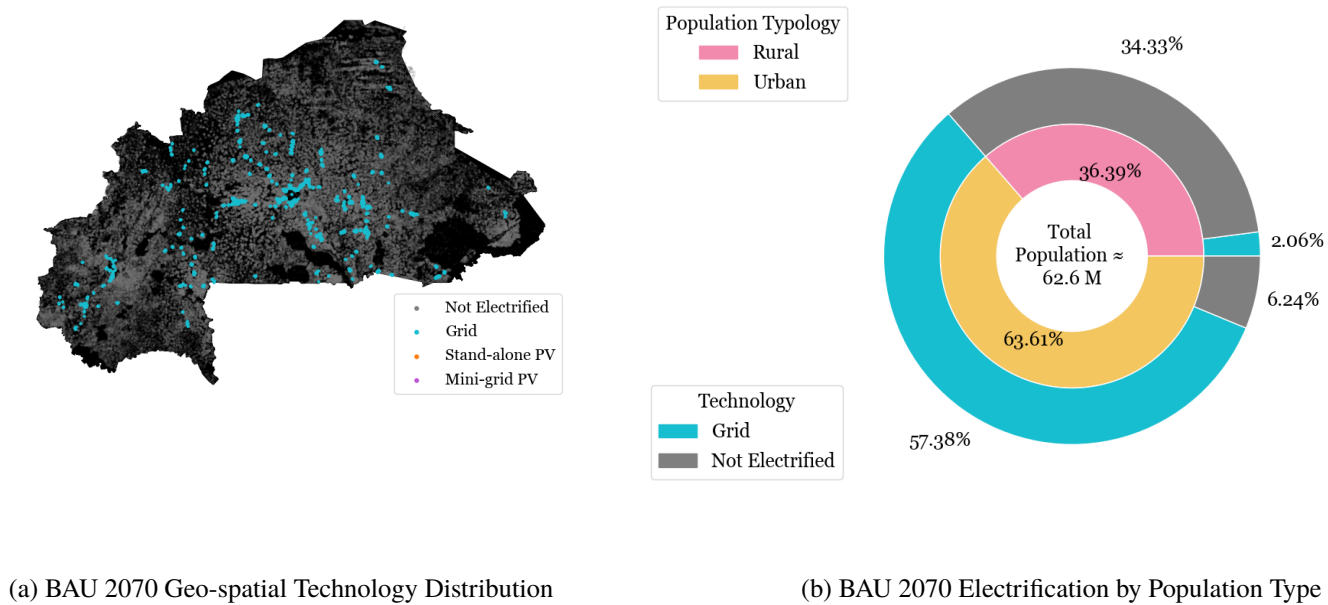


Figure 5.1.11: BAU 2070 Electrification

5.2 Balanced (BAL)

Using OSGEM to evaluate the BAL scenario as explained in the methodology yields the annual electricity generated by source graph seen in Figure 5.2.1. HFO is the main source of electricity early in the modelling period, supported by imported electricity and slight amounts of electricity from diesel power plants. Electricity from solar enters the mix in 2024 and its increase is accompanied by the increase of electricity generated from biomass power plants and the decrease in electricity from HFO. Wind becomes significant in 2035, hitting its peak, which comes at further expense to HFO. By 2040, HFO is phased out entirely and natural gas takes its place to become the dominant source of electricity for the rest of the modelling period. Electricity from hydropower plants are very minuscule, with solar and natural gas sources being used to meet rising demand, as biomass holds steady until 2055 where it gradually decreases to be phased out just before the end of the modelling period. Nuclear is never used in the BAL scenario despite its availability after 2040. The penetration of renewable electricity into the grid was low early in the modelling period, at a value of 10% in 2023, which then rose drastically to 50% in 2028. 2038 saw an even greater increase to 60% in 2034, before jumping to the model's highest penetration of 78% in 2039. From 2040 on, a general decrease in the penetration level was observed in the results, dropping to 56% in 2050 and to 33% in 2070.

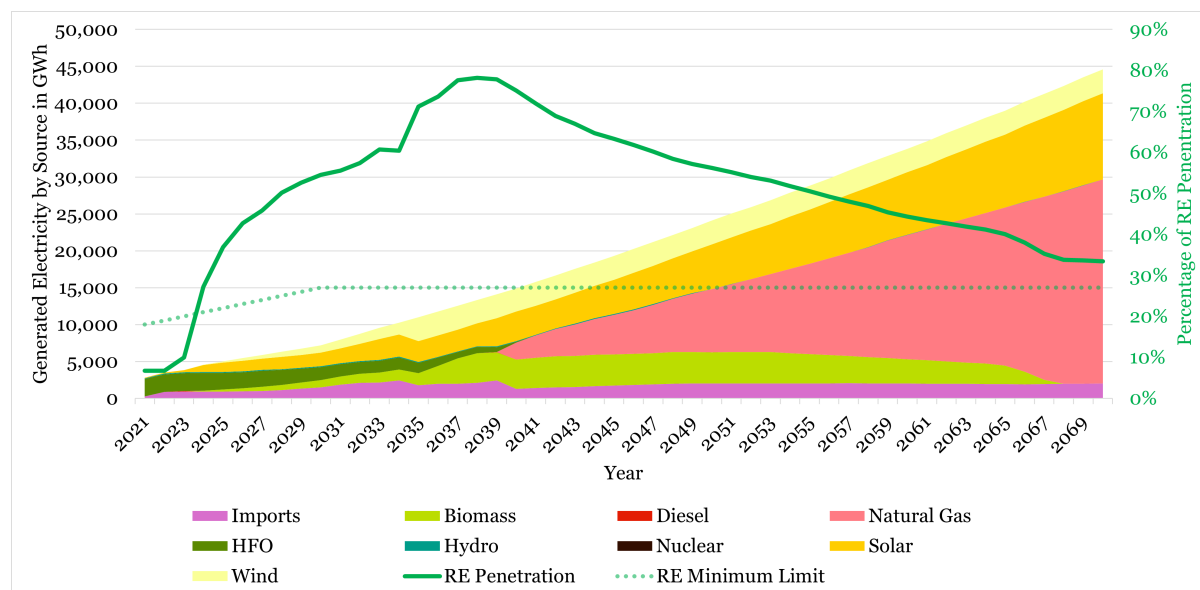


Figure 5.2.1: BAL Annual Production by Technology

The evaluation also presented us with the total annual capacity by generation technology in the BAL scenario as seen in Figure 5.2.2. Most of the installed capacity in 2021 is HFO power plants, with trace amounts of diesel, solar and hydro. By 2024, solar power plant capacity grows to outstrip HFO capacity and become the dominant plant type. From here until 2040, solar, wind and biomass power plants grow while HFO and diesel capacity shrinks. 2040 sees the introduction of natural gas capacity additions to the grid which continue to grow until 2070. Biomass capacity decreases over this time, completing fading away by 2068. Solar capacity maintains its majority in the face of growing natural gas power plant capacity.

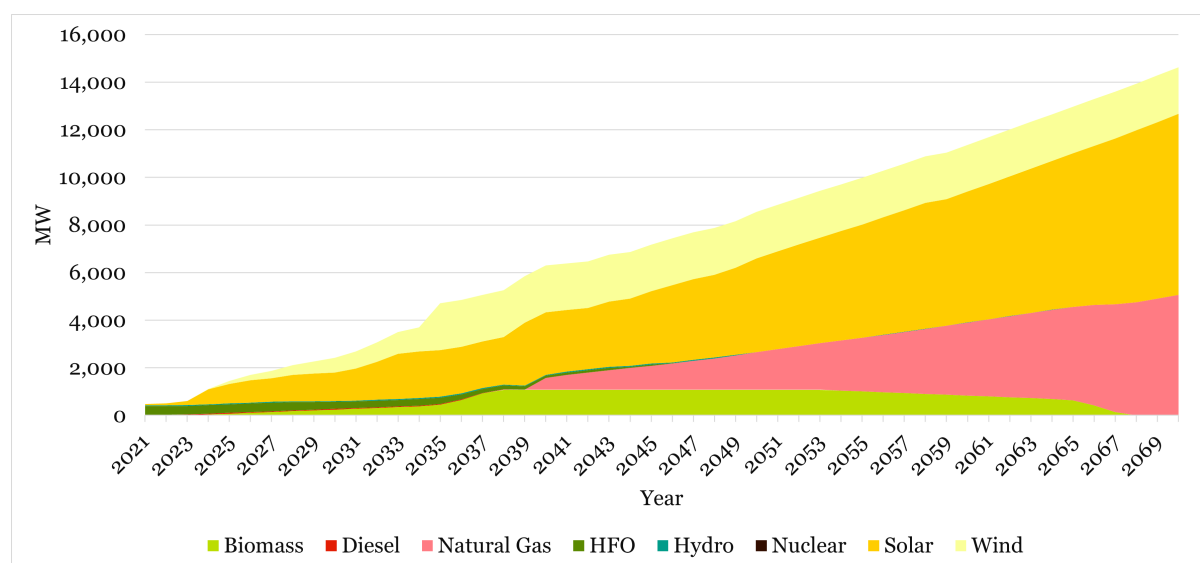


Figure 5.2.2: BAL Total Annual Capacity by Technology

Figure 5.2.3 gives us a closer look at electricity imports in the BAL scenario. Early in the

modelling period, imports come in from Côte d'Ivoire, Ghana and Mali. Côte d'Ivoire import utilization remains fairly constant throughout the modelling period. Imports from Ghana start out at similar level to Côte d'Ivoire but continually increases until 2033 where it remains constant until 2070. Mali imports fluctuate over time, starting relatively high in 2022 and decreasing towards 2029 to once again increase and reaching a maximum level by 2033. In 2040, Mali imports drop to 0, to then rise gradually again until 2049 and maintaining that level for roughly the rest of the modelling period, dropping off only slightly near the end of the period. Imports from Niger are intermittent, with small peaks in the 2030s to then disappear until 2047 where it increases at a gradual pace until 2070.

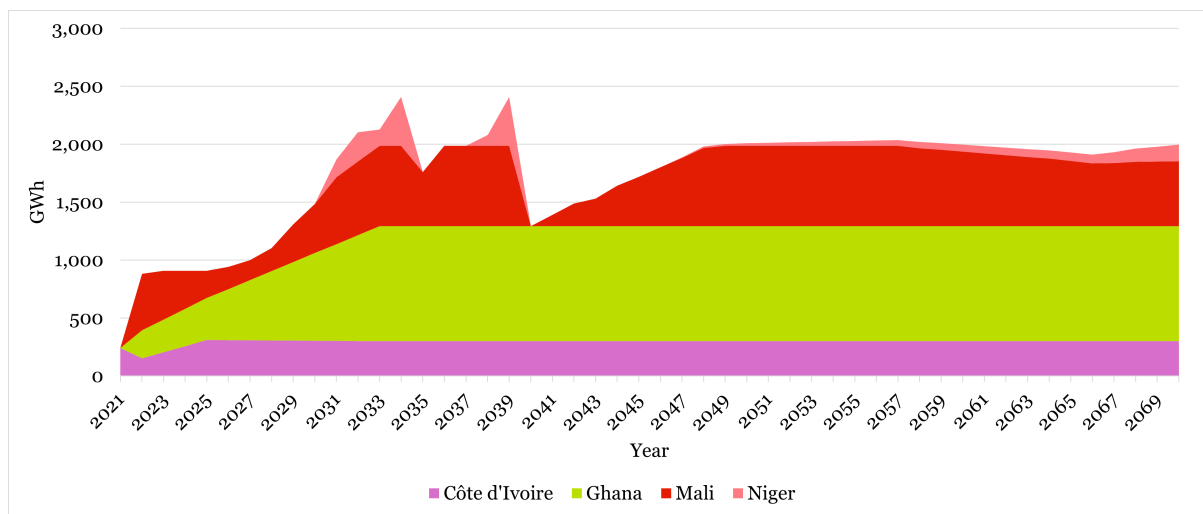


Figure 5.2.3: BAL Electricity Imports by Country

Figure 5.2.4 displays the annual emissions from electricity generated for Burkina Faso in the BAL scenario. From around 2000 kTon CO₂-equivalent in 2021, annual emissions gradually reduce to its lowest value in 2039 of nearly 600 kTon CO₂-equivalent. From 2040 on, the emission see rapid linear growth, maintaining this trend to the end of the modelling period, to an annual emission value of 11665 kTon CO₂-equivalent by 2070. The shape of the carbon intensity of the electricity generated curve is similar to the annual emissions curve. In 2021, Burkina Faso has its highest carbon intensity of 0.68 g CO₂eq per kWh. From then on, the carbon intensity sharply decreases to 0.36 g CO₂eq per kWh by 2025 and 0.13 g CO₂eq per kWh by 2034, but the decrease in carbon intensity decelerates between 2025 and 2034. 2035 to 2039 still sees a decrease to 0.04 g CO₂eq per kWh in 2039, the lowest of the modelling period. 2040 to 2070 sees gradual growth in the carbon intensity of electricity in a linear manner, growing from 0.07 g CO₂eq per kWh in 2040 to 0.26 g CO₂eq per kWh in 2070.

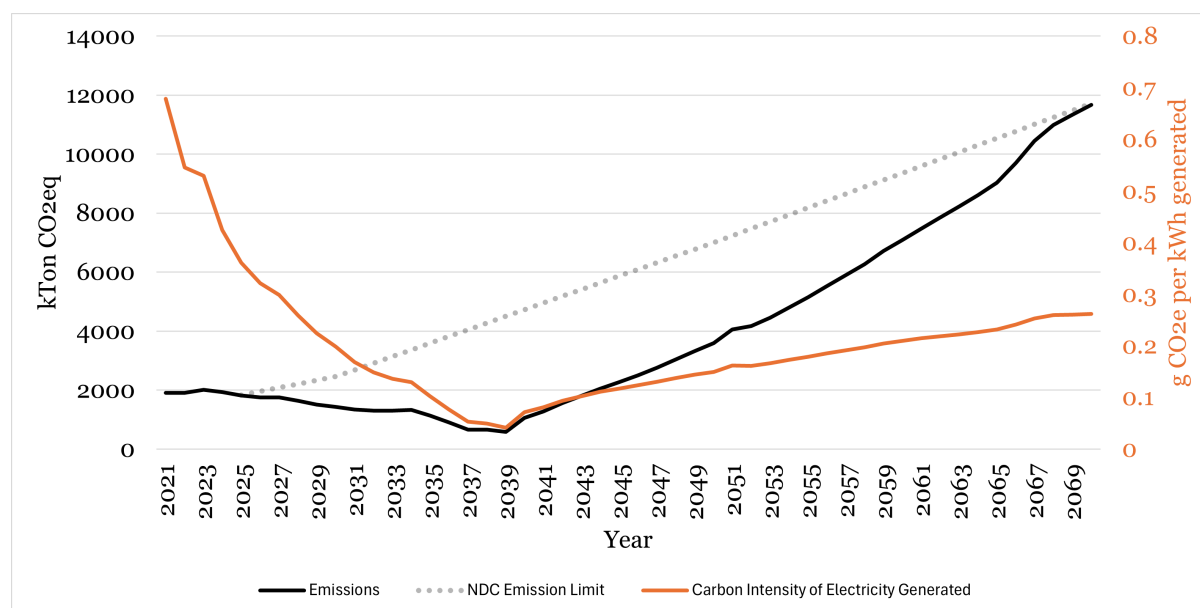


Figure 5.2.4: BAL Electricity Emissions

Figure 5.2.5 shows the annual grid investments in the BAL scenario, where biomass experiences constant investment from 2023 to 2038, peaking in 2037 at nearly 700 million USD. The trend in constant annual investments for thermal power plants is transferred from biomass to natural gas, where after becoming available in 2040 is invested in continually until 2070, with a peak in investments in the first year of availability at nearly 600 million USD, and smaller peak in 2067 at nearly 400 million USD. Solar also is to be invested it for the whole modelling period, occasionally seeing spikes in investment. Wind investment is more intermittent where a batch of investments occur between 2025 and 2035, and then again from 2050 to 2060. The years where total capital investment are the highest occur in year where wind has its highest investments, where in 2035, total capital investment is about 1 billion USD and in 2060, 1.2 billion USD, where wind is responsible for the vast majority of the investments.

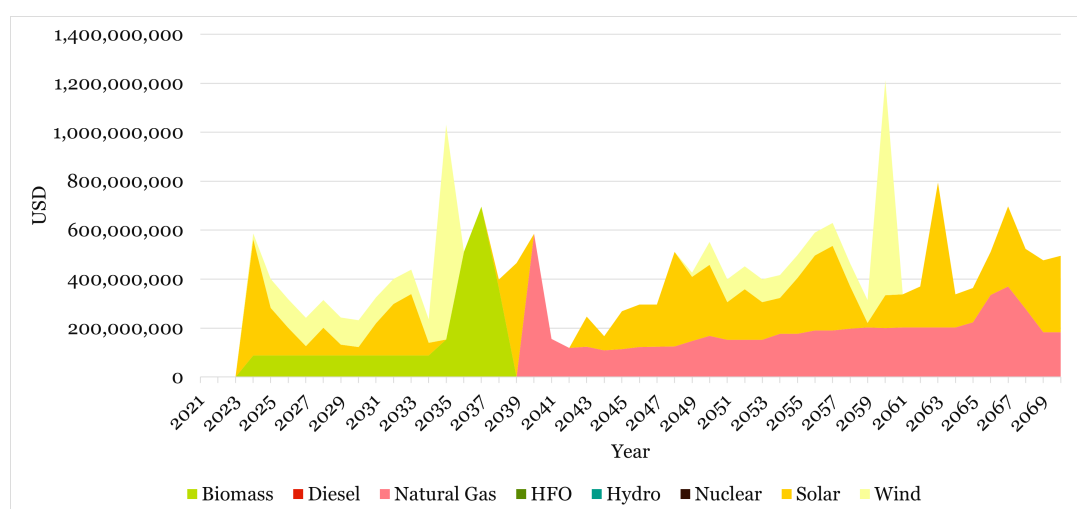


Figure 5.2.5: BAL Grid Capital Investment

The LCOEs representing the grid's cost of generating electricity in each decade of the modelling period is listed in Table 5.2.1. The highest LCOE is in the first period of the optimization and drop by 4 - 5 cents/kWh for the next two periods (until 2050). In the last two periods of the optimization, the LCOE of the grid remains relatively low but increases slightly from 2050 levels.

Table 5.2.1: BAL LCOEs (Grid Cost)

Year	LCOE (USD/kWh)
2021-2030	0.139
2031-2040	0.100
2041-2050	0.048
2051-2060	0.055
2061-2070	0.058

Figure 5.2.6 shows the annual amount of demand that newly electrified residences/settlements have added to the grid for the BAL scenario. Demand again increases almost linearly, accelerating over time, going from 576 GWh in 2025 to 6393 GWh in 2050 and 16267 GWh in 2070. This newly electrified load made up an increasing share of the demand throughout the modelling years, from 12% in 2025 to 28% in 2050 and 38% in 2070. The number of loops that the optimization needed to converge was two before the change in LCOE between runs was less than the threshold for change.

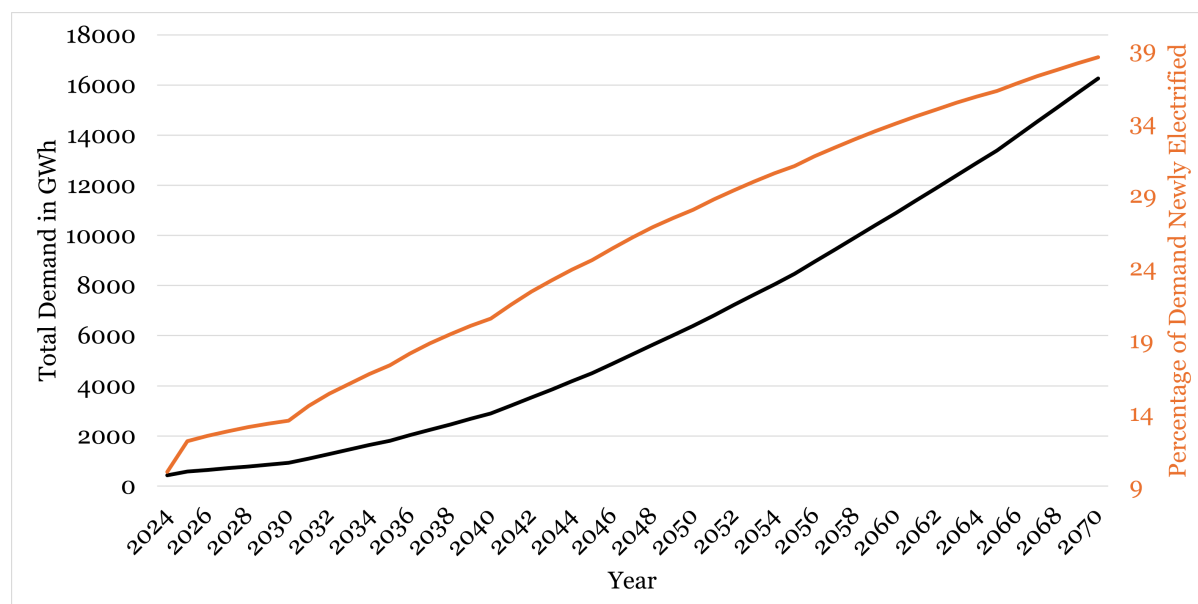


Figure 5.2.6: BAL Electricity Demand for Newly Electrified Settlements

The following charts (Figures 5.2.7 - 5.2.11) show the electrification of Burkina Faso in the BAL scenario geo-spatially and typologically. In 2030, everyone who is electrified, are electrified by

the grid, with most of these people living in urban areas. By 2040, a greater share of the rural population is electrified and mini-grid solar PV systems are used to electrify this fraction of the population, with the grid electrification rate keeping up with its status in 2030. As the total electrification rate increases in 2050, more unelectrified area receive electricity services through the employment of mini-grid solar PV systems. The mini-grid are installed all over the country, from east to west, and north to south. 2060 shows a greater geo-spatial electrification of the grid around transmission lines and even greater prevalence of mini-grid PV systems, to the point where previously unelectrified urban areas are electrified using these mini-grids. By 2070, Burkina is 100% electrified, where the grid and mini-grids have become geo-spatially denser to cover just about everyone, except for less than 40,000 people in rural areas now receive electricity via stand-alone PV systems.

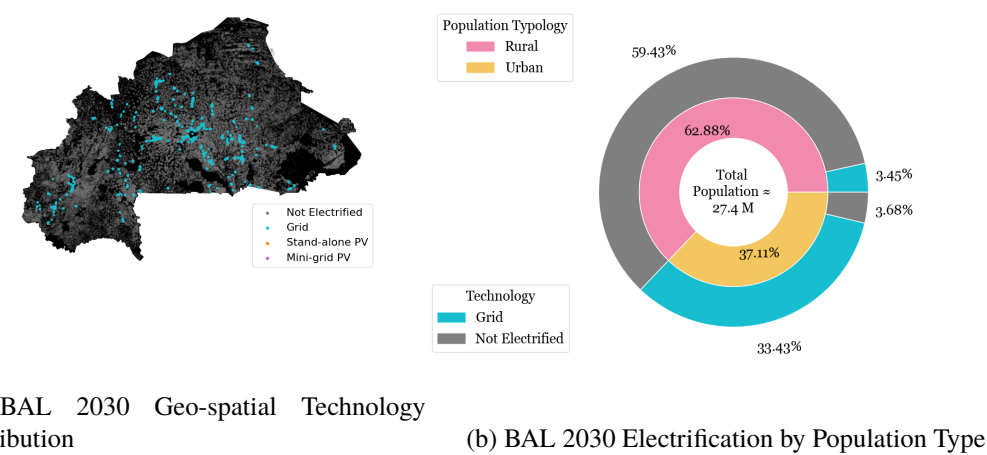


Figure 5.2.7: BAL 2030 Electrification

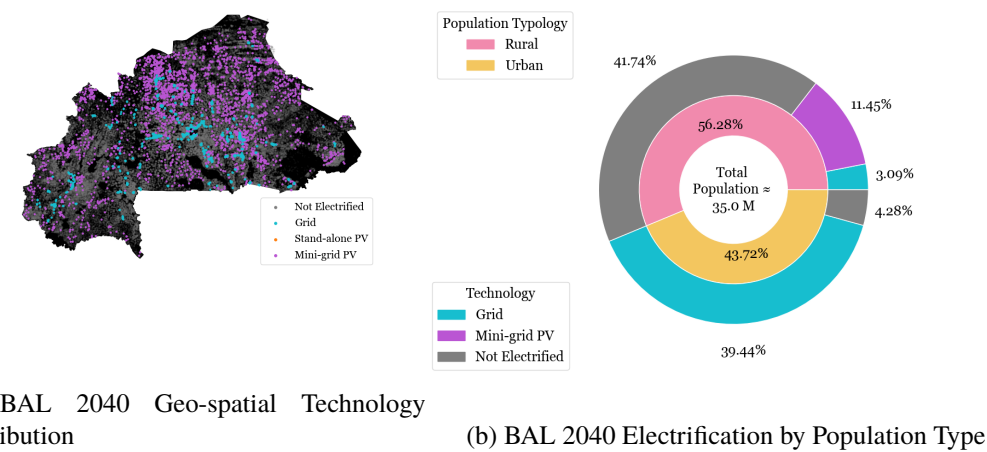


Figure 5.2.8: BAL 2040 Electrification

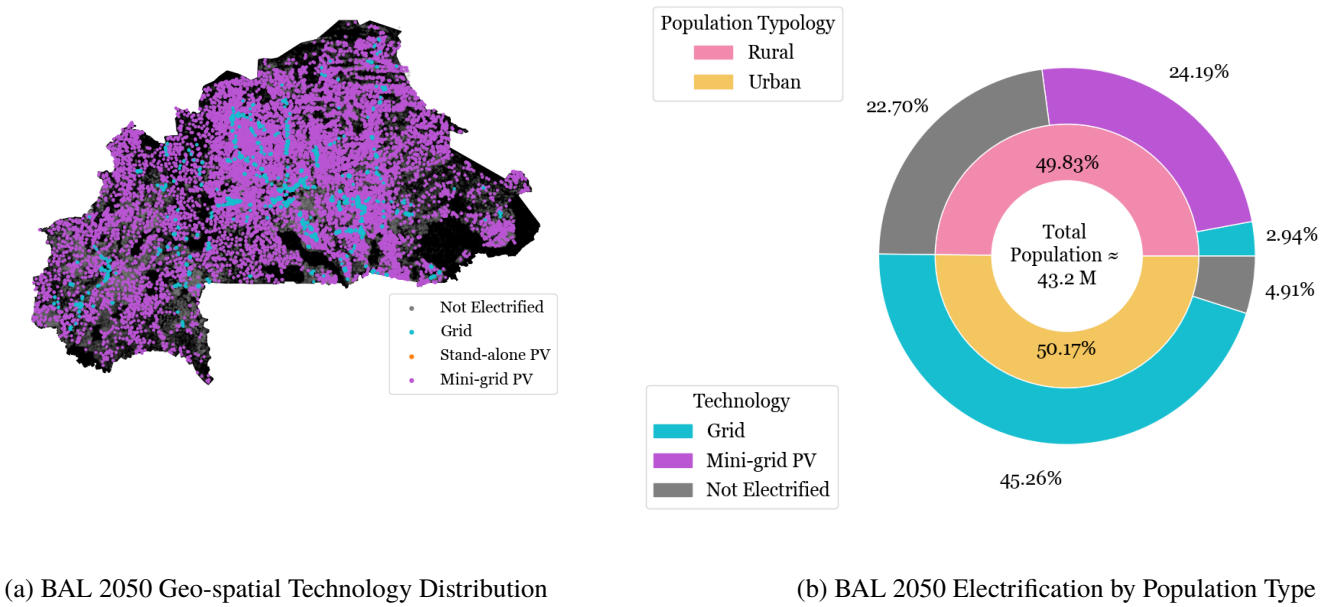


Figure 5.2.9: BAL 2050 Electrification

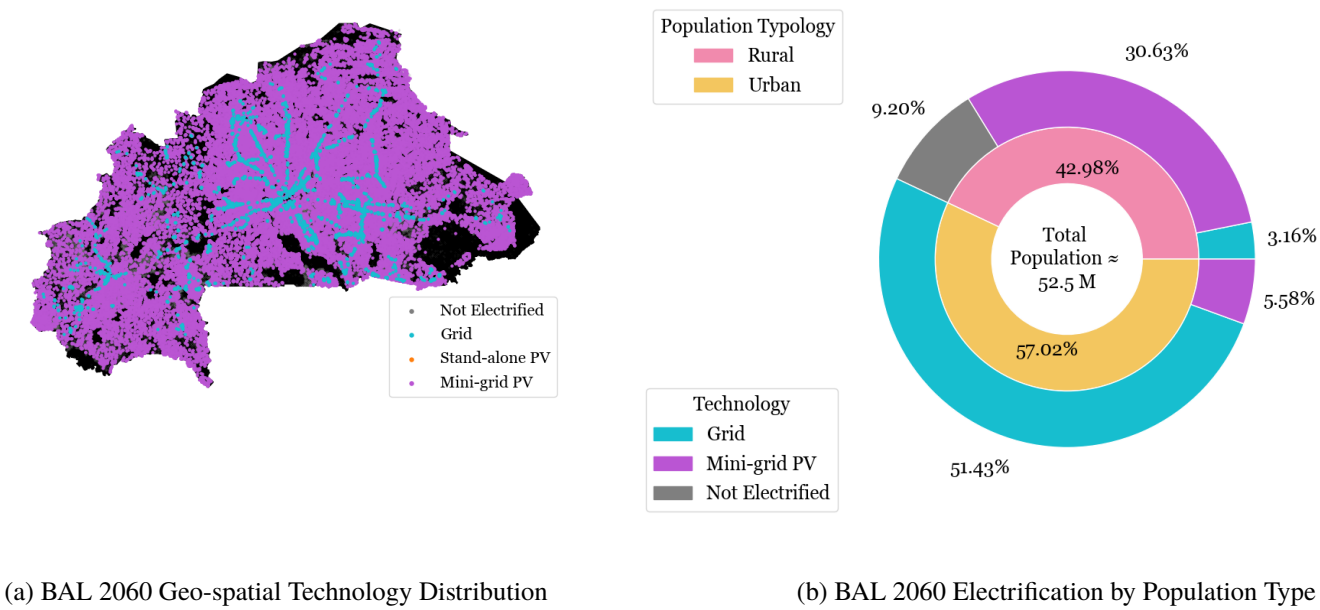


Figure 5.2.10: BAL 2060 Electrification

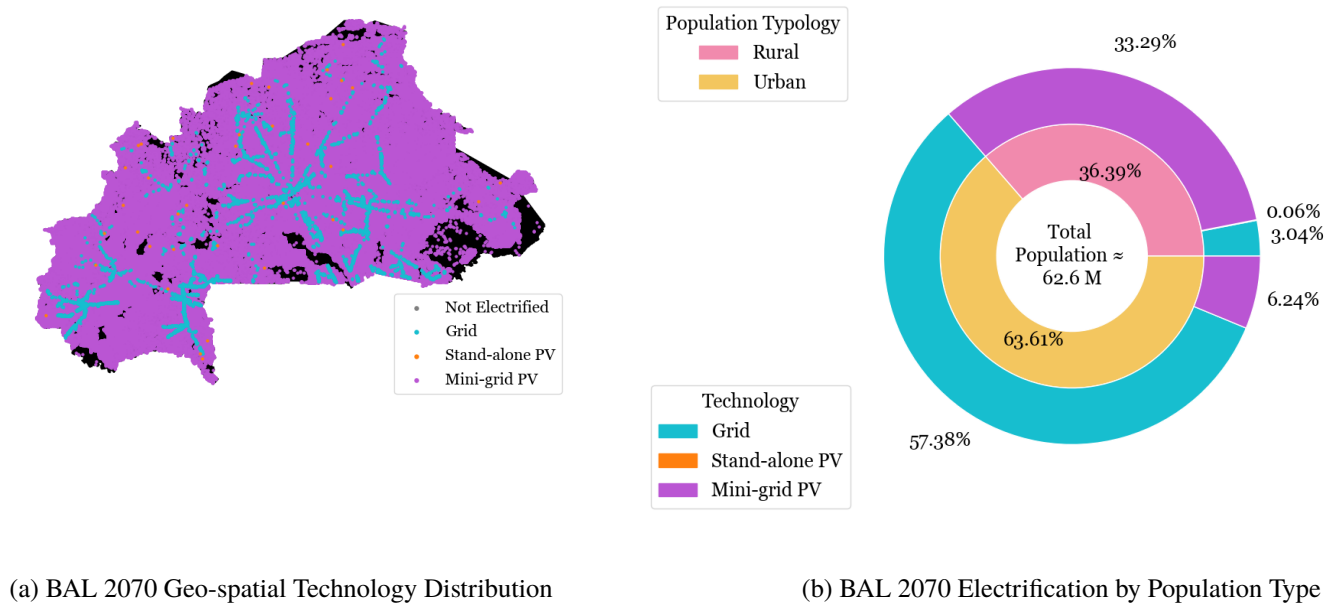


Figure 5.2.11: BAL 2070 Electrification

5.3 Ambitious (AMB)

Applying OSGEM to the AMB scenario, Figure 5.3.1 shows us the annual electricity production by source in this scenario. HFO is the dominant source of electricity in the early years. Heading into the 2030s, solar, more imports, and biomass enter the electricity mix and increase as demand increases. Diesel is briefly used for a few years in the 2020s, 2030s and 2040s at minute levels. In 2035, natural gas becomes available and seems to substitute most of the HFO that was being used to generate electricity previously. The 2030s also see increases in electricity from hydropower plants but the contribution to the overall mix is tiny. By the early 2040s, natural gas use begins to decline, as biomass use stabilizes, imports steadily increase and nuclear energy enters the energy mix. In 2050, natural gas, HFO, and diesel are completely phased out of the grid and a jump in nuclear energy occurs to meet demand, and solar use slightly increases and imports see an abrupt decrease. From 2050 on, the electricity mix stabilizes with solar and nuclear growing, as wind and biomass remains constant and imports shrink. In this time, nuclear becomes the dominant source of electricity in Burkina Faso. The AMB scenario saw the penetration of renewable electricity into the grid skyrocket over time, going from 10% in 2023 to 48% in 2028. The penetration then grew gradually to 64% in 2034 where it plateaued (with some oscillation) until 2042. From then, it decreases to 48% by 2050 and this rate is maintained until the end of the modelling period.

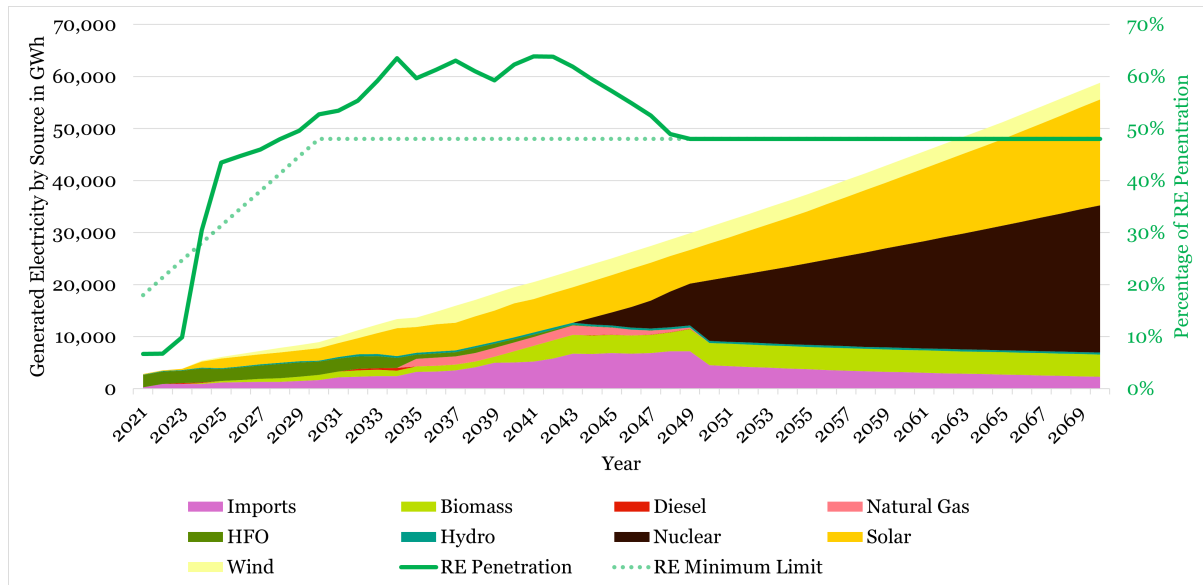


Figure 5.3.1: AMB Annual Production by Technology

The total annual capacity by generation technology for the AMB scenario is charted in Figure 5.3.2. Most of the capacity can be attributed to HFO in the very beginning of the modelling period. By 2025, large amounts of solar capacity are added to the grid and solar dominates in capacity terms for the rest of the modelling period, seeing a large jump in 2035 and a large rate of increase from 2050 on. Natural gas capacity appears in 2035 which is maintained until 2070. Biomass capacity mostly comes in during the 2040s and wind sees additions in the 2030s which maintains until 2070. Nuclear appears in 2044 and grows every year until 2070, with a bump in capacity in 2050.

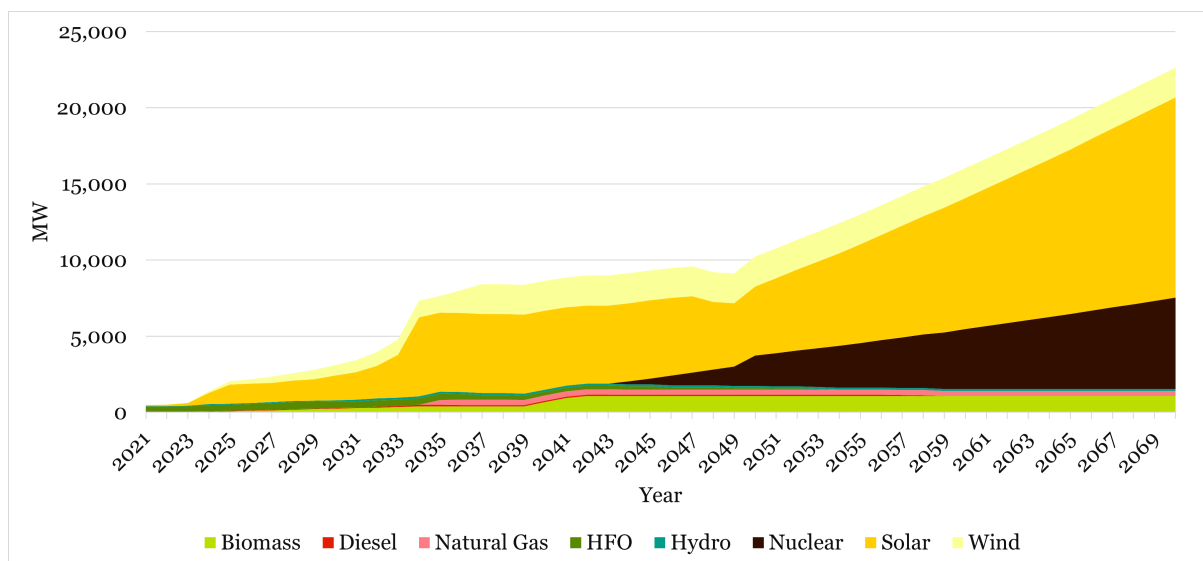


Figure 5.3.2: AMB Total Annual Capacity by Technology

Figure 5.3.3 gives us a detailed look into the electricity imports in the AMB scenario. Imports

from Ghana, Mali, and Niger steadily increase over time, with Ghana increasing a faster rate than Mali, which is in turn faster than Niger. Imports from Côte d'Ivoire are low and are constant. This is the regime until 2035 where imports from Côte d'Ivoire experience a drastic increase and imports from Mali and Niger fall to zero. However, the imports are the highest they have been so far in this scenario. Between 2035 and 2050, imports overall see a high increase, with most of that increase attributed to Niger, as Nigerien imports grow by 2000 GWh in a decade, and imports from other countries seeing a moderate increase with time. 2050 brought a sharp decline in imports with most imports being lost from Niger, but each country sees a decrease. From 2050 until the end of the modelling period, imports fall off over time from all countries, with imports being cut in half between 2050 and 2070.

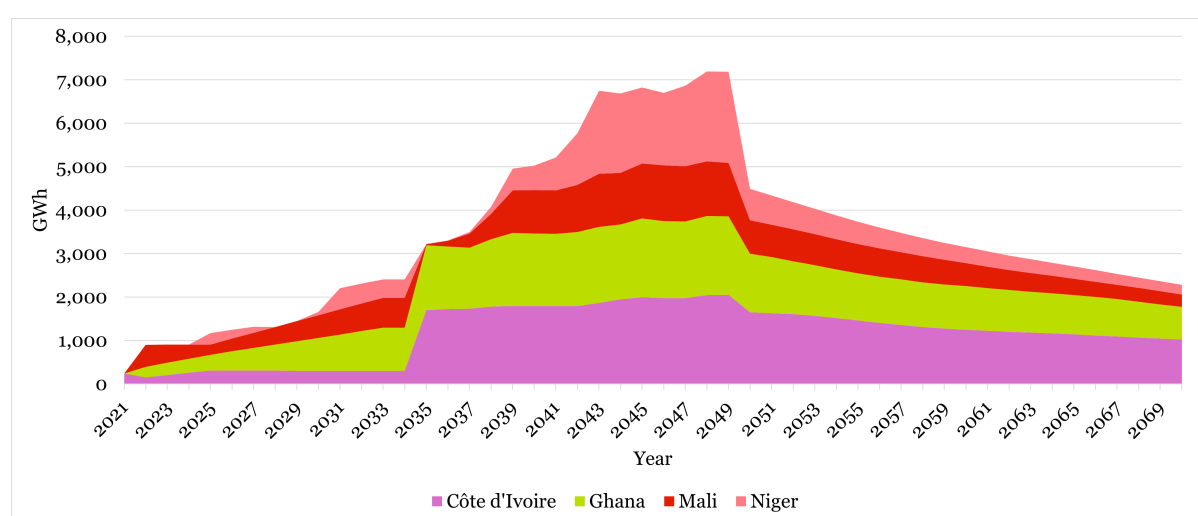


Figure 5.3.3: AMB Electricity Imports by Country

Figure 5.3.4 portrays the evolution of annual emissions from electricity generated over the modelling period for Burkina Faso in the AMB scenario. The nature of emissions can be described as cyclic and erratic until 2034, with several peaks of around 2200 kTon CO₂-equivalent forming in these early years. 2035 then see a sharp decrease in emissions to nearly 1300 kTon CO₂-equivalent with a slight rise until 2038 to 1449 kTon CO₂-equivalent. From 2039 to 2050, there is a straight-line descent in the curve to the 0 kTon CO₂-equivalent in 2050, the net-zero emissions year in this scenario. Until the end of the modelling period, the emissions remain at 0 kTon CO₂-equivalent annually. The carbon intensity of the electricity generated decreases sharply to 0.3 g CO₂eq per kWh in 2025 from 0.68 g CO₂eq per kWh in 2021. After 2025, the intensity still decreases but less steeply to 0.09 g CO₂eq per kWh in 2035. Post-2035, the rate of decrease becomes smaller as annual carbon intensity of generated electricity approaches 0 g CO₂eq per kWh by 2050. The intensity remains at 0 g CO₂eq per kWh until the end of the modelling period in 2070.

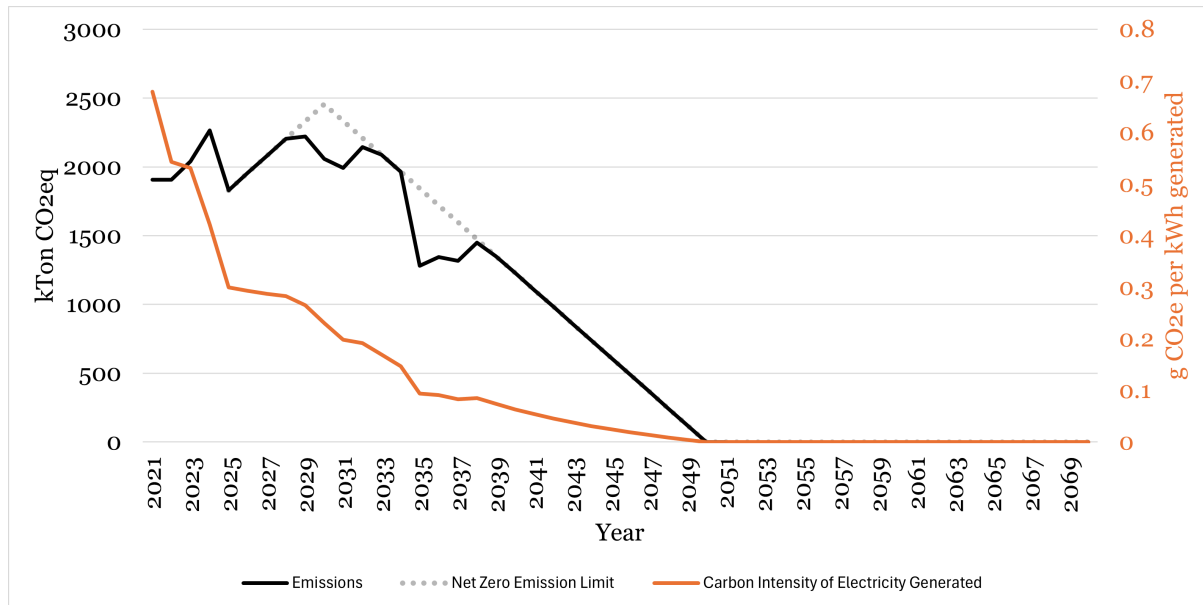


Figure 5.3.4: AMB Electricity Emissions

In the AMB scenario, as seen in Figure 5.3.5, most of the capital expenditure is due the nuclear power plant installations, with the investments being fairly constant over time from 2044 until 2070, except for a spike in 2050 of nearly 4.5 billion USD. At the start of the modelling period, until 2035, most expenditure is due to solar investments, where a smaller peak of nearly 1 billion USD occurs in 2024, and a larger peak of about 2 billion USD occurs in 2034. Investments occurs in other technologies in the same period but pale in comparison to investments in solar. Notable investments in wind and biomass occur in the late 2030s and early 2040s, respectively, ranging between 0.5 and 1 billion USD each year the occur. From 2050 on, in addition to nuclear investments, solar expenditure is common after 2050 every year until the end of the modelling period.

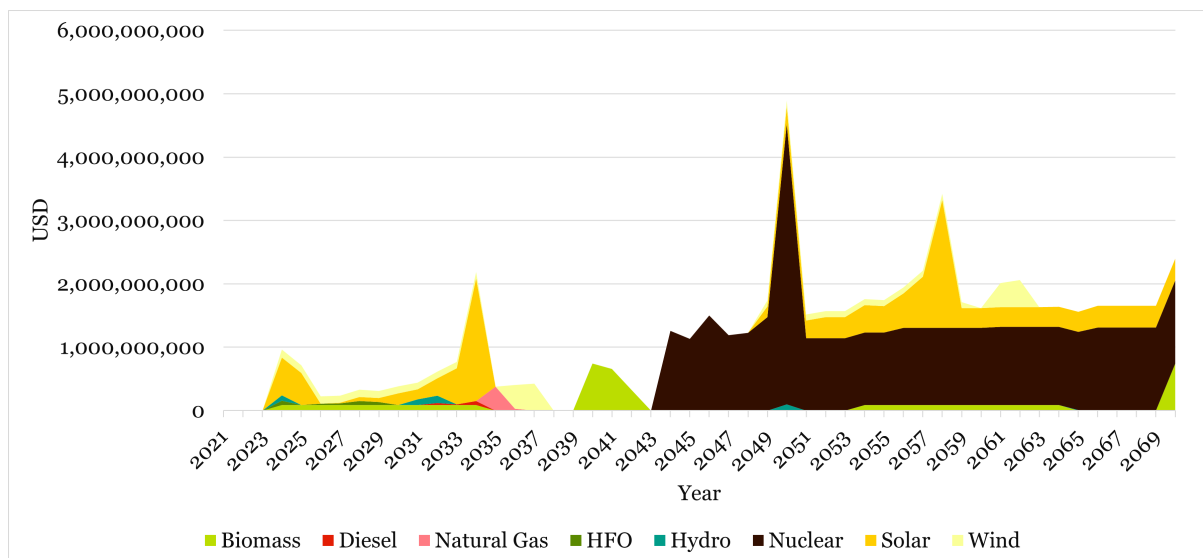


Figure 5.3.5: AMB Grid Capital Investment

Table 5.3.1 lists the LCOEs in each myopic period in the AMB scenario. Each decade, except the last, saw a decrease in the LCOE by either 3 or 4 cents/kWh. The 2060s only saw a 2 cents/kWh decrease from the previous decade. Over the whole modelling period, the LCOE is decreased by 90%.

Table 5.3.1: AMB LCOEs (Grid Cost)

Year	LCOE (USD/kWh)
2021-2030	0.144
2031-2040	0.103
2041-2050	0.071
2051-2060	0.031
2061-2070	0.013

Figure 5.3.6 shows the annual amount of demand that newly electrified residences/settlements have added to the grid for the AMB scenario. Demand increases over time with accelerating linearity going from 1543 GWh in 2025 to 8810 GWh in 2050 and 18707 GWh in 2070. This newly electrified load made up an increasing share of the demand throughout the modelling years, from 27% in 2025 to 30% in 2050 and 35% in 2070. The number of loops that the optimization needed to converge was two before the change in LCOE between runs was less than the threshold for change.

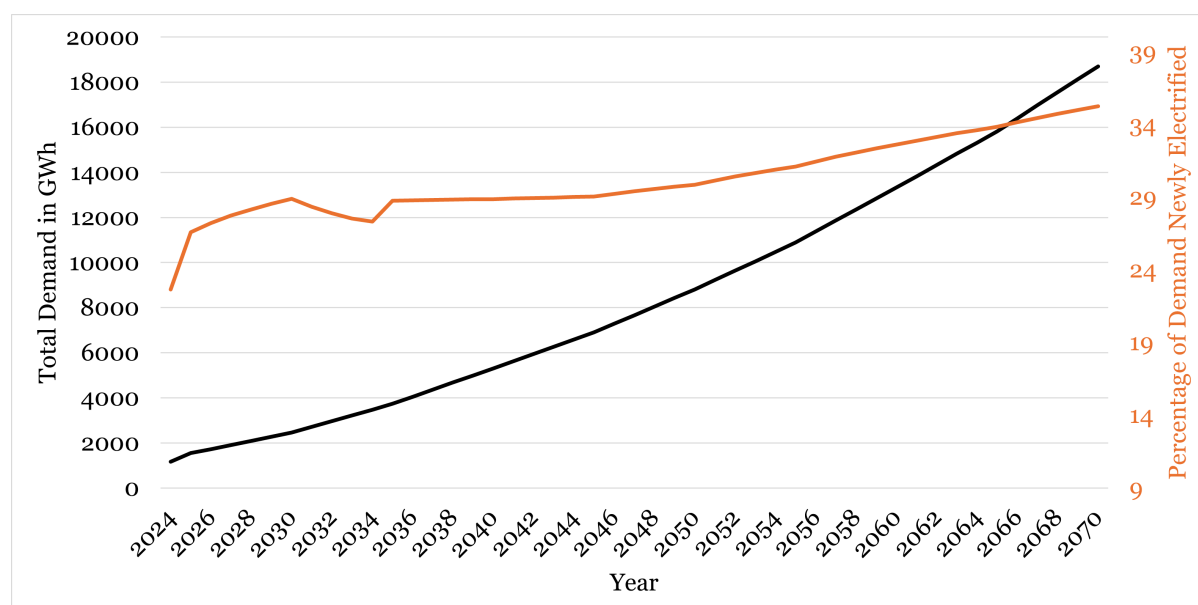


Figure 5.3.6: AMB Electricity Demand for Newly Electrified Settlements

Figures 5.2.7 - 5.2.11 show how electrification evolves in the AMB scenario. By 2030, almost the whole population is electrified, with a massive deployment of mini-grid solar PV systems

in unelectrified rural communities being responsible for that electrification. Stand-alone PV systems also see wide implementation geo-spatially in the northern 3rd of the Burkina Faso, not withstanding that area representing such a tiny portion of the overall population of the country. Most urban dwellers rely on the grid for electricity services, with those living in spaces without grid access, depending on mini-grid PV systems for electricity. By 2040 and 2050, this regime essentially stays the same, with the change in population dynamics leading in an increasing about of people having access to the national grid and a decreasing population using mini-grids. 2060 sees a sizable jump in the percentage of persons with national grid access and geo-spatially denser consumption of grid electricity near the existing transmission lines. 2070 maintains that trend. Stand-alone PV retains its geo-spatial prevalence but it is always electrifying a minute part of the population.

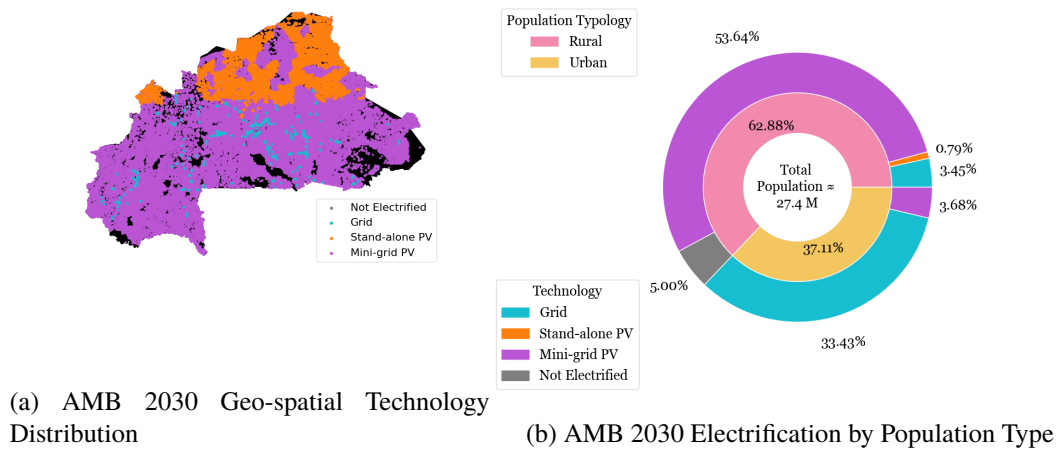


Figure 5.3.7: AMB 2030 Electrification

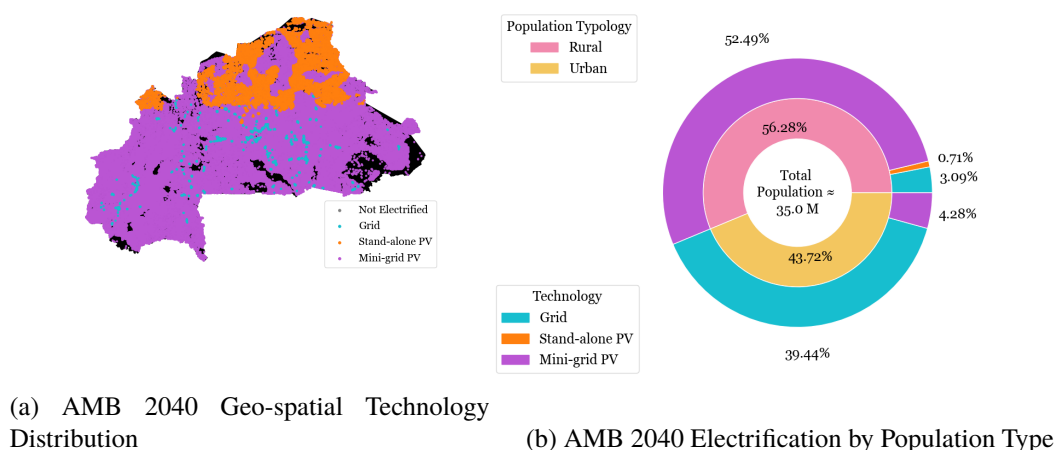
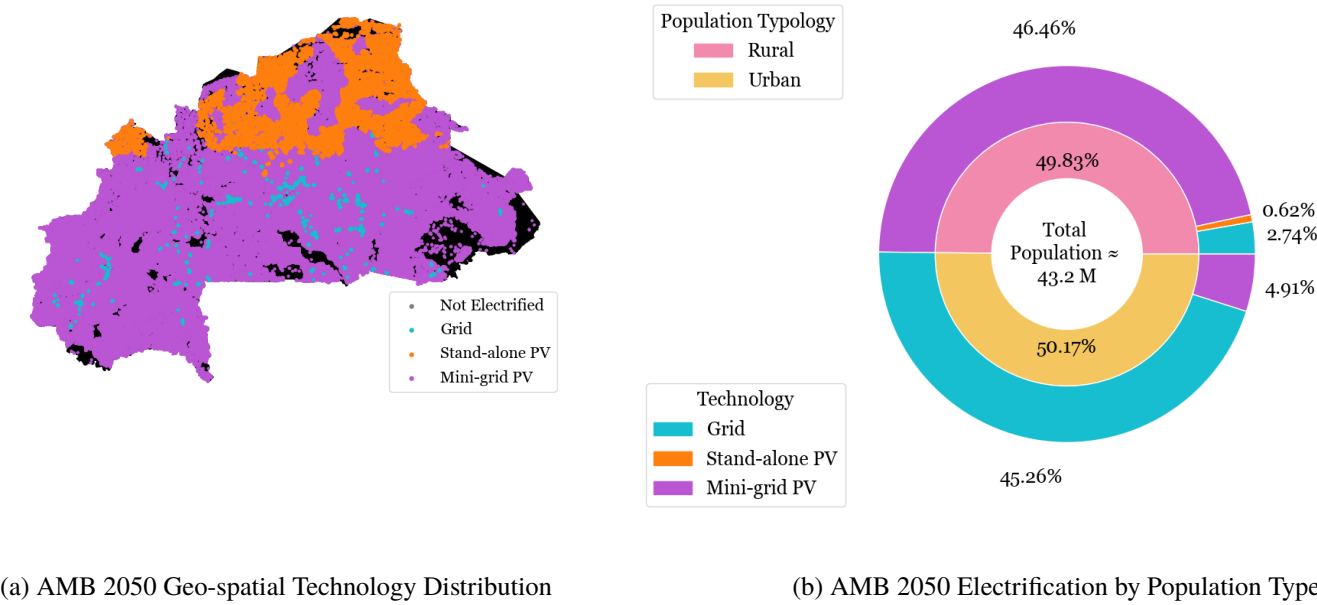


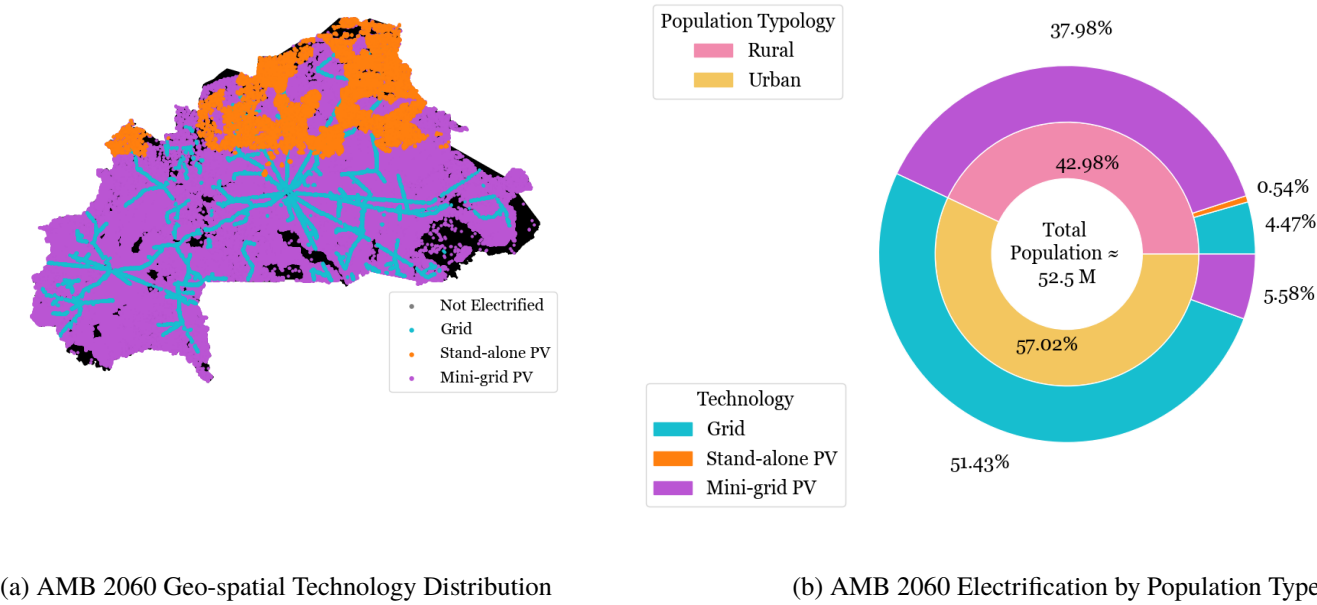
Figure 5.3.8: AMB 2040 Electrification



(a) AMB 2050 Geo-spatial Technology Distribution

(b) AMB 2050 Electrification by Population Type

Figure 5.3.9: AMB 2050 Electrification



(a) AMB 2060 Geo-spatial Technology Distribution

(b) AMB 2060 Electrification by Population Type

Figure 5.3.10: AMB 2060 Electrification

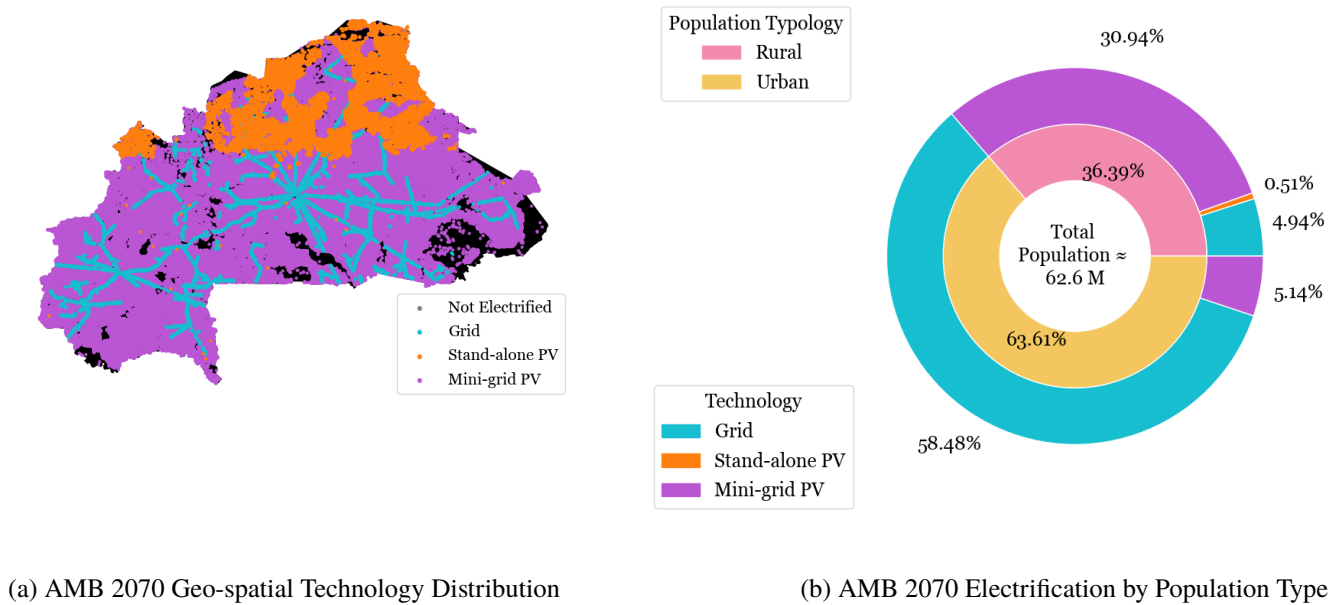


Figure 5.3.11: AMB 2070 Electrification

5.4 Sensitivity Analysis

5.4.1 Net-Zero Year

Figure 5.4.1 shows how the electricity mix changes when the net-zero year is pushed more and more into the future. The changes in the analysis year 2030 are fairly constant when changing the net-zero year which show that a delay causes there to be a bit more HFO at cost of a bit less electricity from solar and Nigerian imports. By 2040, the range of changes are greater. Delaying the net-zero year by 5 years leads to a mix change of $\pm 7\%$ with the increase occurring in natural gas and biomass at the cost of largely solar, but also imports and HFO. As the delay increases, the mix percentage change increases, but increases by a smaller amount the further into the future the net-zero year is pushed. The increase in biomass in the mix still occurs but decreases over time, leaving the lion share of the increase being attributed to natural gas, with small amounts of wind. The sources that decreased in the NZ2055 scenario, also decrease in the others, but the margins are greater, proportional to the increases in natural gas and biomass combined. 2040 sees a mix change of $\pm 15\%$ if net-zero is delayed until 2070. Looking at the analysis year 2050, the character changes. While the range of change still increases over time, from $\pm 11\%$ to $\pm 16\%$, it is nuclear that loses some share in the electricity mix, which is made up by imports from all the importing countries and natural gas. The level of imports stays steady as the delay increases, with natural gas seeing the proportional increase. When looking at the mix

in 2060, the scenarios with net-zero years at or before 2060 see no changes from the base AMB scenario. With the scenarios dealing with after 2060, the changes occur in a similar fashion as they did in 2050 but to a lesser degree. Electricity from nuclear still decreases in favour of imports and natural gas. NZ2065 sees a mix change of $\pm 8\%$ while NZ2070 sees a change of $\pm 11\%$.

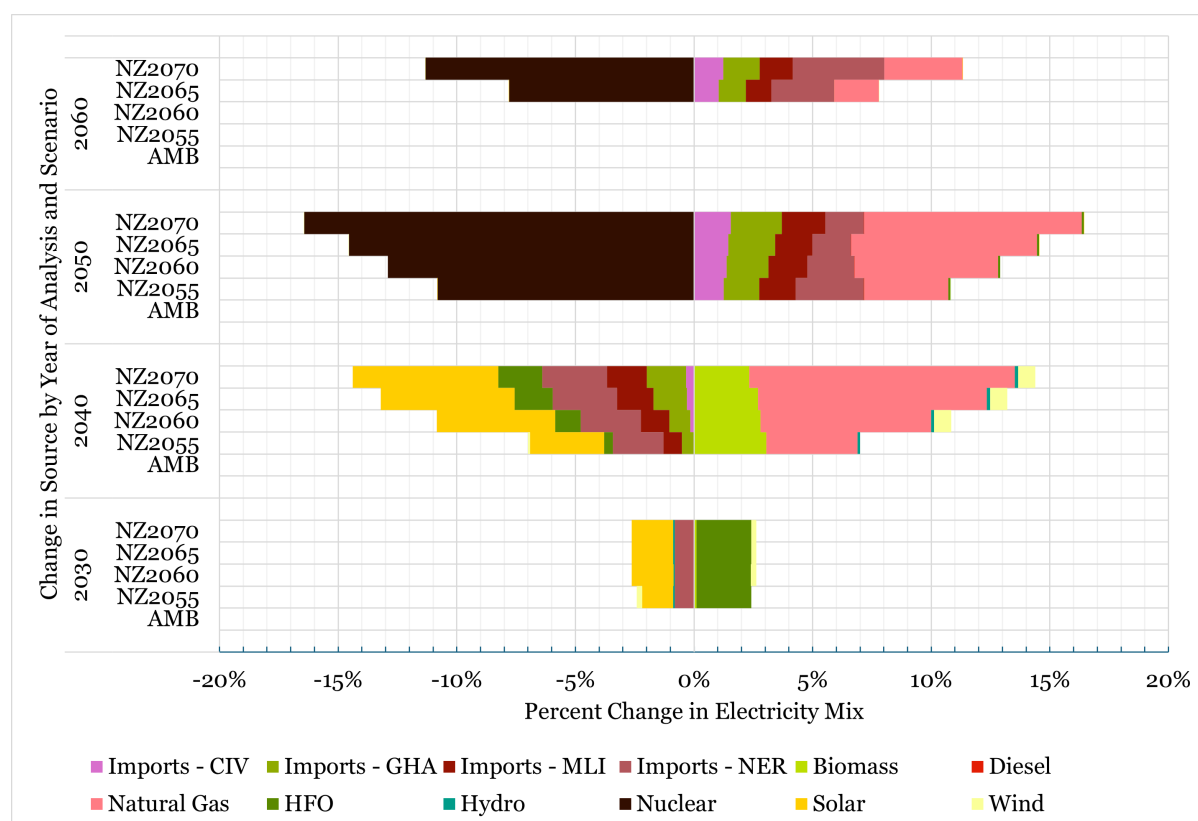


Figure 5.4.1: Change in Electricity Mix by varying the Net-Zero Year

Figure 5.4.2 summarizes how the changes in installed capacity change with the net-zero year in the AMB scenario. 2030 sees little change irrespective of the net-zero year. Solar capacity share dips by 2% and biomass, HFO and wind capacity together increases by 2%. 2040 sees greater changes in the capacity mix with a $\pm 7\%$ mix change seen in the NZ2055 scenario and growing to a 11.5% mix change in the NZ2070 scenario. All sensitivity scenarios in 2040 see 0.5% decrease in the capacity share help by diesel with the characteristic decrease being seen in solar PV capacity. The capacities that are installed in solar PV's stead are a mix of biomass, natural gas, HFO and wind, with natural gas and wind increasing as the net-zero year is delayed. In 2050, the range of change in capacity mix decreases a bit, falling between $\pm 6\%$ and $\pm 10\%$. The installed capacity of nuclear is the subject of most of the decrease in share with respect to the change in the net-zero year. When less nuclear installed, more natural gas and solar capacity takes its place. In 2060, a similar trend continues, however, natural gas is even more responsible for the total change in capacity as opposed to sharing that significance with solar PV capacity

a decade before. The net-zero scenarios of 2055 and 2060 show only small changes but the scenarios with net-zero years after 2060 exhibit wider change.

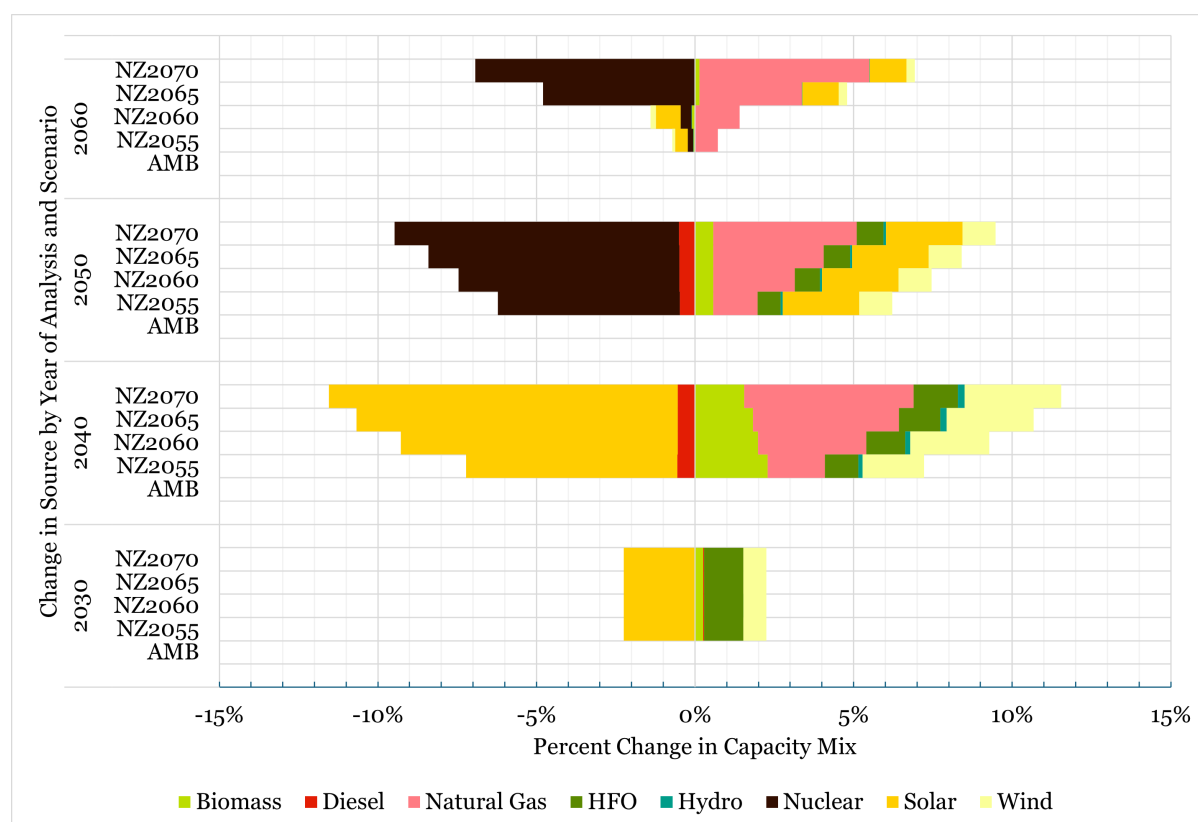


Figure 5.4.2: Change in Installed Capacity by varying the Net-Zero Year

Presented below in Table 5.4.1 are the LCOEs, which represent the cost of providing electricity to the grid, for each decade in each scenario which varies the net-zero year. For the first decade, there is no change in the LCOE. In the second decade, each scenario sees a decrease in the LCOE. As the net-zero year increase, the more the LCOE decreases in this decade. Delaying the net-zero year over a 20-year timespan, allows a variance in the LCOE of about 0.01 USD/kWh. In the third decade of the scenario, each scenario sees a decrease in the LCOE but LCOEs are much closer in value across the scenarios. The LCOEs are within 0.003 USD/kWh of each other with the LCOEs being identical with any delay of the net-zero year. In the decade ending in 2060, the drop in LCOE trend continues across the scenarios but instead of the LCOE decreasing as you delay when net-zero occurs, it increases, with the gap between the lowest and highest LCOEs being roughly one and a half cents per kilowatt-hour. The last decade in the modelling period has the lowest set of LCOEs in the modelling period, with about a 2 cents decrease from the previous decade. The LCOEs still exhibit a trend of increasing as the net-zero year is delayed, like the previous decade. The LCOE nearly doubles in the NZ2070 scenario versus the AMB scenario.

Table 5.4.1: LCOEs from Net-Zero Sensitivity

Sensitivity Scenario	AMB	NZ2055	NZ2060	NZ2065	NZ2070
Period	USD/kWh	USD/kWh	USD/kWh	USD/kWh	USD/kWh
2021-2030	0.144	0.144	0.144	0.144	0.144
2031-2040	0.103	0.098	0.096	0.095	0.094
2041-2050	0.071	0.068	0.068	0.068	0.068
2051-2060	0.031	0.039	0.042	0.043	0.045
2061-2070	0.013	0.013	0.013	0.018	0.024

Figure 5.4.3 presents a summary on how differently the population is electrified by the technologies available to them as the net-zero year changes. Changing the net-zero from 2050 to 2070 in five-year increments has no effect on the distribution of the grid, stand-alone, or mini-grid technologies used for electrification from 2030 to 2050. Changes in the share start to appear in 2060, where the share change is only $\pm 0.2\%$ from grid to mini-grid PV. In 2070, there are greater changes in the share of electrifying technology but that change is only about $\pm 1\%$. For very small amounts of people (communities), the technology of use changes from grid to mini-grid, except for NZ2060 where it does the other way, but fundamentally the changes are not appreciable.

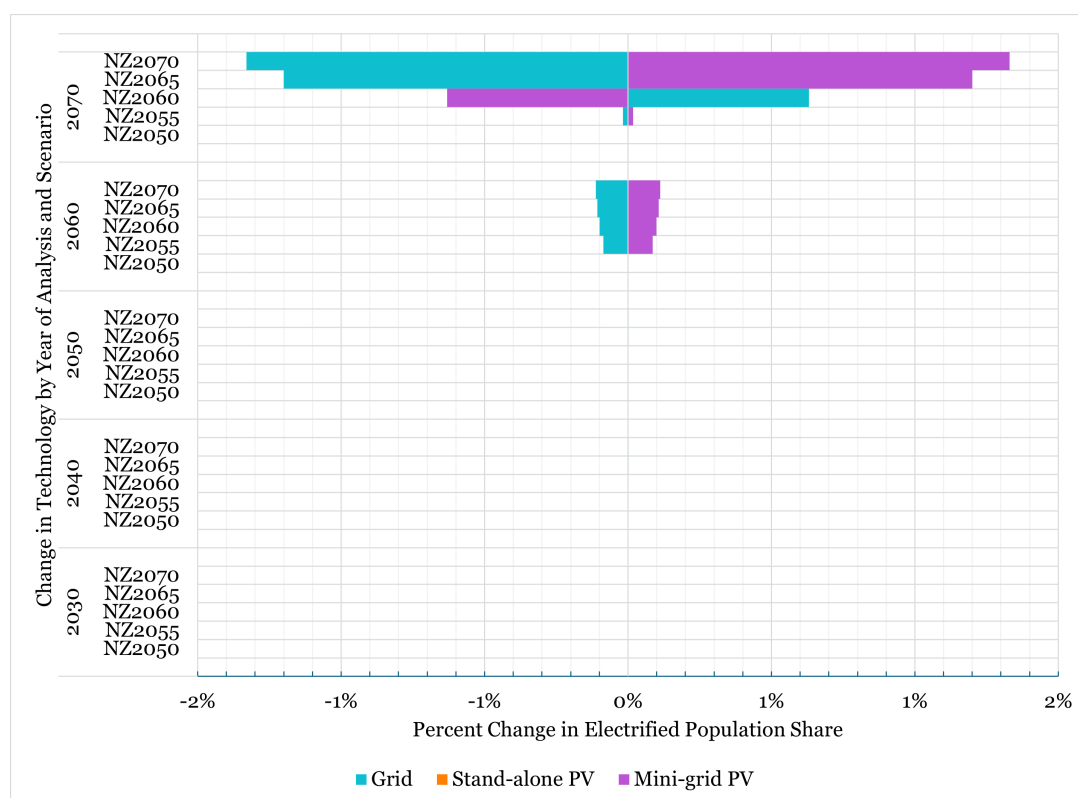


Figure 5.4.3: Change in Electrifying Technology by varying the Net-Year Zero

5.4.2 PV Capital Costs

Figure 5.4.4 shows the effects on the electricity mix from adjusting PV capital costs. 2030 exhibits the greatest change but all the changes are in the $\pm 3\%$ range. Capital costs cheaper than the base AMB scenario, encourage more solar and HFO production into the mix mostly to the detriment of hydro, Nigerien imports and wind. It would appear that cheapening the costs lessens the degree of change. When the capital costs are greater than that in the base AMB scenario., solar production is decreased in favour of hydro and imports. Looking at 2040, the range of changes are in the $\pm 2\%$ range. Decreasing capital costs leads to minute readjustments in the mix with no clear pattern. An increase in capital costs once again means a slightly decrease in solar production and increase in imports and wind production. From 2050 on, there are no changes in the electricity mix over the whole range of price changes.

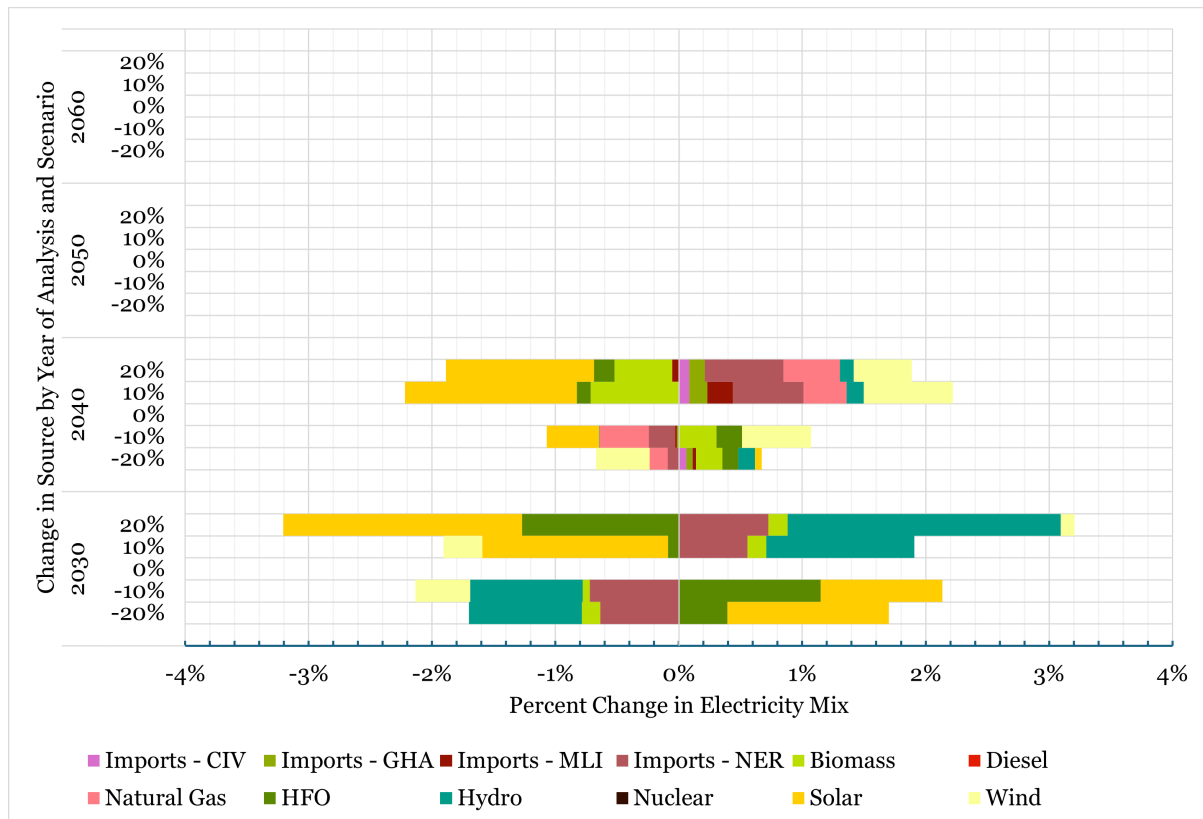


Figure 5.4.4: Change in Electricity Mix by varying PV Capital Costs

The range of changes in the capacity mix occur in a similar range as they did for the electricity mix figure, as portrayed by Figure 5.4.5. The year 2030 displays the greatest change. As solar costs increase, less solar capacity exists in 2030 which is replaced by more hydro and wind capacity, mainly. When the PV capital costs are more favourable, more solar capacity is built at the cost of wind and hydro capacity. In 2040, the changes in capital costs lead to changes in fossil fuel technology. More expensive capital costs decrease slightly the installed capacity of solar but also of biomass and HFO leading to slightly more diesel and wind capacity. Cheaper

capital costs leads to the inverse. 2050 sees even smaller changes, with there being more diesel and less HFO in a higher PV capital costs environment and less diesel and more HFO and solar in a lower PV capital costs environment. 2060 had only the slightest of changes with no appreciable change in the capacity mix.

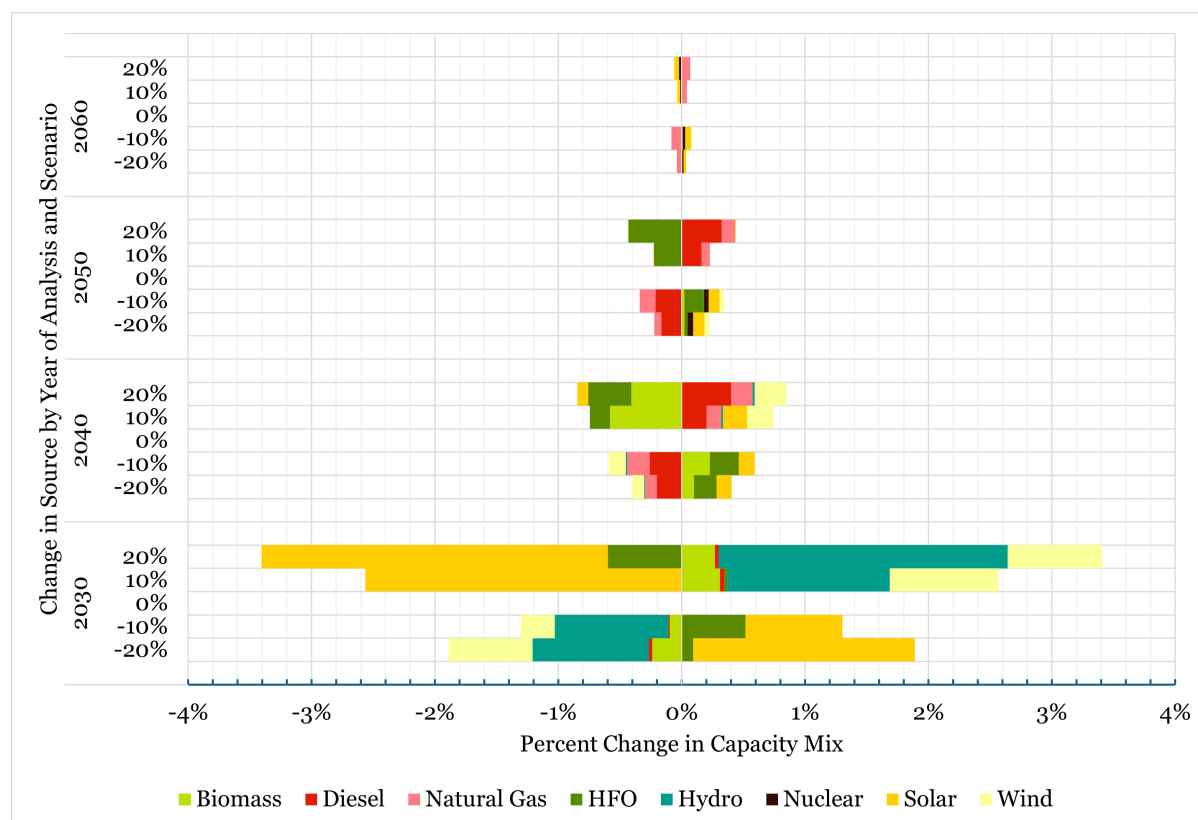


Figure 5.4.5: Change in Installed Capacity by varying PV Capital Costs

Table 5.4.2 shows the LCOEs for each period in the sensitivity scenarios on PV capital costs. Appreciable variation in the LCOE can be seen in the first two decades of the modelling period. In the decade ending 2030, each 10% change in PV costs changes the LCOE by 0.02-0.03 cents/kWh, directly proportional to the increase in costs. By the next decade, the LCOE has dropped universally by 4 cents/kWh and the amount of change between the scenarios is roughly 0.03 cents/kWh. The difference between the lowest and highest LCOEs in the first two decades are roughly one cent/kWh. That range is vastly diminished in the following decades. The decade ending 2050 sees a 2 cents/kWh decrease in the LCOE but across the scenarios, the LCOEs are much closer, with the scenarios with decreased costs being 0.001 cents/kWh lower than the base AMB and the scenarios with increased costs being 0.001 cents/kWh higher than the base AMB scenario. By the decade ending 2060, the LCOE has fallen a further 4 cents/kWh in all scenarios and the LCOE distribution is as it was in the previous decade. By the last decade of the modelling period, each scenario has an LCOE of 0.013 USD/kWh as the changes in PV capital costs have negligible effect on the LCOE.

Table 5.4.2: LCOEs from PV Capital Costs Sensitivity

Sensitivity Scenario	-20%	-10%	AMB	10%	20%
Period	<i>USD/kWh</i>	<i>USD/kWh</i>	<i>USD/kWh</i>	<i>USD/kWh</i>	<i>USD/kWh</i>
2021-2030	0.139	0.142	0.144	0.146	0.148
2031-2040	0.096	0.100	0.103	0.106	0.109
2041-2050	0.070	0.070	0.071	0.072	0.072
2051-2060	0.030	0.030	0.031	0.031	0.032
2061-2070	0.013	0.013	0.013	0.013	0.013

The way that changes in PV capital costs change the degree to which each technology type electrifies the population is shown in Figure 5.4.6. In 2030 (and every other year), increasing the costs by 10% or 20% result in a less than one percentage change in stand-alone PV and mini-grid PV prevalence. Increasing the cost induces a shift from stand-alone to mini-grid, and decreasing the cost induces the reverse operation. The most significant changes throughout the years only occur when the capital costs are decreased by 20%. 2030 sees the most change at this level of cost decrease, at $\pm 6\%$, but the amount of change decreases with time. In 2060, changes in grid subscription begin to appear with fractions of a percent being lost to stand alone PV at lower PV costs, and fractions of a percent being lost to mini-grid at higher PV cost in 2070.

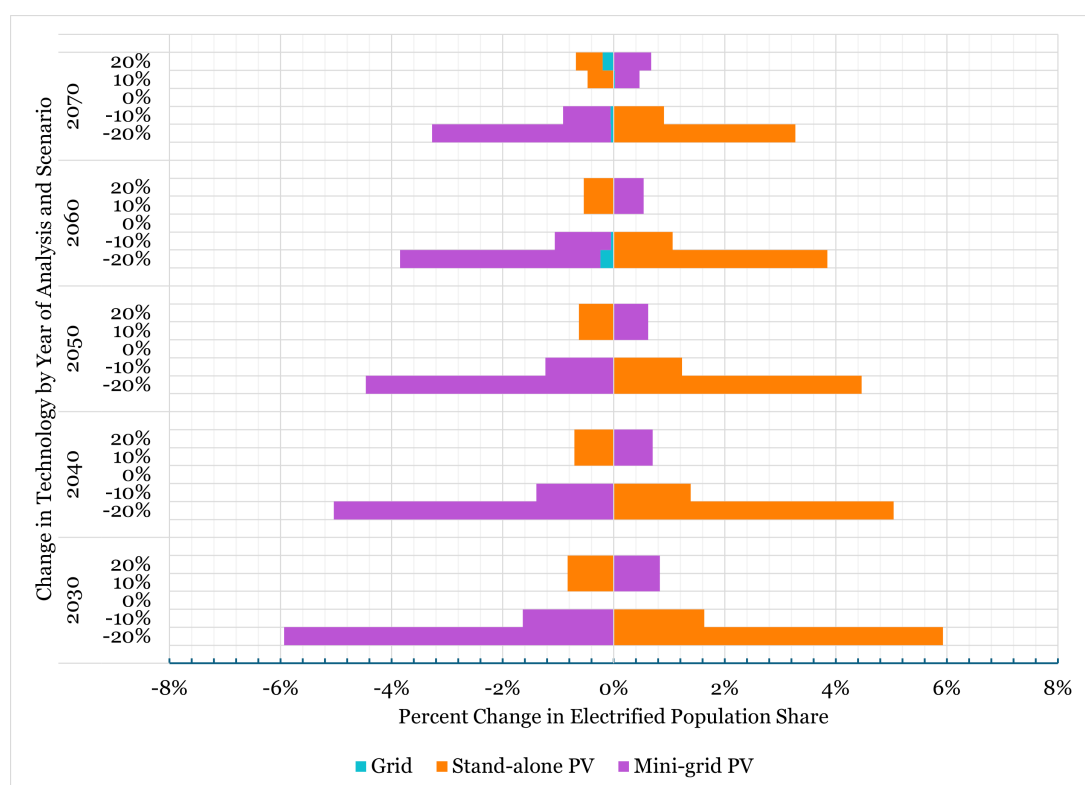


Figure 5.4.6: Change in Electrifying Technology by varying PV Capital Costs

Chapter 6

Discussion

This chapter discusses the results of the three scenarios, BAU, BAL, and AMB along with the sensitivity analyses performed in the OSGEM optimization to glean insights into the electricity sector development of Burkina Faso.

6.1 Business-as-Usual (BAU)

HFO is the dominant fuel source in the early stages of the model due to the initial large capacity that Burkina Faso possesses and the cost effectiveness of HFO fuel. However, in the face of imports and the introduction of solar, which has no variable cost and ever-decreasing capital cost, it becomes more advantageous to import more but also generate more using solar, which is better from a cost perspective as solar experiences declining capital costs. Modelled into the scenario is the natural decline of existing HFO technology due to those assets reaching their end-of-life, but no new HFO generating assets are built until 2047, after the generation inflection point around 2045. Though solar has the cost advantage, it does not cause a sharp decrease in HFO use due to HFO comprising the majority of the baseload operation (accompanied by biomass and hydro (admittedly tiny)), characteristic of thermal generation plants. In this part of the modelling period, output from solar, wind and imports are varied to meet the hourly demand. This can also be understood from looking at the BAU imports figure (Figure 5.1.3), where imports are steady or steadily increasing from Côte d'Ivoire and Ghana, respectively, but varying from Mali as OSGEM determines the least-cost pathway. The choice to vary imports from Mali is due to the dispatch decisions OSGEM made due to the variable costs of importing. Imports from Côte d'Ivoire and Ghana are 2 cents/kWh and 1 cent/kWh, respectively, less expensive than imports from Mali which makes them more likely to be used first, to minimize costs. Although the increase in solar production did not come with a sharp decrease in HFO use, it did come with a large capacity increase in solar PV, which occurred due to a lower capacity factor as compared with conventional generators which typically have higher capacity factors.

In 2035, the electricity imports from Mali briefly fall to zero due to the high influx of electricity from wind sources. It is interesting to see that even though the wind infrastructure was restricted

to gradual growth over a decade (the decade before 2035), the model still chose to implement a jump in wind energy. This is a case-in-point of wind's affordability and ability to compete with other forms of energy, including imports. It can be said that if the model were left to its own devices, that more wind would have been implemented sooner, likely decreasing or eliminating the need to import electricity from Mali. However, this case would require more strategies on the supply side to maintain stability and more human capital earlier on to facilitate this.

After 2045, hydroelectric generation is fully developed and is used until the end of the model period, indicating its cost preference over HFO. However, there is not much energy available from hydro sources. Consideration should be given to whether or not hydroelectricity should be realized in Burkina Faso, where meeting a small portion of electricity needs comes at the cost of water resources that could possibly go towards agriculture and, more specifically, biofuel production from biomass whose growth requires water, maintaining the ecological diversity of the rivers and lakes in the country, or toward the potable water needs of communities for drinking and sanitation purposes.

Politically, it would be prudent for Burkina Faso to maintain close ties with WAPP in order to facilitate this energy trade, especially if the import tariff remains at the same level for Côte d'Ivoire and Ghana, as electricity from these countries is used at all times. It is also interesting to see that even though imports from Niger are available from 2022, imports do not start until 2045, due to the fact that these imports have the highest tariff. In this part of the optimization, there are few sources to select from to meet the demand especially at night, despite the abundance of solar energy totally available to the country, which is not available at that time. As the optimization resorts to HFO to satisfy the future energy requirement, the full utilization of electricity from Niger according to the constraints proves that the Nigerien tariff is less than the overall per kWh cost of electricity from HFO. More work is required to fully understand how tariffs will affect electricity availability to Burkina Faso as WAPP has a market with fluctuating prices in reality, unlike the constant price assumption used in this optimization for simplicity.

It is encouraging to see that a cost-optimal build-out of the grid in Burkina Faso BAU scenario drives emissions down to near zero by the mid-2040s. This shows that, until then, emissions policies are not needed to reduce GHG emissions for a lower demand scenario. The cost-effectiveness of solar, wind, and biomass electricity generation allows for the natural diminution of emissions. However, post-mid-2040s, the available energy from those sources cannot meet the demand at the critical times, and in the absence of other sources, HFO is used and the lack of emission policies allows for unfettered use of the fossil fuel, which is responsible for these emissions. Even though annual emissions at the end of the modelling period are more than twice what they were at the beginning, the carbon intensity of electricity remains low as compared to in 2021. This is indicative of the amount of renewable electricity in the grid. Due to the uncertainty in carbon intensity of neighbouring countries' grid at the time, this was ignored

for imports so that a focus could be put on the pollution due to internal generation, but it is worth investigating from a regional perspective. Additionally, emissions may be higher in the real-life BAU scenario as, from a fuel flexibility perspective, Burkina Faso may choose to also employ diesel generation which will increase emissions, but with the benefit of stronger energy security.

In the BAU scenario, pre-2045, notwithstanding the continued but diminishing use of HFO in the electricity grid, no new investments are made in HFO in this time. Instead, there are massive influxes of capital investments into solar and wind technology due to these technologies possessing a lower capacity factor, necessitating the realization of more installed assets. However, in this same pre-2045 period, the biomass installation rate is more constant over time, as it replaces the load that HFO is carrying. In the 2050s and 2060s, new biomass installations are simply the ones that are replacing the assets that were installed 10 - 20 years prior, as Burkina's capacity to utilize biomass-sourced electricity has been maxed out. HFO assets are only invested in post-2045 when the demand of energy outstrips the capacity for all other available sources to meet demand at all time. The amount of capacity is kept low annually due to the high capacity factor of HFO assets. Burkina Faso should prepare itself for annual investments of 300 to 400 million USD annual, while expecting spikes in solar and wind to maintain the fidelity of its grid.

These investment amounts have direct implications for the LCOE in each decade. In the beginning of the modelling period, the LCOE is high due to high investments in renewable energy, while demand is relatively low for this modelling. As the demand increases throughout the years while the annual investment roughly stays the same, we see the LCOE decrease over time. In the last decade of the modelling period, though the demand is the highest of period, investments in all major contributors to the grid increase to more than any other time during modelling. However, this high amount can be attributed to edge effects from modelling, and the LCOE in a real-life BAU scenario might be lower than what has been modelled here.

Additionally, it is interesting to note that changes in the grid LCOE does not have an effect on the deployment of off-grid technology to electrify the unelectrified population. Because the historical trend dictates a slow electrification rate over time, it is enough to use the existing grid infrastructure to provide electricity to the minimum number of people required to satisfy the prescribed electrification rate. Geo-spatially, grid expansion is not necessary to satisfy the electrification rate as those that will be electrified under this regime already live or will live in an electrified population cluster. If this government or electricity sector stakeholders are slow in the development of the sector, only focus on the grid is required so long as the grid cost does not shoot up rapidly for some reason. Sahlberg et al. (2021) found similar patterns in least-cost access modeling for Burkina Faso where slow electrification strategies favored urban clusters and left rural gaps.

The BAU scenario represents a pathway showing that even following historical trends, renewable electricity is encouraged on the basis of cost, aligning with global trends, though that benefit only accrues to urban dwellers, with rural community members being mostly left out of the electricity developments in-country. Falchetta et al. (2021) also highlight that, without targeted policy for rural electrification, least-cost solutions tend to serve denser urban/peri-urban settlements sooner. However, a future without emission policies causes a resurgence in the use of fossil fuel, leaving the developing country vulnerable to price shocks and a system with negative consequences for climate change. The BAU fails to align with SDG 7 (universal access).

6.2 Balanced (BAL)

Like BAU, the BAL scenario has HFO as the dominant fuel source in the early stages of the model due to the initial large capacity that Burkina Faso possesses and the cost effectiveness of HFO fuel, historically. However, as HFO assets start to be decommissioned, generation from new solar and wind assets fill the gap, along with increasing imports from Côte d'Ivoire, Ghana, and Mali. What differs between BAL and BAU, is that once existing HFO assets are decommissioned, they do not return as new assets are not built. Once again the renewable energy assets are able to out-compete HFO on an advantageous cost basis. The results show that the minimum NDC requirement for renewable energy in the grid does not force renewable energy development as the penetration levels greatly exceed the minimum threshold. Pre-2040, the behaviour in the BAL is similar to the BAU, HFO and biomass play a baseload role while solar, wind and imports are varied to meet demand.

Whereas in the latter years, BAU chose HFO to meet increasingly significant portions of demand, BAL met that demand with natural gas. Pappis et al. (2021) observed similar shifts in energy mix when firm, cheap capacity like natural gas was introduced into African scenarios. Natural gas electricity generation becomes the base supply for the grid. Natural gas is cost-competitive to the point of discouraging the development of biomass assets as the generation is seen to be decreasing in the 2040s, 50s and 60s. The presence of natural gas has also had an effect on the penetration of solar electricity in the grid due to BAU having a 27-30% penetration while BAL had a 25-26% penetration between 2050 and 2070. Toward 2070, the biomass generation completely disappears in this scenario, but in a real-life application of this scenario, as demand grows and in order to maintain the minimum RE penetration threshold, biomass will be required in the absence of energy storage or another source. Also, as opposed to the BAU, no hydroelectric generating assets are built, but existing ones are utilized until the end-of-asset-life in BAL. Thus, the water concerns in the BAU do not come up here but water use must be contemplated for facilitating natural gas-based electricity generation.

Imports are important in the BAL, just as in the BAU. The imports ramp up according to the

uptake of solar energy into the grid in the 2020s. Throughout the 2030s, as HFO is further decreased, imports are used to meet peak demand as seen in the case of Nigerien imports. The two dramatic drops in imports occur in 2035 with the quick expansion of wind energy, and in 2040 with the availability of natural gas, both of which out-price electricity imports from Niger. In 2040, with the level of demand in that year, natural gas also out-prices imports from Mali. Imports from Niger are noticeably seen again toward the end of the modelling period to also meet peak demand. This scenario requires more cooperation from WAPP, especially from Niger, if Burkina Faso is to meet this peak demand at least-cost. Just like in BAU, the behaviour of imports depends on the tariff that can be agreed upon between countries.

In a similar fashion to BAU, the emission limit does not put pressure on the sources used to generate electricity. By cost optimization alone, the emissions trend downward until 2040, due to the decommissioning of HFO assets and up again because of natural gas use. However, the NDC emission limit is likely to become relevant and restrictive in the 2070s as by the late 2060s emission are just shy of the modelled trend. This also suggests that in the 2070s, in order to continue in the steps of this scenario, that biomass use or possible nuclear energy will be needed to provide electricity without Burkina Faso exceeding the emissions limit. Nuclear, though available in this scenario, is not used as it cannot compete on a cost basis with natural gas or other sources, suggesting nuclear would need international support or stringent policy for the realization of nuclear energy in-country. It is good to see that the carbon intensity of grid electricity in BAL falls rapidly toward 2040 and does not rise to even half of the 2021 value by 2070, showing that even in a higher demand situation natural gas does not have a negative impact of the emission per generated electricity, thanks to the amount of renewable energy present in the grid. The mix in the latter years could again change if accounting for the carbon intensity of imports. Compared to BAU, BAL is a more environmentally friendly scenario as the low point in emission is reached 6 years earlier in BAL than in BAU.

As in BAU, BAL sees no new investments in HFO technology and allows the use of HFO to decay over time. Instead, throughout the modelling period, solar investments are made, which made up the bulk of investments due to solar's lower capacity factor but also its availability and zero variable cost factors. Wind is invested to its maximum amount, which is advantageous to achieving the least-cost pathway in Burkina Faso, but generally does not require much capital over the modelling lifetime. In the pre-2040 dispensation, the firm capacity of choice is biomass, which has constant installations over time until the late 2030, where a spike in installations occur in preparation for the post-2040 dispensation and to replace HFO assets that have been reaching end-of-life, similar to what occurs in BAU. However, in the post-2040 dispensation, natural gas is the firm capacity of choice, which sees an initial uptick in installations to meet demand which the last of HFO goes away. After that, the installations occur gradually over time to meet the annually rising demand. In the late 2060s, there is another small spike in natural gas (and solar) asset installations to replace the last of biomass assets that were installed in the late

2030s. Wind sees a large set of new installations in 2060, which replace wind assets built in 2035, proving that in the era of natural gas baseloading, wind is still a cost-competitive option to supply power. Burkina Faso should again prepare itself for annual investments of 300 to 400 million USD annual, while expecting spikes in solar, wind, biomass and natural gas to maintain the fidelity of its grid.

The investment choices led to a more favourable set of LCOEs for the BAL scenario than the BAU. Similarly, in the beginning of the modelling period, the LCOE is high due to investments in primarily solar assets, in the lowest demand regime for the modelling period. The 2030s BAL LCOE is higher than the 2030s BAU LCOE due to the higher investments in solar, wind and biomass in the late 2030s. However, the low-cost character of natural gas, combined with the investments made in the decades prior, is able to drop the grid cost below 5 cents/kWh in the 2040s. The 2050s and 2060s see a rising grid cost due to the replacement of wind assets made in the 2020s and increasing amounts of solar and natural to meet the higher demand in this period. Edge effects may once again contribute to a higher LCOE at the end of the period than if realistically implemented.

Given the trajectory of reaching universal access by 2065 in BAL, we see that the grid access increases over time, but the increase seems to be loosely related to the change in the grid cost and more closely related/proportional to the urbanization rate of the population. Geo-spatially, the grid develops along presently existing medium voltage lines, with the density of connections increasing over the decades. Notwithstanding the low geographical density, grid connections supply most people with access due to urban areas having a higher population compared to rural areas. Rural electrification depends on the build-out of mini-grid PV systems as the country enjoys strong solar irradiation, while the cost of the grid power and the cost of extending the grid are higher than mini-grid PV systems. Sahlberg et al. (2021) and Peña Balderrama et al. (2020) both show that mini-grid PV is often the least-cost solution in low density rural contexts. The decentralized systems allow for electricity services to be provided all over the country, and the growth in systems match the set electricity access rate for that decade. By 2070, all people would enjoy electricity services, with ubiquitous mini-grid services, and a sparse amount of stand-alone PV systems used to achieve that 100% electrification rate due in part to mini-grid and stand-alone PV systems having a lower connection cost than to the grid.

The BAL scenario represents a case where even in the presence of policy ambitions to meet emission and renewable energy goals, the cost-effectiveness of new technologies are sufficient to meet the goals, without the policies being relevant until late in the century. Solar and wind development remain important in a higher demand scenario, and the availability of natural gas makes electricity more affordable. Nuclear is not a natural option in this case where the costs are higher than other options to implement this technology, while biomass acts as a transitional fuel but over the long term with a possible resurgence later in the century. Similar conclusions about

the cost-effectiveness and development challenges of renewables and nuclear energy are found in Pappis et al. (2021) and Sahlberg et al. (2021). Essentially, all unelectrified rural dwellers should employ mini-grid PV systems for their electrification for a final technology mix by 2070 of roughly 60% grid and 40% mini-grid PV. A comparable emphasis on the role of mini-grids in least-cost universal access pathways is made in Falchetta et al. (2021) and Peña Balderrama et al. (2020). The BAL fails to align with SDG 7 (universal access), but achieves Burkina Faso's NDC targets, and shows a pathway to achieving Agenda 2063 for electricity.

6.3 Ambitious (AMB)

In a similar fashion to BAU and BAL, AMB has HFO as the dominant fuel source during the 2020s, due to high installed capacity from before the modelling period and inertia in using this source. AMB diverges from the other scenarios in that HFO remains strong going into the mid-2030s and does not drop off right away. This behaviour is encouraged by the higher demand in this scenario than in the others. HFO provides baseload power while wind and solar provide intermediate load and imports provide peak load. Use of solar energy in the grid rapidly increases to meet growing demand. Wind and biomass remain repressed due to the self-imposed limits on growth by the user. The introduction of natural gas in 2035 replaces much of the electricity sourced from HFO between 2035 and 2050, a testament to the cheapness with which electricity can be produced with natural gas technology over HFO. Diesel is also used in peak demand times. It is interesting to note that biomass uptake in AMB coincides with the decrease in fossil fuel-sourced electricity. This is a sustainable switch in firm capacity, driven by the emission limit.

Toward 2050, both the RE and emission limits come into play, by guiding generation toward net-zero emissions, and a strong renewable energy presence. In the latter part of 2040, a surge in nuclear energy is injected into the grid. Under normal circumstances, as seen in BAL, nuclear would not appear as it is not cost-effective against other sources. However, with the model's requirement of no emissions after 2050, the limit encourages this zero-emission source. It is interesting to see that even though nuclear is capital intensive, it keeps growing until the end of the modelling period. This behaviour can be attributed to the very low cost of nuclear fuel, which is much lower than the import tariff from neighbouring countries. The overall cost of electricity from nuclear energy is therefore low, and when combined with nuclear's ability to provide constant baseload power, leads to its prevalence in a zero emission environment. This leads to its competition with renewable sources in the AMB scenario. From 2050 on, the RE penetration remains at 48% despite the expectation of RE proliferation as in BAU and BAL. The RE minimum limit is responsible for the preservation of solar, wind, hydro, and biomass in the grid. Without this limit, while solar may still increase, the other sources may suffer.

The character of imports is markedly different in AMB versus BAU and BAL. In the limited

regime (pre-2035), the higher demand necessitates constant use of all imports except for Niger to meet intermediate demand with Nigerien imports used to help meet demand at peak times. 2035 and onward sees a more expected trend in imports, where the two cheapest sources of imports immediately take over from the two most expensive sources from the period before (Niger and Mali). Imports until 2050 grow as demand in general grows, but also as other firm capacity sources like natural gas, diesel, and HFO are limited from the emissions limit. This scenario is the more import dependent in absolute terms of the three scenarios and further consideration must be given to the energy needs of the others in WAPP and their true capacity of possessing this excess energy. Also, it is questionable whether the current political climate will deem West African cooperation as favourable to Burkina Faso's overall interest, which leads to modifications of the implementation of an electrification pathway like this one. Imports decrease from 2050 on because as demand increases, nuclear is the cost-preferred option but nuclear fuel gets increasingly more expensive until the end of the modelling period, and since more has to be spent on nuclear (which is still the best option cost-wise), a little less is available to be spent on imports annual, which drives this decrease. This is combined with the renewable energy minimum requirement which forces RE in the grid. Solar still grows incrementally in this period, and this soft increase also leads to the soft decrease in imports.

In contrast to BAL, the emission limit does put pressure on what is allowed to generate in AMB. While the emission limit is increasing, renewable penetration has yet to ramp up to its highest level, which leads to increased reliance on HFO. By the time the late 2020s and early 2030s roll in, renewables contribute more to meeting demand, which decreases the emissions. 2035 sees a massive injection of imports and introduction of natural gas, which is less polluting than HFO case a sudden decrease in emissions. Emissions rise subsequently due to HFO use holding steady while more natural gas is used in the grid. By 2040, the emission limit is effective in consistently limiting fossil fuel use as higher demand places upward pressure on generation. Natural gas electricity being cheaper than HFO electricity, as seen in BAL, still increases until 2043 but at the cost of HFO, which begins to have its output curtailed. After 2043, nuclear penetration grows which makes it possible to balance the grid at a time where fossil fuel generation is forcibly decreased, and when fossil fuel generation is zero. The carbon intensity of the grid is always decreasing in this scenario, showing the power of renewable energy and emission limits in bringing a low carbon grid to fruition in Burkina Faso. Nuclear energy is crucially important to achieving net zero in this scenario but it also highlight the possibility of energy storage or increased biomass output or input in meeting this demand.

Given the demand growth in AMB, there are massive capital investments in infrastructure, particularly in solar and nuclear power plants, which provide the bulk of electricity throughout this scenario. The low capacity factor and large energy availability of solar makes it a source of choice for development. Nuclear's zero-emission nature in generation along with fuel cheaper than electricity imports allow for its explosion in the last three decades of this scenario. Wind,

though now representing a relatively small fraction of total investment, is still invested in to its maximum capacity due to its zero-cost fuel and general wind availability in Burkina Faso. The large growth in solar is possible through the relatively small investments in firm capacity for diesel, HFO, natural gas, biomass and hydro in the pre-2040 period. Diesel investments occur in the few years immediately before natural gas investments occur. Seeing the affordability of natural gas from the BAL scenario, an earlier availability of natural gas could result in the lack of necessity to invest in diesel or HFO, resulting in a further lowering of cost. Biomass' relatively lower capital investment costs compared to its constant and maximized use and indigenous qualities continue to make it an attractive source to generate power. The earlier availability of natural gas could also further displace HFO in the 2020s as well. This scenario faces the enormous hurdle of cost with nuclear seeing over one billion USD in investment every year with a peak of almost 4.5 billion USD, and solar seeing an average of 400 million USD annually with a peak of 2 billion USD. A lower-income nation like Burkina Faso will require access to concessionary funding and international support to achieve an electricity sector structured in this way. However, an avenue exists given the strengthening of ties between Burkina and Russia, a long-time user of nuclear energy.

The AMB scenario has even more favourable LCOE costs later in the modelling period than BAU or BAL, but the set up for that leads to higher costs in the earlier decades. The 2020s' and 2030s' LCOE is half a cent and three-tenths of a cent higher in the AMB, respectively, over the BAL due to higher investments in solar caused by the higher demand found in this scenario. The LCOE in the 2040s is higher than those of the other scenarios due to investments in nuclear energy, forced by an emission limit that decreased the fossil fuel penetration. However, there is a reversal in affordability in the 2050s and 2060s. The large supply of solar and nuclear energy into the grid in a high demand environment allows that high investment cost to be spread out over a rate-paying base, leading to the lowest LCOEs of the 2050s and 2060s in any scenario. This suggests that though it may be costly to build such a system, the average price to consumers will be low, an encouragement to electrification and economic growth. This is the only scenario where the grid cost trends down into the future as more nuclear penetration is realized. It could be argued that if not for the renewable energy penetration requirement, the grid cost could be even lower.

Universal access is achieved 30 years earlier (in 2035) in the AMB scenario compared to the BAL scenario, and geo-spatially, that has translated to the northern third of the country being covered in stand-alone PV system, which meets the decentralized electricity demand of some 200,000 Burkina Faso residents by 2030. The solar irradiation here is the strongest in the country. This fact, combined with the small incremental number of people who need to be electrified, makes stand-alone PV systems the technology of choice. Where there are larger population clusters, a larger system type in the form of mini-grid PV systems is chosen. Grid extension does not occur as the cost to do so is greater than the cost to erect other system

types. Due to population shifts, the number of people connected to the grid grows while mini-grid penetration sinks, as seen in the BAL scenario as well, but geographically, the general distribution remains the same. This pattern holds until 2060, when a massive extension of the grid network takes place, converting persons using stand-alone and mini-grid system to grid electricity, which occurs due to the great drop in the grid cost for generating electricity. This happens through the 2060s until 2070 as well. It is interesting to note that this is the only time in any scenario where OSGEM decides to extend the grid, which speaks who the cost prohibitive nature of increasing access. If the general security of grid connections are desired for the population, policies should exist to decrease the cost barriers to grid connection.

The AMB scenario lays out a pathway to national electrification given rapid rural and urban electrification, high renewables and net-zero emission goals and policies for Burkina Faso. It requires rapid on- and off-grid solar energy development, high use of other renewables and nuclear, and transitional fossil fuel technology, highlighting a high degree of energy independence. AMB does align with IRENA recommendations for high solar penetration, but also bring to light the barriers in achieving a grid dominated by nuclear energy. While nuclear is challenging, it makes the scenario the most resilient to fossil price volatility and import dependency of all the scenarios. Organizing the requisite financing, policy frameworks, and international cooperation is key to achieving this strong scenario for Burkina Faso. AMB very nearly meets SDG 7 goals, but only slight tweaking would be needed to achieve it on time.

6.4 Sensitivity Analysis

6.4.1 Net-Zero Year

Delaying the net-zero year has implications on energy independence and costs for Burkina Faso. Natural gas use increases over time as net-zero emission is delayed, which requires an increase in biomass use around 2040 and increase in imports between 2050 and 2070. This results in the taxing of the natural resource earlier than in the reference scenario (AMB), while encountering more imports, decreasing the independence of the nation. These changes come at the cost of solar and nuclear penetration, reducing the endemic sources of generation.

Burkina Faso's weakened position is highlighted in the volatility in the installed capacity over the course of the scenario. The changes in the net-zero target cause there to be an increasing reduction in solar capacity in 2040. While this is replaced by firm capacity options like natural gas, HFO, biomass, and wind, a renewable source with a more constant output than solar, the solar capacity still needs to be made up to some degree in the 2050s and 2060s, as the pathway must still abide by the minimum RE penetration limit.

These changes are negatively reflected in the costs. Though the relaxation of the net-zero year

reduces near-term pressure on decarbonization and reduces the cost of generating electricity in the near term, the effect of this is felt in later decades. The LCOE in the 2050s increases by 45% over a 20-year delay and in the 2060s by 85% over the same delay period. Although changes in grid cost barely affect the distribution of grid usage and off-grid technology usage, these cost signals underscore the long-term cost of delaying decarbonization and demonstrate the role of effective policymaking.

6.4.2 PV Capital Costs

Changes in the capital costs of solar power plants and smaller PV systems have a small effect on energy security and energy costs in the AMB scenario for Burkina Faso. The greatest changes seen in the years analyzed are in 2030. As expected, as capital costs increase, installed capacity decreases as OSGEM minimizes the cost. This capacity change is inversely proportional to the growth of more energy-stable sources such as hydro and wind, though these changes are relatively tiny. These early changes cause small perturbations in the later decades, a demonstration of how early action has lasting impact when developing the electricity grid.

It is interesting to see that no changes in the generated electricity mix occur at the 2050 and 2060 marks despite the change in PV capital costs. This effect is due to the RE minimum limit mandating the inclusion of renewable energy into the grid at 48% and since we see that the limit is strictly maintained at 48% in the AMB scenario, we know that PV (and other renewable energies) are not cost-preferred to the rise in nuclear energy in the grid. This suggests that either solar capital costs would have to be much lower than 20% to see a change in solar uptake in the grid post-2050 or that the intermittency and/or diurnal nature of solar energy puts in a step behind the baseload capabilities of nuclear energy and imports. This is a demonstration that strong policy implementation can achieve renewable energy goals where costs themselves do not encourage growth. This situation also demonstrates that changes to this degree have little impact on energy security.

When it comes to the grid cost (LCOE) itself, the changes are small to almost nil in every decade of the modelling period, which shows how cost resilient the Burkina Faso grid could be to capital cost fluctuations, a good sign for energy security from an affordability standpoint. This kind of response to solar, which is heavily featured in AMB is a good sign for both investors and ratepayers and this lowers the uncertainty in price, raising confidence for investment and participation in the electricity sector.

Significant changes occur among the decentralized energy systems when PV capital costs change. The share of people electrified by stand-alone PV systems increases by 700% in 2030 (with diminishing returns thereafter) when the costs decrease by 20%. However, the changes are much smaller when the capital costs increase. This bodes well where the volatility exists when the costs are in the consumer's/investor's favor. However, this presents a planning challenge

in general for policymakers as they account for the number of people to be electrified by which technology. A policy that is able to harmonize as much as possible the common aspects of stand-alone and mini-grid system could be a beneficial policy to manage the realization of these systems in the coming decades. These analyses highlight the sensitivity of Burkina Faso's energy transition to external factors, emphasizing the importance of stable policies and international support.

6.5 Sustainability Analysis

The scenarios and their analyses show electrification pathways with varying levels of environmental stewardship. All pathways feature declining emissions at least into the 2040s, with AMB achieving net-zero emission by 2050. Overall, BAU's roughly 90,000 kTon CO₂eq scenario emissions was less than half of BAL's 200,000 kTon CO₂ eq scenario emissions and more than double of AMB's roughly 40,000 kTon CO₂ eq scenario emissions, highlighting how growing demand without policy intervention could easily lead to ballooning emissions. However, while AMB has less direct air pollution than the other pathways, consideration should be given to the treatment and handling of an estimated 260 cubic metres of high-level nuclear fuel waste (the equivalent of 3.5 standard shipping containers) over the whole modelling period [61]. This tiny amount of space can easily be accounted for in a facility on the premises of the utility. Both BAU and BAL both keep a relatively low carbon intensity toward the end of the modelling period, but if Burkina wishes to drive that intensity lower into the future without using nuclear energy, it might investigate the use of CCS, but the cost of implementation would have to be studied. In terms of fuel use, natural gas is preferable in emissions and cost to HFO, and the faster the fuel substitution, the more environmentally sustainable the solution.

Although these electrification pathways thankfully rely little on hydropower, these scenarios raise concerns about water stress. Without contributing much to the overall energy mix, the development of hydropower assets could alter the ecology of the Volta River system and affect the agricultural yield for not only the Burkinabé population, but also transnational disputes may arise from Ghana and Côte d'Ivoire about water use and water rights. This concern is most relevant in the BAU scenario, and less so in the BAL and AMB scenarios which minimize hydro use. Land use should also be considered in the context of solar and wind generation. An estimated 260 - 400 km² would be needed to support the highest annual installed capacity of solar PV plants and 220 km² would be needed to support wind turbines in Burkina Faso, a country that has orders of magnitude more land than required [62]. In the implementation of these plants, care is required to not compete with use requirements for food, biomass use to support grid electricity, or proliferate deforestation.

All of the pathways have strong renewable energy grid penetration rates, meeting and exceeding national penetration targets. BAU, thanks to its lower demand, is able to have the higher overall

penetration rate among the scenarios. All pathways heavily depend on solar PV electricity which avoids emissions by decreasing fossil fuel reliance while providing power at a low cost. This transition is vital to achieving the country's Nationally Determined Contributions (NDCs) and aligns with Sustainable Development Goal (SDG) 13—Climate Action. Sustainable use of solar and other renewable energy sources can be strengthened by the development of energy storage systems.

The scenarios give insight into social sustainability through the lens of equity in energy access. The scenarios demonstrate how slothful action, indicative of lack of pointed policies, results in communities going without electricity for their self-development and in the perpetuation of urban-rural divides (BAU). When goals are achieved, as in the BAL and AMB scenarios, energy access is granted in mixed modes; with the centralized grid for those near the existing infrastructure and decentralized mini-grids (PV) for those out of the grid's reach, showing that multiple solutions are needed to increase access to all at least-cost.

The proliferation of electricity through grid/mini-grid expansion should also result in the reduction of disadvantageous cooking methods, responsible for respiratory diseases, with a disproportionate impact on women and girls in terms of health, time, education, and income. With the introduction of this clean cooking fuel, policies to encourage the adoption of clean cooking surfaces should especially target rural communities to ensure the reduction of poverty, the increase in health outcomes, and the advancements in education, consistent with SDGs 3,4,5,6 and 7: Good Health and Wellbeing, Quality Education, Gender Equality, Clean Water and Sanitation, and Affordable and Clean Energy, respectively, to avoid geographic and socio-economic disparities.

While the scenarios give least-cost pathways, the distribution of employed technologies could raise geo-spatial fairness concerns about who and what area *should* receive what type of system. Northern Burkina is dominated by decentralized systems (in BAL and AMB) where those people may not benefit from the energy security the grid provides. Southern Burkina cities have more grid access but may lose out on the flexibility and agency that decentralized systems can bring. The different locations for deploying the different types of system may affect the speed of energy access to communities and the time it takes for the benefits of electricity to reach the masses. Policies should be tailored in a two-pronged approach, having a systematic roll-out of grid extension and decentralized system development to ensure fairness.

Economically, the scenarios bode well for affordability in all scenarios, the LCOEs decrease over time generally. This decreasing cost regime should be a positive signal to investors for grid development, and these lower costs can hopefully be passed on to consumers, to encourage grid access. This behaviour is supported by strong solar inclusion paired with low cost HFO, natural gas or nuclear, depending on the scenario. Policies and strategies geared toward financing these technologies (especially in the case of nuclear with very high upfront costs) should be

implemented to secure this energy transition. AMB in particular can enjoy lower operational costs once the massive capital costs are overcome, while BAL and BAU have cheap fuel and imports to support these pathways.

Both imports and nuclear use expose the country to increased risk if over-relied upon. Diversifying renewables (e.g., wind, biomass) and strengthening regional grids in WAPP could enhance/secure stability. When examining the sensitivity of the net-zero year on cost, it was found that it is more sustainable to adhere to the policy instead of pushing it off into the future, which incurs more cost. In addition, rural electrification stands to benefit from lowering PV capital costs, and the pathways should be flexible in design to take advantage of this decrease should it arise. Moreover in regards to these changes, the system exhibits resilience to fluctuations in technology costs. The LCOE remains relatively stable even under 20% changes in solar PV capital costs, indicating low financial risk and high investment confidence. Job creation from infrastructure deployment, particularly decentralized energy systems, is another key economic benefit. The cost-effectiveness of distributed technologies in sparsely populated areas further enhances the long-term affordability and sustainability of the system, provided that adequate financing mechanisms, training programs and public-private partnerships are established.

Overall, the scenarios expose trade-offs between cost, sustainability, and resilience. BAU is cheapest in the short term but exposes Burkina Faso to import risks and prolonged fossil dependency, BAL represents a feasible policy path with moderate costs and sustainability benefits, and AMB is most aligned with climate targets but requires strong institutional support and investment. Solar PV is indispensable in all cases. However, without diversification (e.g., gas in BAL or batteries), energy reliability may suffer. It is also important to follow policies as laid out like the net-zero target. Delaying net-zero ambitions leads to higher system costs later on, emphasizing the need for early action.

This analysis underscores the necessity of stable, forward-looking policies. To support the most sustainable outcomes:-

1. Modernize grid infrastructure, especially to accommodate solar and wind. Ensure that trade policies complement the development of renewable energy storage to maximize electrification and investment efficiency, managing cost and reliability.
2. Develop regulatory frameworks for the prioritization of decentralized energy systems. Scale mini-grid PV for rapid rural access. Support local capacity building and workforce development, especially in rural energy deployment.
3. Leverage international climate finance for infrastructure investment in large-scale renewable and nuclear energy. Especially in the case of nuclear financing, seek concessional loans/IAEA partnerships to overcome high upfront costs in a high demand

scenario.

4. Ensure justice in the energy transition by pairing any biomass phase-out with reforestation and clean cooking programs. Prioritize infrastructure that supports energy justice, by electrifying schools and community health centers, ensuring that no population segment is left behind.

6.6 Limitations

While this study makes important contributions to electrification pathways for Burkina Faso and long-term geo-spatial energy systems modelling, it is not without limitations. The timeslices used in this study were selected for modelling efficiency but as a result neglect the month to month and hour to hour changes that occur in the electricity grid. Therefore, it leaves something to be desired in terms of operationalizing this study. In addition, the myopic periods in the OnSSET in this study at best give results every 5 years and while that may allow the nation's administration in that time to set achievable goals in that mandated period, it doesn't say how the development should be made year to year.

This study also does not contemplate the stochasticity of supply or demand. While the solution involves a reserve margin to handle the failure of assets, it lacks a detailed reliability analysis to show how random changes in the output of intermittent sources like solar PV and wind have an effect on the rest of the system. The demand curve as implemented in OSeMOSYS is also unable to handle random variability and expected losses of load.

Additionally, this study does not consider the effect of energy storage systems to meet demand and make best use of intermittent sources of energy. Depending on the cost and size of storage, this could affect what technologies are used at peak times and storage's ability to support renewable energy output. Even if storage was implemented, it would suffer from the previously described limitations of not capturing the stochasticity of supply and demand, and not having enough timeslices to develop an optimization algorithm for energy storage employment.

This study leaves room for refinement of the demand projection over time. In the absence of data, the proportion of demand for the various electricity-consuming sectors was carried forward from the initial year, which does not take into account changes in the prominence of residential or commercial demand as Burkina Faso develops. This can have an effect on the OnSSET portion of the optimization as perturbations in this demand can affect the convergence of the model and the results thereof. In the same vein, electricity imports and exports are taken into account with detail until 2033, after which are assumed to be constant. While that is unlikely to happen, data was lacking in this regard and is hard to project over the long term.

The thesis is also limited in its insights into how Burkina Faso's electricity sector will be

impacted by climate change. Changes in rain patterns and cloud cover can change the availability of energy for solar PV and hydroelectric plants. As the sun is a dominant source of energy in all scenario, changes in solar irradiation could impact the energy mix of the grid and the prevalence of distributed energy sources. While changes in rainfall affect hydropower, even more pertinent for this thesis is the effect of changing water availability on agriculture, especially biomass for energy, another important resource in this study. The lack of accounting for climate change leaves the uncertainty surrounding it unmitigated.

Moreover, the geo-spatial work in this thesis depends on many population clusters whose relative density (persons/area) does not change throughout the modelling period, even though the bulk urban/rural population distribution does change. This constant nature ignores the development possibilities of persons moving to new towns to take advantage of new opportunities, that might be captured in the OSeMOSYS demand projections, but have no geographical location assigned to that demand. Taking this into account makes the scenarios more robust in crafting strategies at the community level, namely changing the local grid/off-grid energy technology distribution.

6.7 Future Work

The limitations of this thesis provide opportunities for further growth in this work. To account for the stochasticity of supply and demand, OSGEM can be improved to handle Monte Carlo simulations in the OSeMOSYS-part of OSGEM to model many variations of expected annual patterns of irradiation and rainfall to obtain more accurate insights on solar PV generation, especially. In the same breath, this work could be strengthened by integrating this work into a climate-water-energy nexus model to fully capture the system interactions and effects of energy on other systems in Burkina Faso.

It would also be interesting to perform a more detailed demand projection for this type of model to capture the true growth in demand for each individual sector, including imported and exported electricity. While doing this, hourly (or as defined as possible) demand curves should be formed for electricity imports and exports to improve the generation estimation hour by hour (or at least as detailed as the timeslices).

Future work on the OnSSET side of OSGEM can include adding code to alter input files to account for shifting population densities to be able to model different movements of people according to climate change, economic development or disaster. Additionally, to address one of the drawbacks of the myopic modelling where there is no memory of the chosen sources to electrify then-unelectrified areas, code can be added so that OnSSET knows what sources are currently providing electricity to an area, to know the remaining operational life of assets and ascertain the true capacity to be built in a development year, accounting for assets that expire

since the start of modelling.

Chapter 7

Conclusion

This chapter summarizes the key motivations, findings and contributions of this thesis.

This thesis presented an analysis of electrification pathways available to Burkina Faso, over the long-term and with geo-specific detail. The study found its importance in supporting a country, that to a large degree, lacks electricity access and has low renewable energy use, which is in a region also largely lacking electricity access and use of renewable energy. A study on one country in the Sub-Saharan African region supports efforts to develop the entire region and alleviate poverty, increase economic and social welfare, increase energy sovereignty, and increase natural and built environmental management. Particular interest was taken in the energy systems that can be used for this development and distributed energy systems seem promising for reaching far-flung area, despite most electricity sector investment being funneled into the national grid. This thesis therefore provides insights into how Burkina Faso can leverage its natural resources and manage its policies to bring forth this needed development, specifically looking at the change in grid and distributed energy system use over time and the effect of changing costs on this system.

To develop these electrification pathways, the hybrid modeller, Open Source Geo-spatial Energy Modeller (OSGEM), was created to combine the strengths of OSeMOSYS (long term energy modelling) and OnSSET (geospatial electrification modelling) and cover each other's weaknesses, to examine the interaction of grid development on off-grid electricity sector development. The creation and use of OSGEM addressed some research gaps in the field of electricity planning as little work has been done using both tools for one country and performing long term geospatial analyses. In this thesis, three electrification pathways were developed for Burkina Faso for the years 2021 - 2070: BAU, BAL and AMB scenarios. The BAU (Business-as-Usual) scenario represented the evolution of Burkina Faso's electricity sector according to historical trends, existing energy trade plans aligning with regional (West African) ambitions, and low demand growth. New renewable technologies undergo gradual inclusion into the grid, and natural gas and nuclear were excluded as being suitable/indicative of more aggressive national growth.

The BAL (Balanced) scenario uses BAU as a base, but represents a Burkina Faso with

moderate demand growth, portraying a higher rate of development, such that universal access is attained by 2065, just after the deadline of Africa's Agenda 2063. BAL also portrays a moderate approach to sustainable development by realizing a 27% minimum renewable energy penetration rate, availability for natural gas and nuclear generating assets, and NDC emission limits. The AMB (Ambitious) scenario takes the BAL scenario a step further by modelling even higher demand, higher minimum requirement for renewable electricity (48%), net zero emissions by 2050, no electricity exports and as needed imports from 2035 on, and universal access by 2035, indicative of aggressive growth. Altogether, this scenario represents a strong, energy-independent Burkina Faso.

If the historical pattern dictates the future of Burkina Faso (BAU), then the future is categorized by fossil fuel dependence, coupled with high (but decreasing into the future) renewable energy penetration and high imports (RE: 62%, Imports: 10%, HFO: 28% in 2070). While slower growth and limited options result in overall costs that the country in its current state has the easiest time achieving, it also results in higher costs borne per person/entity to access the service. The situation changes, according to BAL, to lower renewable energy penetration, but lower imports and lower costs, while increasing the fossil fuel dependence (RE: 33%, Imports: 5%, Natural Gas: 62% in 2070). Lower costs, imports, and renewable energy penetration can be attributed to the cheap costs and firm capacity of natural gas which allows it to hold the dominance it has in this scenario. While investments are higher compared to BAU, the higher overall demand allows that cost to be spread out over the rate base. AMB is characterized by medium renewable energy penetration, relatively low imports, very high upfront costs, and nuclear energy domination (RE: 50%, Imports: 4%, Nuclear: 46% in 2070). While natural gas proved very cost effective in BAL, it is just a transition fuel to bridge the high-HFO era to the high-nuclear energy era. Increased imports also assisted in the transition. Nuclear's dominance does not come solely from its cheap fuel (as its very high capital cost negates the benefit normally), which competes marginally with the cost of imports in AMB, but also from its zero carbon emission fuel which allows it to operate under the net zero emissions limit. Sensitivity analyses on the AMB scenario showed that early action on net-zero targets pays off in terms of emissions and system cost reductions and that change in PV capital cost have a small effect of the grid energy mix.

Geospatially, the results show that when development is slow, the grid is the most affordable and able choice to electrify communities under those conditions (BAU: 100% of the 42% of the population electrified by the grid by 2070). Because the rate of electrification is slow, to achieve least-cost development, rural communities are disadvantaged as it is not beneficial to extend the grid and thus most rural communities miss out on electricity. However, when electrification rate policies are introduced (BAL and AMB), in order to fulfill the goals of the policy on time, grid, mini-grid, and stand-alone energy systems are required to meet the demand (BAL: 100% of population electrified by 2070 [60% grid, 40% mini-grid PV, minuscule stand-alone PV];

AMB: 100% of population electrified by 2070 [63% grid, 36% mini-grid PV, minuscule stand-alone PV]). The earlier the desired date for universal access, the more stand-alone PV is used (as seen in AMB) but nearly all distributed energy systems used to increase electrification are mini-grid PV systems. The decrease in grid cost had the positive effect in BAL and AMB of inducing more connections to the grid, an effect of affordability increasing accessibility. The decrease in PV capital costs by up to 20% shifted the mini-grid/stand-alone PV electrification share by between 3 to 6 percentage points, depending on the decade, with the deeper of the effects being felt earlier on in the modelling period.

In all pathways, solar is an indispensable part of the energy mix, both on- and off-grid and its development and management is critical for securing Burkina Faso's energy future. However, in the absence of storage, solar (and wind) must be paired with the most affordable firm capacity option to maintain the stability of the grid. BAL and AMB though requiring larger upfront costs enjoy long-term benefits such as lower emissions and improved quality of life. AMB is the pathway that comes closest to meeting SDG 7 in Burkina Faso, indicative of the significant financial and institutional commitment needed to achieve the SDGs in this shortening time frame. However, BAL strikes a middle ground between BAU and AMB and may be a more realistic choice for policy adoption. These pathways show how cost optimization and renewable energy and emissions goals can work together to decarbonize Burkina Faso's electricity sector, while supporting its people socioeconomically.

The policy outcomes from this work include modernizing grid infrastructure to make best use of imports and storage, developing regulatory frameworks to prioritize decentralized energy system in both infrastructure and human capacity development, leveraging international finance to large scale renewable and nuclear works, and ensuring energy justice by electrifying critical infrastructure for the upward mobility of Burkinabé citizens and residents. This study should not be used for ascertaining hourly operation of the electricity grid as the granularity in the thesis did not allow for this, neglecting possible shortfalls in supply, or balancing strategies with energy storage systems. The thesis work also does not consider the impacts of climate change on the electricity sector, such as changes in rainfall, solar irradiation, or water availability, which could affect hydropower, solar, and biomass resources. Additionally, the geo-spatial modeling assumes static population clusters, ignoring the possibility of migration, urbanization, or new settlements, which could affect the optimal distribution of grid and off-grid technologies. Future works include improving OSGEM's capacity for Monte Carlo simulations to allow for more realistic modeling of annual patterns in solar and hydro availability, and integrating the electricity planning model into a broader climate-water-energy nexus framework to better understand system interactions and cross-sector impacts in Burkina Faso. Additionally, the modeller could be enhanced to allow for dynamic population densities, enabling the simulation of population movements and to address the limitation of myopic modeling in OnSSET by developing code that tracks the sources currently supplying electricity to each area.

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Appendix A

Enumerated Results

A.1 BAU

Table A.1.1: BAU Annual Production by Technology Table in GigaWatt-hours (GWh)

Year	Imports – CIV	Imports – GHA	Imports – MLI	Imports – NER	Biomass	Diesel	Natural Gas	HFO	Hydro	Nuclear	Solar	Wind	Total
2021	241.58	0.00	0.00	0.00	0.00	0.00	0.00	2,378.65	100.38	0.00	87.59	0.00	2,808.20
2022	152.61	240.79	471.63	0.00	0.00	0.00	0.00	2,378.65	100.38	0.00	134.86	0.00	3,478.91
2023	205.14	281.28	420.92	0.00	0.00	67.40	0.00	2,414.34	106.18	0.00	273.30	0.00	3,768.56
2024	257.67	321.77	328.29	0.00	139.66	0.00	0.00	2,414.34	106.18	0.00	932.37	0.00	4,500.29
2025	310.20	362.26	235.66	0.00	279.32	0.00	0.00	2,202.98	106.18	0.00	1,337.15	103.55	4,937.30
2026	308.89	441.20	292.68	0.00	418.99	0.00	0.00	2,177.48	106.18	0.00	1,471.47	132.71	5,349.58
2027	307.58	520.14	312.06	0.00	558.65	0.00	0.00	2,186.27	106.18	0.00	1,545.60	225.38	5,761.86
2028	306.27	599.08	326.72	0.00	698.31	0.00	0.00	2,041.13	106.18	0.00	1,705.78	390.67	6,174.14
2029	304.97	678.02	442.97	0.00	837.97	0.00	0.00	1,887.61	106.18	0.00	1,772.74	555.96	6,586.42
2030	303.66	756.96	520.73	0.00	977.63	0.00	0.00	1,772.59	106.18	0.00	1,849.78	711.18	6,998.70
2031	302.35	835.90	577.74	4.51	1,102.08	0.00	0.00	1,645.19	106.18	0.00	1,949.24	886.55	7,409.73
2032	301.05	914.84	634.75	0.00	1,256.96	0.00	0.00	1,672.07	106.18	0.00	1,976.06	958.88	7,820.77
2033	299.74	993.77	349.38	0.00	1,386.12	0.00	0.00	1,679.75	106.18	0.00	2,319.64	1,106.97	8,241.55
2034	299.74	993.77	386.21	0.00	1,536.28	0.00	0.00	1,686.64	106.18	0.00	2,293.41	1,299.13	8,601.35
2035	299.74	993.77	0.00	0.00	1,462.47	0.00	0.00	1,383.21	106.18	0.00	2,249.59	2,466.19	8,961.15
2036	299.74	993.77	279.27	0.00	1,443.59	0.00	0.00	1,171.41	106.18	0.00	2,283.73	2,759.17	9,336.87
2037	299.74	993.77	583.24	0.00	1,739.15	0.00	0.00	914.88	106.18	0.00	2,305.64	2,769.98	9,712.58
2038	299.74	993.77	607.76	0.00	1,937.32	0.00	0.00	904.39	106.18	0.00	2,534.31	2,704.83	10,088.29
2039	299.74	993.77	691.77	0.00	2,261.25	0.00	0.00	691.46	106.18	0.00	2,649.85	2,769.98	10,464.01
2040	299.74	993.77	691.77	0.00	2,806.49	0.00	0.00	479.09	106.18	0.00	2,692.70	2,769.98	10,839.72
2041	299.74	993.77	691.77	0.00	3,169.69	0.00	0.00	477.41	106.18	0.00	2,709.61	2,769.98	11,218.14
2042	299.74	993.77	691.77	0.00	3,540.97	0.00	0.00	475.84	106.18	0.00	2,718.32	2,769.98	11,596.56
2043	299.74	993.77	691.77	0.00	3,913.06	0.00	0.00	474.26	106.18	0.00	2,726.22	2,769.98	11,974.98
2044	299.74	993.77	691.77	0.00	3,784.92	0.00	0.00	245.78	106.18	0.00	3,079.05	3,161.03	12,362.23
2045	299.74	993.77	691.77	182.63	3,799.06	0.00	0.00	247.63	106.18	0.00	3,187.37	3,223.67	12,731.82
2046	299.74	993.77	691.77	423.20	3,794.33	0.00	0.00	24.87	328.16	0.00	3,353.05	3,239.76	13,148.66
2047	299.74	993.77	691.77	423.20	3,833.25	0.00	0.00	207.41	385.84	0.00	3,541.59	3,145.99	13,522.55
2048	299.74	993.77	691.77	423.20	3,858.01	0.00	0.00	443.64	385.84	0.00	3,729.31	3,215.08	14,040.34
2049	299.74	993.77	691.77	423.20	3,875.91	0.00	0.00	686.71	385.84	0.00	3,862.89	3,238.18	14,458.01
2050	299.74	993.77	691.77	423.20	3,887.35	0.00	0.00	925.58	385.84	0.00	3,998.38	3,239.76	14,845.39
2051	299.74	993.77	691.77	423.20	3,796.97	0.00	0.00	1,094.05	385.84	0.00	4,368.90	3,129.28	15,183.51
2052	299.74	993.77	691.77	423.20	3,797.59	0.00	0.00	1,350.79	385.84	0.00	4,500.11	3,223.72	15,666.52
2053	299.74	993.77	691.77	423.20	3,795.74	0.00	0.00	1,610.00	385.84	0.00	4,651.04	3,239.76	16,090.86
2054	299.74	993.77	691.77	423.20	3,793.89	0.00	0.00	1,869.22	385.84	0.00	4,882.63	3,187.59	16,527.64
2055	299.74	993.77	691.77	423.20	3,839.43	0.00	0.00	2,166.73	385.84	0.00	4,890.04	3,239.76	16,930.28
2056	299.74	993.77	691.77	423.20	3,888.71	0.00	0.00	2,480.26	385.84	0.00	4,805.86	3,239.76	17,208.91
2057	299.74	993.77	691.77	423.20	3,932.56	0.00	0.00	2,789.41	385.84	0.00	4,880.19	3,239.76	17,636.24
2058	299.74	993.77	691.77	423.20	3,936.44	0.00	0.00	3,066.26	385.84	0.00	4,972.94	3,239.76	18,009.71
2059	299.74	993.77	691.77	423.20	3,927.05	0.00	0.00	3,332.39	385.84	0.00	5,213.94	3,239.76	18,507.45
2060	299.74	993.77	691.77	423.20	3,944.19	0.00	0.00	3,619.95	385.84	0.00	5,399.83	3,129.28	18,887.56
2061	299.74	993.77	691.77	423.20	3,948.11	0.00	0.00	3,899.88	385.84	0.00	5,408.17	3,239.76	19,290.23
2062	299.74	993.77	691.77	423.20	3,907.18	0.00	0.00	4,143.56	385.84	0.00	5,710.50	3,239.76	19,795.32
2063	299.74	993.77	691.77	423.20	3,776.43	0.00	0.00	4,314.66	385.84	0.00	6,025.06	3,239.76	20,150.22
2064	299.74	993.77	691.77	423.20	3,786.70	0.00	0.00	4,599.72	385.84	0.00	6,159.73	3,239.76	20,580.21
2065	299.74	993.77	691.77	423.20	3,776.75	0.00	0.00	4,866.00	385.84	0.00	6,523.32	3,239.76	21,200.14
2066	299.74	993.77	691.77	423.20	3,774.68	0.00	0.00	5,158.24	385.84	0.00	6,612.69	3,239.76	21,579.69
2067	299.74	993.77	691.77	423.20	3,772.62	0.00	0.00	5,450.49	385.84	0.00	6,668.03	3,239.76	21,925.22
2068	299.74	993.77	691.77	423.20	3,770.56	0.00	0.00	5,742.74	385.84	0.00	6,832.73	3,239.76	22,380.10
2069	299.74	993.77	691.77	423.20	3,770.84	0.00	0.00	6,034.98	385.84	0.00	7,122.70	3,239.76	22,962.60
2070	299.74	993.77	691.77	423.20	3,771.33	0.00	0.00	6,327.23	385.84	0.00	7,157.22	3,239.76	23,289.86

Table A.1.2: BAU Total Annual Capacity by Technology Table in MegaWatts (MW)

Year	Imports – CIV	Imports – GHA	Imports – MLI	Imports – NER	Biomass	Diesel	Natural Gas	HFO	Hydro	Nuclear	Solar	Wind	Total
2022	327.4	327.0	250.0	760.0	0.0	34.1	0.0	355.1	34.6	0.0	87.3	0.0	2,175.5
2023	327.4	327.0	250.0	760.0	0.0	34.6	0.0	360.4	36.6	0.0	176.9	0.0	2,272.9
2024	327.4	327.0	250.0	760.0	35.0	34.6	0.0	360.4	36.6	0.0	603.5	0.0	2,734.5
2025	327.4	327.0	250.0	760.0	70.0	34.6	0.0	360.4	36.6	0.0	867.7	62.6	3,096.3
2026	327.4	327.0	250.0	760.0	105.0	34.6	0.0	360.4	36.6	0.0	957.3	80.3	3,238.6
2027	327.4	327.0	250.0	760.0	140.0	34.6	0.0	360.4	36.6	0.0	1,006.6	136.3	3,378.9
2028	327.4	327.0	250.0	760.0	175.0	33.3	0.0	346.6	36.6	0.0	1,123.2	236.3	3,615.4
2029	327.4	327.0	250.0	760.0	210.0	30.5	0.0	318.3	36.6	0.0	1,169.8	336.3	3,766.0
2030	327.4	327.0	250.0	760.0	245.0	28.5	0.0	297.3	36.6	0.0	1,216.4	436.3	3,924.6
2031	327.4	327.0	250.0	760.0	280.0	26.6	0.0	277.2	36.6	0.0	1,295.1	536.3	4,116.3
2032	327.4	327.0	250.0	760.0	315.0	26.6	0.0	277.2	36.6	0.0	1,301.2	585.9	4,207.0
2033	327.4	327.0	250.0	760.0	350.0	26.5	0.0	276.3	36.6	0.0	1,501.5	685.9	4,541.3
2034	327.4	327.0	250.0	760.0	385.0	26.5	0.0	276.3	36.6	0.0	1,501.4	785.9	4,676.2
2035	327.4	327.0	250.0	760.0	385.0	26.5	0.0	276.3	36.6	0.0	1,501.4	1,492.0	5,382.2
2036	327.4	327.0	250.0	760.0	385.0	21.1	0.0	219.8	36.6	0.0	1,529.4	1,669.3	5,525.5
2037	327.4	327.0	250.0	760.0	458.2	16.1	0.0	167.7	36.6	0.0	1,531.6	1,675.8	5,550.4
2038	327.4	327.0	250.0	760.0	515.0	16.1	0.0	167.7	36.6	0.0	1,644.2	1,675.8	5,719.9
2039	327.4	327.0	250.0	760.0	610.0	12.8	0.0	133.1	36.6	0.0	1,776.3	1,675.8	5,908.9
2040	327.4	327.0	250.0	760.0	761.0	9.3	0.0	96.6	36.6	0.0	1,793.3	1,675.8	6,036.9
2041	327.4	327.0	250.0	760.0	852.4	9.3	0.0	96.6	36.6	0.0	1,794.1	1,675.8	6,129.1
2042	327.4	327.0	250.0	760.0	943.9	9.3	0.0	96.6	36.6	0.0	1,794.9	1,675.8	6,221.4
2043	327.4	327.0	250.0	760.0	1,035.4	9.3	0.0	96.6	36.6	0.0	1,795.7	1,675.8	6,313.7
2044	327.4	327.0	250.0	760.0	1,075.0	4.9	0.0	51.0	36.6	0.0	1,995.7	1,960.0	6,787.6
2045	327.4	327.0	250.0	760.0	1,075.0	4.9	0.0	51.0	36.6	0.0	2,130.0	1,960.0	6,921.9
2046	327.4	327.0	250.0	760.0	1,075.0	0.5	0.0	5.3	113.1	0.0	2,249.5	1,960.0	7,067.9
2047	327.4	327.0	250.0	760.0	1,075.0	0.5	0.0	43.1	133.0	0.0	2,328.4	1,960.0	7,204.4
2048	327.4	327.0	250.0	760.0	1,075.0	0.0	0.0	89.3	133.0	0.0	2,424.0	1,960.0	7,345.6
2049	327.4	327.0	250.0	760.0	1,075.0	0.0	0.0	136.7	133.0	0.0	2,519.5	1,960.0	7,488.7
2050	327.4	327.0	250.0	760.0	1,075.0	0.0	0.0	183.6	133.0	0.0	2,625.0	1,960.0	7,641.0
2051	327.4	327.0	250.0	760.0	1,075.0	0.0	0.0	219.7	133.0	0.0	2,897.4	1,960.0	7,949.5
2052	327.4	327.0	250.0	760.0	1,075.0	0.0	0.0	268.1	133.0	0.0	3,004.6	1,960.0	8,105.1
2053	327.4	327.0	250.0	760.0	1,075.0	0.0	0.0	316.6	133.0	0.0	3,111.8	1,960.0	8,260.8
2054	327.4	327.0	250.0	760.0	1,075.0	0.0	0.0	365.1	133.0	0.0	3,218.9	1,960.0	8,416.4
2055	327.4	327.0	250.0	760.0	1,075.0	0.0	0.0	419.7	133.0	0.0	3,247.1	1,960.0	8,499.2
2056	327.4	327.0	250.0	760.0	1,075.0	0.0	0.0	477.5	133.0	0.0	3,274.2	1,960.0	8,584.1
2057	327.4	327.0	250.0	760.0	1,075.0	0.0	0.0	535.2	133.0	0.0	3,310.4	1,960.0	8,677.9
2058	327.4	327.0	250.0	760.0	1,075.0	0.0	0.0	590.3	133.0	0.0	3,413.2	1,960.0	8,835.9
2059	327.4	327.0	250.0	760.0	1,075.0	0.0	0.0	644.6	133.0	0.0	3,538.1	1,960.0	9,015.1
2060	327.4	327.0	250.0	760.0	1,075.0	0.0	0.0	701.7	133.0	0.0	3,618.8	1,960.0	9,152.9
2061	327.4	327.0	250.0	760.0	1,075.0	0.0	0.0	758.8	133.0	0.0	3,722.8	1,960.0	9,314.0
2062	327.4	327.0	250.0	760.0	1,075.0	0.0	0.0	814.3	133.0	0.0	3,901.5	1,960.0	9,548.2
2063	327.4	327.0	250.0	760.0	1,075.0	0.0	0.0	866.4	133.0	0.0	4,230.0	1,960.0	9,928.7
2064	327.4	327.0	250.0	760.0	1,075.0	0.0	0.0	923.7	133.0	0.0	4,323.3	1,960.0	10,079.5
2065	327.4	327.0	250.0	760.0	1,075.0	0.0	0.0	980.2	133.0	0.0	4,457.5	1,960.0	10,270.1
2066	327.4	327.0	250.0	760.0	1,075.0	0.0	0.0	1,040.0	133.0	0.0	4,577.0	1,960.0	10,449.4
2067	327.4	327.0	250.0	760.0	1,075.0	0.0	0.0	1,099.8	133.0	0.0	4,696.5	1,960.0	10,628.7
2068	327.4	327.0	250.0	760.0	1,075.0	0.0	0.0	1,159.6	133.0	0.0	4,816.0	1,960.0	10,808.0
2069	327.4	327.0	250.0	760.0	1,075.0	0.0	0.0	1,219.4	133.0	0.0	4,935.5	1,960.0	10,987.3
2070	327.4	327.0	250.0	760.0	1,075.0	0.0	0.0	1,279.5	133.0	0.0	5,055.0	1,960.0	11,167.0

Table A.1.3: BAU Grid Capital Investment by Technology Table in 10⁶ USD (MUSD)

Year	Imports – CIV	Imports – GHA	Imports – MLI	Imports – NER	Biomass	Diesel	Natural Gas	HFO	Hydro	Nuclear	Solar	Wind	Total
2021	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
2022	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
2023	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
2024	0.0	0.0	0.0	0.0	87.5	0.0	0.0	0.0	0.0	0.0	453.4	0.0	540.9
2025	0.0	0.0	0.0	0.0	87.5	0.0	0.0	0.0	0.0	0.0	259.9	74.6	422.0
2026	0.0	0.0	0.0	0.0	87.5	0.0	0.0	0.0	0.0	0.0	86.4	20.6	194.6
2027	0.0	0.0	0.0	0.0	87.5	0.0	0.0	0.0	0.0	0.0	46.6	64.4	198.5
2028	0.0	0.0	0.0	0.0	87.5	0.0	0.0	0.0	0.0	0.0	108.0	112.9	308.4
2029	0.0	0.0	0.0	0.0	87.5	0.0	0.0	0.0	0.0	0.0	42.2	110.8	240.5
2030	0.0	0.0	0.0	0.0	87.5	0.0	0.0	0.0	0.0	0.0	41.3	108.7	237.5
2031	0.0	0.0	0.0	0.0	87.5	0.0	0.0	0.0	0.0	0.0	68.3	105.6	261.4
2032	0.0	0.0	0.0	0.0	87.5	0.0	0.0	0.0	0.0	0.0	5.2	50.9	143.6
2033	0.0	0.0	0.0	0.0	87.5	0.0	0.0	0.0	0.0	0.0	167.2	99.5	354.1
2034	0.0	0.0	0.0	0.0	87.5	0.0	0.0	0.0	0.0	0.0	0.0	96.4	183.9
2035	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	658.7	658.7
2036	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	22.4	165.4	187.8
2037	0.0	0.0	0.0	0.0	182.9	0.0	0.0	0.0	0.0	0.0	2.1	6.1	191.1
2038	0.0	0.0	0.0	0.0	142.2	0.0	0.0	0.0	0.0	0.0	85.2	0.0	227.4
2039	0.0	0.0	0.0	0.0	237.4	0.0	0.0	0.0	0.0	0.0	98.2	0.0	335.6
2040	0.0	0.0	0.0	0.0	377.4	0.0	0.0	0.0	0.0	0.0	12.6	0.0	390.0
2041	0.0	0.0	0.0	0.0	228.7	0.0	0.0	0.0	0.0	0.0	26.0	0.0	254.6
2042	0.0	0.0	0.0	0.0	228.7	0.0	0.0	0.0	0.0	0.0	11.8	0.0	240.4
2043	0.0	0.0	0.0	0.0	228.7	0.0	0.0	0.0	0.0	0.0	1.8	0.0	230.5
2044	0.0	0.0	0.0	0.0	99.1	0.0	0.0	0.0	0.0	0.0	144.9	265.2	509.2
2045	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	97.7	0.0	97.7
2046	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	229.6	0.0	108.5	0.0	338.1
2047	0.0	0.0	0.0	0.0	0.0	0.0	0.0	54.8	59.6	0.0	121.8	0.0	236.3
2048	0.0	0.0	0.0	0.0	0.0	0.0	0.0	74.6	0.0	0.0	377.6	0.0	452.2
2049	0.0	0.0	0.0	0.0	0.0	0.0	0.0	68.8	0.0	0.0	260.1	0.0	328.9
2050	0.0	0.0	0.0	0.0	0.0	0.0	0.0	68.0	96.0	0.0	141.0	58.4	363.5
2051	0.0	0.0	0.0	0.0	0.0	0.0	0.0	52.3	0.0	0.0	232.6	16.5	301.4
2052	0.0	0.0	0.0	0.0	0.0	0.0	0.0	70.3	0.0	0.0	161.9	52.3	284.5
2053	0.0	0.0	0.0	0.0	0.0	0.0	0.0	70.3	0.0	0.0	111.2	93.3	274.7
2054	0.0	0.0	0.0	0.0	87.5	0.0	0.0	70.3	0.0	0.0	111.2	93.3	362.3
2055	0.0	0.0	0.0	0.0	87.5	0.0	0.0	79.3	0.0	0.0	77.2	93.3	337.3
2056	0.0	0.0	0.0	0.0	87.5	0.0	0.0	83.8	0.0	0.0	24.0	93.3	288.6
2057	0.0	0.0	0.0	0.0	87.5	0.0	0.0	83.6	0.0	0.0	171.0	46.3	388.4
2058	0.0	0.0	0.0	0.0	87.5	0.0	0.0	79.9	0.0	0.0	74.3	93.3	335.1
2059	0.0	0.0	0.0	0.0	87.5	0.0	0.0	78.8	0.0	0.0	90.3	93.3	349.9
2060	0.0	0.0	0.0	0.0	87.5	0.0	0.0	82.7	0.0	0.0	79.0	658.7	907.9
2061	0.0	0.0	0.0	0.0	87.5	0.0	0.0	82.9	0.0	0.0	77.1	165.4	412.9
2062	0.0	0.0	0.0	0.0	87.5	0.0	0.0	80.4	0.0	0.0	211.0	6.1	385.0
2063	0.0	0.0	0.0	0.0	87.5	0.0	0.0	75.5	0.0	0.0	333.7	0.0	496.7
2064	0.0	0.0	0.0	0.0	87.5	0.0	0.0	83.2	0.0	0.0	80.1	0.0	250.8
2065	0.0	0.0	0.0	0.0	0.0	0.0	0.0	81.9	7.8	0.0	123.0	0.0	212.6
2066	0.0	0.0	0.0	0.0	0.0	0.0	0.0	86.7	0.0	0.0	98.2	0.0	184.9
2067	0.0	0.0	0.0	0.0	182.9	0.0	0.0	86.7	0.0	0.0	88.2	0.0	357.9
2068	0.0	0.0	0.0	0.0	142.2	0.0	0.0	86.7	0.0	0.0	231.3	0.0	460.2
2069	0.0	0.0	0.0	0.0	237.4	0.0	0.0	86.7	0.0	0.0	184.1	265.2	773.4
2070	0.0	0.0	0.0	0.0	377.4	0.0	0.0	87.2	0.0	0.0	194.9	0.0	659.5

Table A.1.4: BAU Grid Annual Fixed Operating Costs by Technology Table in 10⁶ USD (MUSD)

Year	Imports – CIV	Imports – GHA	Imports – MLI	Imports – NER	Biomass	Diesel	Natural Gas	HFO	Hydro	Nuclear	Solar	Wind	Total
2021	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
2022	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
2023	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
2024	0.0	0.0	0.0	0.0	2.6	0.0	0.0	0.0	0.0	0.0	5.9	0.0	8.5
2025	0.0	0.0	0.0	0.0	2.6	0.0	0.0	0.0	0.0	0.0	3.4	3.0	9.0
2026	0.0	0.0	0.0	0.0	2.6	0.0	0.0	0.0	0.0	0.0	1.1	0.8	4.6
2027	0.0	0.0	0.0	0.0	2.6	0.0	0.0	0.0	0.0	0.0	0.6	2.6	5.8
2028	0.0	0.0	0.0	0.0	2.6	0.0	0.0	0.0	0.0	0.0	1.4	4.5	8.5
2029	0.0	0.0	0.0	0.0	2.6	0.0	0.0	0.0	0.0	0.0	0.5	4.4	7.6
2030	0.0	0.0	0.0	0.0	2.6	0.0	0.0	0.0	0.0	0.0	0.5	4.3	7.5
2031	0.0	0.0	0.0	0.0	2.6	0.0	0.0	0.0	0.0	0.0	0.9	4.2	7.7
2032	0.0	0.0	0.0	0.0	2.6	0.0	0.0	0.0	0.0	0.0	0.1	2.0	4.7
2033	0.0	0.0	0.0	0.0	2.6	0.0	0.0	0.0	0.0	0.0	2.2	4.0	8.8
2034	0.0	0.0	0.0	0.0	2.6	0.0	0.0	0.0	0.0	0.0	0.0	3.9	6.5
2035	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	26.3	26.3
2036	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.3	6.6	6.9
2037	0.0	0.0	0.0	0.0	5.5	0.0	0.0	0.0	0.0	0.0	0.0	0.2	5.8
2038	0.0	0.0	0.0	0.0	4.3	0.0	0.0	0.0	0.0	0.0	1.1	0.0	5.4
2039	0.0	0.0	0.0	0.0	7.1	0.0	0.0	0.0	0.0	0.0	1.3	0.0	8.4
2040	0.0	0.0	0.0	0.0	11.3	0.0	0.0	0.0	0.0	0.0	0.2	0.0	11.5
2041	0.0	0.0	0.0	0.0	6.9	0.0	0.0	0.0	0.0	0.0	0.3	0.0	7.2
2042	0.0	0.0	0.0	0.0	6.9	0.0	0.0	0.0	0.0	0.0	0.2	0.0	7.0
2043	0.0	0.0	0.0	0.0	6.9	0.0	0.0	0.0	0.0	0.0	0.0	0.0	6.9
2044	0.0	0.0	0.0	0.0	3.0	0.0	0.0	0.0	0.0	0.0	1.9	10.6	15.5
2045	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	1.3	0.0	1.3
2046	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	6.9	0.0	1.4	0.0	8.3
2047	0.0	0.0	0.0	0.0	0.0	0.0	0.0	1.7	1.8	0.0	1.6	0.0	5.1
2048	0.0	0.0	0.0	0.0	0.0	0.0	0.0	2.3	0.0	0.0	4.9	0.0	7.2
2049	0.0	0.0	0.0	0.0	0.0	0.0	0.0	2.1	0.0	0.0	3.4	0.0	5.5
2050	0.0	0.0	0.0	0.0	0.0	0.0	0.0	2.1	2.9	0.0	1.8	2.3	9.2
2051	0.0	0.0	0.0	0.0	0.0	0.0	0.0	1.6	0.0	0.0	3.0	0.7	5.3
2052	0.0	0.0	0.0	0.0	0.0	0.0	0.0	2.2	0.0	0.0	2.1	2.1	6.4
2053	0.0	0.0	0.0	0.0	0.0	0.0	0.0	2.2	0.0	0.0	1.4	3.7	7.4
2054	0.0	0.0	0.0	0.0	2.6	0.0	0.0	2.2	0.0	0.0	1.4	3.7	10.0
2055	0.0	0.0	0.0	0.0	2.6	0.0	0.0	2.5	0.0	0.0	1.0	3.7	9.8
2056	0.0	0.0	0.0	0.0	2.6	0.0	0.0	2.6	0.0	0.0	0.3	3.7	9.3
2057	0.0	0.0	0.0	0.0	2.6	0.0	0.0	2.6	0.0	0.0	2.2	1.9	9.3
2058	0.0	0.0	0.0	0.0	2.6	0.0	0.0	2.5	0.0	0.0	1.0	3.7	9.8
2059	0.0	0.0	0.0	0.0	2.6	0.0	0.0	2.4	0.0	0.0	1.2	3.7	10.0
2060	0.0	0.0	0.0	0.0	2.6	0.0	0.0	2.6	0.0	0.0	1.0	26.3	32.6
2061	0.0	0.0	0.0	0.0	2.6	0.0	0.0	2.6	0.0	0.0	1.0	6.6	12.8
2062	0.0	0.0	0.0	0.0	2.6	0.0	0.0	2.5	0.0	0.0	2.7	0.2	8.1
2063	0.0	0.0	0.0	0.0	2.6	0.0	0.0	2.3	0.0	0.0	4.3	0.0	9.3
2064	0.0	0.0	0.0	0.0	2.6	0.0	0.0	2.6	0.0	0.0	1.0	0.0	6.2
2065	0.0	0.0	0.0	0.0	0.0	0.0	0.0	2.5	0.2	0.0	1.6	0.0	4.4
2066	0.0	0.0	0.0	0.0	0.0	0.0	0.0	2.7	0.0	0.0	1.3	0.0	4.0
2067	0.0	0.0	0.0	0.0	5.5	0.0	0.0	2.7	0.0	0.0	1.1	0.0	9.3
2068	0.0	0.0	0.0	0.0	4.3	0.0	0.0	2.7	0.0	0.0	3.0	0.0	10.0
2069	0.0	0.0	0.0	0.0	7.1	0.0	0.0	2.7	0.0	0.0	2.4	10.6	22.8
2070	0.0	0.0	0.0	0.0	11.3	0.0	0.0	2.7	0.0	0.0	2.5	0.0	16.6

Table A.1.5: BAU Grid Annual Variable Operating Costs by Technology Table in 10⁶ USD (MUSD)

Year	Imports – CIV	Imports – GHA	Imports – MLI	Imports – NER	Biomass	Diesel	Natural Gas	HFO	Hydro	Nuclear	Solar	Wind	Total
2021	26.1	0.0	0.0	0.0	0.0	0.0	0.0	219.3	0.0	0.0	0.0	0.0	245.4
2022	16.5	27.9	60.4	0.0	0.0	0.0	0.0	221.3	0.0	0.0	0.0	0.0	326.1
2023	22.2	32.6	53.9	0.0	0.0	10.5	0.0	226.6	0.0	0.0	0.0	0.0	345.8
2024	27.8	37.3	42.0	0.0	2.3	0.0	0.0	228.7	0.0	0.0	0.0	0.0	338.1
2025	33.5	42.0	30.2	0.0	4.6	0.0	0.0	210.5	0.0	0.0	0.0	0.0	320.8
2026	33.4	51.2	37.5	0.0	6.9	0.0	0.0	213.0	0.0	0.0	0.0	0.0	341.9
2027	33.2	60.3	39.9	0.0	9.2	0.0	0.0	218.8	0.0	0.0	0.0	0.0	361.5
2028	33.1	69.5	41.8	0.0	11.5	0.0	0.0	208.8	0.0	0.0	0.0	0.0	364.7
2029	32.9	78.7	56.7	0.0	13.8	0.0	0.0	197.4	0.0	0.0	0.0	0.0	379.5
2030	32.8	87.8	66.7	0.0	16.1	0.0	0.0	189.4	0.0	0.0	0.0	0.0	392.7
2031	32.7	97.0	74.0	0.7	18.2	0.0	0.0	179.0	0.0	0.0	0.0	0.0	401.5
2032	32.5	106.1	81.2	0.0	20.7	0.0	0.0	185.2	0.0	0.0	0.0	0.0	425.8
2033	32.4	115.3	44.7	0.0	22.8	0.0	0.0	189.4	0.0	0.0	0.0	0.0	404.6
2034	32.4	115.3	49.4	0.0	25.3	0.0	0.0	193.5	0.0	0.0	0.0	0.0	415.9
2035	32.4	115.3	0.0	0.0	24.1	0.0	0.0	161.3	0.0	0.0	0.0	0.0	333.1
2036	32.4	115.3	35.7	0.0	23.8	0.0	0.0	138.9	0.0	0.0	0.0	0.0	346.1
2037	32.4	115.3	74.7	0.0	28.7	0.0	0.0	110.3	0.0	0.0	0.0	0.0	361.3
2038	32.4	115.3	77.8	0.0	31.9	0.0	0.0	110.8	0.0	0.0	0.0	0.0	368.2
2039	32.4	115.3	88.5	0.0	37.3	0.0	0.0	86.1	0.0	0.0	0.0	0.0	359.6
2040	32.4	115.3	88.5	0.0	46.2	0.0	0.0	60.6	0.0	0.0	0.0	0.0	343.0
2041	32.4	115.3	88.5	0.0	52.2	0.0	0.0	61.3	0.0	0.0	0.0	0.0	349.8
2042	32.4	115.3	88.5	0.0	58.3	0.0	0.0	62.1	0.0	0.0	0.0	0.0	356.6
2043	32.4	115.3	88.5	0.0	64.5	0.0	0.0	62.8	0.0	0.0	0.0	0.0	363.5
2044	32.4	115.3	88.5	0.0	62.4	0.0	0.0	33.0	0.0	0.0	0.0	0.0	331.6
2045	32.4	115.3	88.5	29.2	62.6	0.0	0.0	33.8	0.0	0.0	0.0	0.0	361.8
2046	32.4	115.3	88.5	67.7	62.5	0.0	0.0	3.4	0.0	0.0	0.0	0.0	369.9
2047	32.4	115.3	88.5	67.7	63.1	0.0	0.0	29.1	0.0	0.0	0.0	0.0	396.1
2048	32.4	115.3	88.5	67.7	63.6	0.0	0.0	63.1	0.0	0.0	0.0	0.0	430.6
2049	32.4	115.3	88.5	67.7	63.9	0.0	0.0	99.0	0.0	0.0	0.0	0.0	466.8
2050	32.4	115.3	88.5	67.7	64.0	0.0	0.0	135.3	0.0	0.0	0.0	0.0	503.3
2051	32.4	115.3	88.5	67.7	62.5	0.0	0.0	162.1	0.0	0.0	0.0	0.0	528.6
2052	32.4	115.3	88.5	67.7	62.6	0.0	0.0	202.8	0.0	0.0	0.0	0.0	569.3
2053	32.4	115.3	88.5	67.7	62.5	0.0	0.0	244.9	0.0	0.0	0.0	0.0	611.4
2054	32.4	115.3	88.5	67.7	62.5	0.0	0.0	288.1	0.0	0.0	0.0	0.0	654.5
2055	32.4	115.3	88.5	67.7	63.2	0.0	0.0	338.0	0.0	0.0	0.0	0.0	705.2
2056	32.4	115.3	88.5	67.7	64.1	0.0	0.0	391.8	0.0	0.0	0.0	0.0	759.8
2057	32.4	115.3	88.5	67.7	64.8	0.0	0.0	446.2	0.0	0.0	0.0	0.0	814.9
2058	32.4	115.3	88.5	67.7	64.8	0.0	0.0	496.5	0.0	0.0	0.0	0.0	865.3
2059	32.4	115.3	88.5	67.7	64.7	0.0	0.0	546.2	0.0	0.0	0.0	0.0	914.8
2060	32.4	115.3	88.5	67.7	65.0	0.0	0.0	600.5	0.0	0.0	0.0	0.0	969.4
2061	32.4	115.3	88.5	67.7	65.0	0.0	0.0	654.3	0.0	0.0	0.0	0.0	1023.3
2062	32.4	115.3	88.5	67.7	64.4	0.0	0.0	703.1	0.0	0.0	0.0	0.0	1071.3
2063	32.4	115.3	88.5	67.7	62.2	0.0	0.0	740.3	0.0	0.0	0.0	0.0	1106.4
2064	32.4	115.3	88.5	67.7	62.4	0.0	0.0	797.9	0.0	0.0	0.0	0.0	1164.2
2065	32.4	115.3	88.5	67.7	62.2	0.0	0.0	854.9	0.0	0.0	0.0	0.0	1221.0
2066	32.4	115.3	88.5	67.7	62.2	0.0	0.0	916.5	0.0	0.0	0.0	0.0	1282.5
2067	32.4	115.3	88.5	67.7	62.1	0.0	0.0	979.2	0.0	0.0	0.0	0.0	1345.2
2068	32.4	115.3	88.5	67.7	62.1	0.0	0.0	1043.0	0.0	0.0	0.0	0.0	1409.0
2069	32.4	115.3	88.5	67.7	62.1	0.0	0.0	1108.0	0.0	0.0	0.0	0.0	1474.0
2070	32.4	115.3	88.5	67.7	62.1	0.0	0.0	1174.2	0.0	0.0	0.0	0.0	1540.2

Table A.1.6: BAU Emissions in kilotons CO_{2eq} and Emission Intensity in grams CO_{2eq}/kWh electricity generated

Year	kTon CO _{2eq}	g CO _{2eq} / kWh electricity generated
2021	1906.2	0.679
2022	1906.2	0.548
2023	1986.5	0.527
2024	1934.8	0.430
2025	1765.4	0.358
2026	1745.0	0.326
2027	1752.0	0.304
2028	1635.7	0.265
2029	1512.7	0.230
2030	1420.5	0.203
2031	1318.4	0.178
2032	1340.0	0.171
2033	1346.1	0.163
2034	1351.6	0.157
2035	1108.5	0.124
2036	938.7	0.101
2037	733.2	0.075
2038	724.8	0.072
2039	554.1	0.053
2040	449.0	0.041
2041	810.4	0.072
2042	802.0	0.069
2043	690.4	0.058
2044	329.2	0.027
2045	410.4	0.032
2046	64.4	0.005
2047	332.6	0.025
2048	355.5	0.025
2049	550.3	0.038
2050	741.7	0.050
2051	876.7	0.058
2052	1082.5	0.069
2053	1290.2	0.080
2054	1497.9	0.091
2055	1736.4	0.103
2056	1987.6	0.115
2057	2235.4	0.127
2058	2457.2	0.136
2059	2670.5	0.144
2060	2900.9	0.154
2061	3125.3	0.162
2062	3320.6	0.168
2063	3457.7	0.172
2064	3686.1	0.179
2065	3899.5	0.184
2066	4133.7	0.192
2067	4367.9	0.199
2068	4602.1	0.206
2069	4836.3	0.211
2070	5070.5	0.218

A.2 BAL

Table A.2.1: BAL Annual Production by Technology Table in GigaWatt-hours (GWh)

Year	Imports – CIV	Imports – GHA	Imports – MLI	Imports – NER	Biomass	Diesel	Natural Gas	HFO	Hydro	Nuclear	Solar	Wind	Total
2021	241.6	0.0	0.0	0.0	0.0	0.0	0.0	2,378.6	100.4	0.0	87.6	0.0	2,808.2
2022	152.6	240.8	488.5	0.0	0.0	0.0	0.0	2,378.6	100.4	0.0	134.9	0.0	3,495.8
2023	205.1	281.3	420.9	0.0	0.0	101.2	0.0	2,414.3	106.2	0.0	273.3	0.0	3,802.4
2024	257.7	321.8	328.3	0.0	139.7	0.0	0.0	2,414.3	106.2	0.0	964.8	31.1	4,563.8
2025	310.2	362.3	235.7	0.0	279.3	0.0	0.0	2,262.0	106.2	0.0	1,270.0	196.4	5,022.0
2026	308.9	441.2	190.2	0.0	419.0	0.0	0.0	2,183.7	106.2	0.0	1,441.0	361.7	5,451.8
2027	307.6	520.1	171.8	0.0	558.6	0.0	0.0	2,188.8	106.2	0.0	1,501.5	527.0	5,881.6
2028	306.3	599.1	199.0	0.0	698.3	0.0	0.0	2,039.3	106.2	0.0	1,679.7	683.6	6,311.4
2029	305.0	678.0	326.7	0.0	838.0	0.0	0.0	1,887.3	106.2	0.0	1,742.6	857.5	6,741.2
2030	303.7	757.0	423.0	0.0	977.6	0.0	0.0	1,779.7	106.2	0.0	1,844.7	979.2	7,171.0
2031	302.4	835.9	577.7	151.9	1,102.6	0.0	0.0	1,669.9	106.2	0.0	2,036.0	1,173.6	7,956.2
2032	301.0	914.8	634.8	253.8	1,213.3	0.0	0.0	1,619.7	106.2	0.0	2,344.4	1,353.4	8,741.4
2033	299.7	993.8	691.8	142.8	1,351.7	0.0	0.0	1,618.6	106.2	0.0	2,813.0	1,518.7	9,536.3
2034	299.7	993.8	691.8	423.2	1,491.3	0.8	0.0	1,654.4	106.2	0.0	2,982.6	1,626.6	10,270.3
2035	299.7	993.8	465.8	0.0	1,653.2	0.0	0.0	1,407.6	106.2	0.0	2,838.2	3,239.8	11,004.2
2036	299.7	993.8	691.8	0.0	2,415.9	0.0	0.0	1,122.5	106.2	0.0	2,910.9	3,239.8	11,780.4
2037	299.7	993.8	691.8	0.0	3,445.4	0.0	0.0	827.8	106.2	0.0	2,952.1	3,239.8	12,556.6
2038	299.7	993.8	691.8	93.4	4,052.8	0.0	0.0	826.8	106.2	0.0	3,090.3	3,178.0	13,332.7
2039	299.7	993.8	691.8	423.2	3,811.1	57.6	0.0	665.5	106.2	0.0	3,820.3	3,239.8	14,108.9
2040	299.7	993.8	0.0	0.0	3,948.1	0.0	2,307.1	110.2	106.2	0.0	3,982.1	3,137.8	14,885.0
2041	299.7	993.8	97.2	0.0	4,090.2	0.0	3,034.0	0.0	106.2	0.0	3,906.8	3,239.8	15,767.6
2042	299.7	993.8	195.6	0.0	4,211.6	0.0	3,684.6	0.0	106.2	0.0	3,918.9	3,239.8	16,650.2
2043	299.7	993.8	237.2	0.0	4,213.6	0.0	4,264.6	0.0	106.2	0.0	4,177.9	3,239.8	17,532.8
2044	299.7	993.8	349.8	0.0	4,258.8	0.0	4,859.0	0.0	106.2	0.0	4,368.9	3,179.2	18,415.4
2045	299.7	993.8	424.9	0.0	4,239.4	0.0	5,364.1	0.0	106.2	0.0	4,630.2	3,239.8	19,298.0
2046	299.7	993.8	506.8	0.0	4,226.2	0.0	5,923.4	0.0	106.2	0.0	4,948.5	3,239.8	20,244.3
2047	299.7	993.8	589.1	4.5	4,238.5	0.0	6,541.0	0.8	106.2	0.0	5,177.3	3,239.8	21,190.7
2048	299.7	993.8	675.0	12.8	4,275.8	0.0	7,216.1	0.0	106.2	0.0	5,382.9	3,174.8	22,137.1
2049	299.7	993.8	691.8	17.5	4,281.4	0.0	7,883.0	0.0	106.2	0.0	5,636.1	3,173.9	23,083.4
2050	299.7	993.8	691.8	23.7	4,246.0	0.0	8,534.8	0.0	13.3	0.0	6,066.4	3,219.3	24,088.9
2051	299.7	993.8	691.8	27.7	4,243.8	0.0	9,210.7	0.0	13.3	0.0	6,330.6	3,217.5	25,028.9
2052	299.7	993.8	691.8	31.5	4,243.8	0.0	9,896.7	0.0	13.3	0.0	6,593.2	3,146.0	25,909.8
2053	299.7	993.8	691.8	35.4	4,243.8	0.0	10,582.7	0.0	13.3	0.0	6,762.4	3,239.8	26,862.6
2054	299.7	993.8	691.8	39.3	4,106.0	0.0	11,406.3	0.0	13.3	0.0	7,083.9	3,216.0	27,850.1
2055	299.7	993.8	691.8	43.1	3,968.3	0.0	12,230.1	0.0	13.3	0.0	7,335.0	3,239.8	28,814.8
2056	299.7	993.8	691.8	47.1	3,830.5	0.0	13,109.4	0.0	13.3	0.0	7,606.4	3,213.1	29,805.1
2057	299.7	993.8	691.8	51.1	3,692.8	0.0	13,988.8	0.0	13.3	0.0	7,877.8	3,239.8	30,848.8
2058	299.7	993.8	673.1	54.2	3,555.0	0.0	14,887.8	0.0	13.3	0.0	8,149.2	3,239.8	31,865.9
2059	299.7	993.8	658.3	56.6	3,451.6	0.0	15,946.9	0.0	13.3	0.0	8,208.5	3,239.8	32,868.6
2060	299.7	993.8	643.2	59.3	3,312.0	0.0	16,833.6	0.0	13.3	0.0	8,489.9	3,165.5	33,810.4
2061	299.7	993.8	628.0	62.0	3,172.3	0.0	17,730.6	0.0	13.3	0.0	8,707.5	3,239.8	34,847.0
2062	299.7	993.8	612.8	64.7	3,032.7	0.0	18,627.6	0.0	13.3	0.0	9,063.9	3,239.8	35,948.3
2063	299.7	993.8	597.6	67.4	2,893.0	0.0	19,524.6	0.0	13.3	0.0	9,351.0	3,239.8	36,980.1
2064	299.7	993.8	582.3	70.0	2,753.3	0.0	20,421.6	0.0	13.3	0.0	9,651.8	3,239.8	38,025.7
2065	299.7	993.8	563.7	72.7	2,509.6	0.0	21,425.3	0.0	5.8	0.0	9,849.9	3,239.8	38,960.2
2066	299.7	993.8	540.7	75.4	1,693.1	0.0	23,011.8	0.0	5.8	0.0	10,326.5	3,239.8	40,186.5
2067	299.7	993.8	544.3	94.9	582.8	0.0	24,802.6	0.0	5.8	0.0	10,724.5	3,239.8	41,288.1
2068	299.7	993.8	554.2	114.0	0.0	0.0	26,091.6	0.0	5.8	0.0	11,053.1	3,239.8	42,352.0
2069	299.7	993.8	556.6	128.5	0.0	0.0	26,890.4	0.0	5.8	0.0	11,400.0	3,239.8	43,514.6
2070	299.7	993.8	559.1	143.1	0.0	0.0	27,689.1	0.0	5.8	0.0	11,646.4	3,239.8	44,576.8

Table A.2.2: BAL Total Annual Capacity by Technology Table in MegaWatts (MW)

Year	Imports – CIV	Imports – GHA	Imports – MLI	Imports – NER	Biomass	Diesel	Natural Gas	HFO	Hydro	Nuclear	Solar	Wind	Total
2021	327.4	327.0	0.0	0.0	0.0	34.1	0.0	355.1	34.6	0.0	56.7	0.0	1,134.9
2022	327.4	327.0	250.0	760.0	0.0	34.1	0.0	355.1	34.6	0.0	87.3	0.0	2,175.5
2023	327.4	327.0	250.0	760.0	0.0	34.6	0.0	360.4	36.6	0.0	176.9	0.0	2,272.9
2024	327.4	327.0	250.0	760.0	35.0	34.6	0.0	360.4	36.6	0.0	624.5	18.8	2,774.3
2025	327.4	327.0	250.0	760.0	70.0	34.6	0.0	360.4	36.6	0.0	822.1	118.8	3,106.9
2026	327.4	327.0	250.0	760.0	105.0	34.6	0.0	360.4	36.6	0.0	940.2	218.8	3,360.0
2027	327.4	327.0	250.0	760.0	140.0	34.6	0.0	360.4	36.6	0.0	981.5	318.8	3,536.3
2028	327.4	327.0	250.0	760.0	175.0	33.3	0.0	346.6	36.6	0.0	1,104.8	418.8	3,779.4
2029	327.4	327.0	250.0	760.0	210.0	30.5	0.0	318.3	36.6	0.0	1,154.3	518.8	3,933.0
2030	327.4	327.0	250.0	760.0	245.0	28.5	0.0	297.3	36.6	0.0	1,194.1	618.8	4,084.7
2031	327.4	327.0	250.0	760.0	280.0	26.6	0.0	277.2	36.6	0.0	1,345.8	718.8	4,349.4
2032	327.4	327.0	250.0	760.0	315.0	26.6	0.0	277.2	36.6	0.0	1,593.0	818.8	4,731.6
2033	327.4	327.0	250.0	760.0	350.0	26.5	0.0	276.3	36.6	0.0	1,895.0	918.8	5,167.7
2034	327.4	327.0	250.0	760.0	385.0	26.5	0.0	276.3	36.6	0.0	1,958.7	1,018.8	5,366.4
2035	327.4	327.0	250.0	760.0	446.1	26.5	0.0	276.3	36.6	0.0	1,958.7	1,960.0	6,368.6
2036	327.4	327.0	250.0	760.0	650.7	21.1	0.0	219.8	36.6	0.0	1,958.1	1,960.0	6,510.6
2037	327.4	327.0	250.0	760.0	928.9	16.1	0.0	167.7	36.6	0.0	1,957.6	1,960.0	6,731.4
2038	327.4	327.0	250.0	760.0	1,075.0	16.1	0.0	167.7	36.6	0.0	2,000.4	1,960.0	6,920.2
2039	327.4	327.0	250.0	760.0	1,075.0	12.8	0.0	133.1	36.6	0.0	2,629.8	1,960.0	7,511.7
2040	327.4	327.0	250.0	760.0	1,075.0	9.3	487.8	96.6	36.6	0.0	2,629.4	1,960.0	7,959.1
2041	327.4	327.0	250.0	760.0	1,075.0	9.3	617.1	96.6	36.6	0.0	2,594.3	1,960.0	8,053.3
2042	327.4	327.0	250.0	760.0	1,075.0	9.3	716.6	96.6	36.6	0.0	2,578.9	1,960.0	8,137.3
2043	327.4	327.0	250.0	760.0	1,075.0	9.3	819.3	96.6	36.6	0.0	2,748.2	1,960.0	8,409.4
2044	327.4	327.0	250.0	760.0	1,075.0	4.9	910.1	51.0	36.6	0.0	2,828.0	1,960.0	8,530.0
2045	327.4	327.0	250.0	760.0	1,075.0	4.9	1,005.0	51.0	36.6	0.0	3,041.0	1,960.0	8,837.9
2046	327.4	327.0	250.0	760.0	1,075.0	0.5	1,107.2	5.3	36.6	0.0	3,250.8	1,960.0	9,099.9
2047	327.4	327.0	250.0	760.0	1,075.0	0.5	1,210.7	5.3	36.6	0.0	3,398.3	1,960.0	9,350.8
2048	327.4	327.0	250.0	760.0	1,075.0	0.0	1,315.3	0.0	36.6	0.0	3,484.4	1,960.0	9,535.7
2049	327.4	327.0	250.0	760.0	1,075.0	0.0	1,438.0	0.0	36.6	0.0	3,648.3	1,960.0	9,822.3
2050	327.4	327.0	250.0	760.0	1,075.0	0.0	1,578.0	0.0	4.6	0.0	3,932.0	1,960.0	10,213.9
2051	327.4	327.0	250.0	760.0	1,075.0	0.0	1,704.4	0.0	4.6	0.0	4,104.5	1,960.0	10,513.0
2052	327.4	327.0	250.0	760.0	1,075.0	0.0	1,831.2	0.0	4.6	0.0	4,267.9	1,960.0	10,803.0
2053	327.4	327.0	250.0	760.0	1,075.0	0.0	1,957.9	0.0	4.6	0.0	4,431.2	1,960.0	11,093.1
2054	327.4	327.0	250.0	760.0	1,040.0	0.0	2,104.8	0.0	4.6	0.0	4,594.5	1,960.0	11,368.3
2055	327.4	327.0	250.0	760.0	1,005.0	0.0	2,252.2	0.0	4.6	0.0	4,757.9	1,960.0	11,644.1
2056	327.4	327.0	250.0	760.0	970.0	0.0	2,410.4	0.0	4.6	0.0	4,934.4	1,960.0	11,943.8
2057	327.4	327.0	250.0	760.0	935.0	0.0	2,568.6	0.0	4.6	0.0	5,111.0	1,960.0	12,243.6
2058	327.4	327.0	250.0	760.0	900.0	0.0	2,732.8	0.0	4.6	0.0	5,287.5	1,960.0	12,549.3
2059	327.4	327.0	250.0	760.0	865.0	0.0	2,901.3	0.0	4.6	0.0	5,313.4	1,960.0	12,708.7
2060	327.4	327.0	250.0	760.0	830.0	0.0	3,068.3	0.0	4.6	0.0	5,498.4	1,960.0	13,025.6
2061	327.4	327.0	250.0	760.0	795.0	0.0	3,237.2	0.0	4.6	0.0	5,685.7	1,960.0	13,347.0
2062	327.4	327.0	250.0	760.0	760.0	0.0	3,406.2	0.0	4.6	0.0	5,873.1	1,960.0	13,668.3
2063	327.4	327.0	250.0	760.0	725.0	0.0	3,575.2	0.0	4.6	0.0	6,060.4	1,960.0	13,989.6
2064	327.4	327.0	250.0	760.0	690.0	0.0	3,744.2	0.0	4.6	0.0	6,247.7	1,960.0	14,310.9
2065	327.4	327.0	250.0	760.0	628.9	0.0	3,930.5	0.0	2.0	0.0	6,441.5	1,960.0	14,627.3
2066	327.4	327.0	250.0	760.0	424.3	0.0	4,210.5	0.0	2.0	0.0	6,684.4	1,960.0	14,945.6
2067	327.4	327.0	250.0	760.0	146.1	0.0	4,518.3	0.0	2.0	0.0	6,966.2	1,960.0	15,256.9
2068	327.4	327.0	250.0	760.0	0.0	0.0	4,750.4	0.0	2.0	0.0	7,225.6	1,960.0	15,602.4
2069	327.4	327.0	250.0	760.0	0.0	0.0	4,902.6	0.0	2.0	0.0	7,418.2	1,960.0	15,947.3
2070	327.4	327.0	250.0	760.0	0.0	0.0	5,054.9	0.0	2.0	0.0	7,610.9	1,960.0	16,292.2

Table A.2.3: BAL Grid Capital Investment by Technology Table in 10⁶ USD (MUSD)

Year	Imports – CIV	Imports – GHA	Imports – MLI	Imports – NER	Biomass	Diesel	Natural Gas	HFO	Hydro	Nuclear	Solar	Wind	Total
2021	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
2022	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
2023	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
2024	0.0	0.0	0.0	0.0	87.5	0.0	0.0	0.0	0.0	0.0	475.7	23.5	586.7
2025	0.0	0.0	0.0	0.0	87.5	0.0	0.0	0.0	0.0	0.0	194.4	119.1	401.0
2026	0.0	0.0	0.0	0.0	87.5	0.0	0.0	0.0	0.0	0.0	113.9	117.0	318.4
2027	0.0	0.0	0.0	0.0	87.5	0.0	0.0	0.0	0.0	0.0	39.0	114.9	241.4
2028	0.0	0.0	0.0	0.0	87.5	0.0	0.0	0.0	0.0	0.0	114.1	112.9	314.5
2029	0.0	0.0	0.0	0.0	87.5	0.0	0.0	0.0	0.0	0.0	44.9	110.8	243.2
2030	0.0	0.0	0.0	0.0	87.5	0.0	0.0	0.0	0.0	0.0	35.2	108.7	231.4
2031	0.0	0.0	0.0	0.0	87.5	0.0	0.0	0.0	0.0	0.0	131.8	105.6	324.9
2032	0.0	0.0	0.0	0.0	87.5	0.0	0.0	0.0	0.0	0.0	210.5	102.5	400.5
2033	0.0	0.0	0.0	0.0	87.5	0.0	0.0	0.0	0.0	0.0	252.0	99.5	439.0
2034	0.0	0.0	0.0	0.0	87.5	0.0	0.0	0.0	0.0	0.0	52.1	96.4	236.0
2035	0.0	0.0	0.0	0.0	152.7	0.0	0.0	0.0	0.0	0.0	0.0	878.1	1,030.9
2036	0.0	0.0	0.0	0.0	511.5	0.0	0.0	0.0	0.0	0.0	0.0	0.0	511.5
2037	0.0	0.0	0.0	0.0	695.6	0.0	0.0	0.0	0.0	0.0	0.0	0.0	695.6
2038	0.0	0.0	0.0	0.0	365.1	0.0	0.0	0.0	0.0	0.0	32.6	0.0	397.8
2039	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	465.5	0.0	465.5
2040	0.0	0.0	0.0	0.0	0.0	0.0	585.4	0.0	0.0	0.0	0.0	0.0	585.4
2041	0.0	0.0	0.0	0.0	0.0	0.0	155.2	0.0	0.0	0.0	0.0	0.0	155.2
2042	0.0	0.0	0.0	0.0	0.0	0.0	119.4	0.0	0.0	0.0	0.0	0.0	119.4
2043	0.0	0.0	0.0	0.0	0.0	0.0	123.3	0.0	0.0	0.0	123.7	0.0	246.9
2044	0.0	0.0	0.0	0.0	0.0	0.0	108.9	0.0	0.0	0.0	58.1	0.0	167.0
2045	0.0	0.0	0.0	0.0	0.0	0.0	113.9	0.0	0.0	0.0	154.6	0.0	268.5
2046	0.0	0.0	0.0	0.0	0.0	0.0	122.7	0.0	0.0	0.0	173.8	0.0	296.4
2047	0.0	0.0	0.0	0.0	0.0	0.0	124.1	0.0	0.0	0.0	171.4	0.0	295.5
2048	0.0	0.0	0.0	0.0	0.0	0.0	125.6	0.0	0.0	0.0	385.9	0.0	511.5
2049	0.0	0.0	0.0	0.0	0.0	0.0	147.2	0.0	0.0	0.0	261.4	17.5	426.1
2050	0.0	0.0	0.0	0.0	0.0	0.0	168.0	0.0	0.0	0.0	290.5	93.3	551.8
2051	0.0	0.0	0.0	0.0	0.0	0.0	151.8	0.0	0.0	0.0	154.6	93.3	399.7
2052	0.0	0.0	0.0	0.0	0.0	0.0	152.1	0.0	0.0	0.0	207.3	93.3	452.6
2053	0.0	0.0	0.0	0.0	0.0	0.0	152.1	0.0	0.0	0.0	153.9	93.3	399.3
2054	0.0	0.0	0.0	0.0	0.0	0.0	176.3	0.0	0.0	0.0	146.8	93.3	416.4
2055	0.0	0.0	0.0	0.0	0.0	0.0	176.9	0.0	0.0	0.0	227.8	93.3	498.0
2056	0.0	0.0	0.0	0.0	0.0	0.0	189.8	0.0	0.0	0.0	306.3	93.3	589.5
2057	0.0	0.0	0.0	0.0	0.0	0.0	189.8	0.0	0.0	0.0	346.0	93.3	629.2
2058	0.0	0.0	0.0	0.0	0.0	0.0	197.0	0.0	0.0	0.0	173.8	93.3	464.1
2059	0.0	0.0	0.0	0.0	0.0	0.0	202.2	0.0	0.0	0.0	18.7	93.3	314.2
2060	0.0	0.0	0.0	0.0	0.0	0.0	200.4	0.0	0.0	0.0	133.7	878.1	1,212.2
2061	0.0	0.0	0.0	0.0	0.0	0.0	202.8	0.0	0.0	0.0	135.4	0.0	338.2
2062	0.0	0.0	0.0	0.0	0.0	0.0	202.8	0.0	0.0	0.0	166.7	0.0	369.5
2063	0.0	0.0	0.0	0.0	0.0	0.0	202.8	0.0	0.0	0.0	591.2	0.0	794.0
2064	0.0	0.0	0.0	0.0	0.0	0.0	202.8	0.0	0.0	0.0	135.4	0.0	338.2
2065	0.0	0.0	0.0	0.0	0.0	0.0	223.6	0.0	0.0	0.0	140.1	0.0	363.6
2066	0.0	0.0	0.0	0.0	0.0	0.0	336.0	0.0	0.0	0.0	175.7	0.0	511.6
2067	0.0	0.0	0.0	0.0	0.0	0.0	369.4	0.0	0.0	0.0	327.3	0.0	696.7
2068	0.0	0.0	0.0	0.0	0.0	0.0	278.5	0.0	0.0	0.0	245.7	0.0	524.1
2069	0.0	0.0	0.0	0.0	0.0	0.0	182.7	0.0	0.0	0.0	293.8	0.0	476.6
2070	0.0	0.0	0.0	0.0	0.0	0.0	182.7	0.0	0.0	0.0	313.1	0.0	495.8

Table A.2.4: BAL Grid Annual Fixed Operating Costs by Technology Table in 10⁶ USD (MUSD)

Year	Imports – CIV	Imports – GHA	Imports – MLI	Imports – NER	Biomass	Diesel	Natural Gas	HFO	Hydro	Nuclear	Solar	Wind	Total
2021	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
2022	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
2023	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
2024	0.0	0.0	0.0	0.0	2.6	0.0	0.0	0.0	0.0	0.0	6.2	0.9	9.7
2025	0.0	0.0	0.0	0.0	2.6	0.0	0.0	0.0	0.0	0.0	2.5	4.8	9.9
2026	0.0	0.0	0.0	0.0	2.6	0.0	0.0	0.0	0.0	0.0	1.5	4.7	8.8
2027	0.0	0.0	0.0	0.0	2.6	0.0	0.0	0.0	0.0	0.0	0.5	4.6	7.7
2028	0.0	0.0	0.0	0.0	2.6	0.0	0.0	0.0	0.0	0.0	1.5	4.5	8.6
2029	0.0	0.0	0.0	0.0	2.6	0.0	0.0	0.0	0.0	0.0	0.6	4.4	7.6
2030	0.0	0.0	0.0	0.0	2.6	0.0	0.0	0.0	0.0	0.0	0.5	4.3	7.4
2031	0.0	0.0	0.0	0.0	2.6	0.0	0.0	0.0	0.0	0.0	1.7	4.2	8.6
2032	0.0	0.0	0.0	0.0	2.6	0.0	0.0	0.0	0.0	0.0	2.7	4.1	9.5
2033	0.0	0.0	0.0	0.0	2.6	0.0	0.0	0.0	0.0	0.0	3.3	4.0	9.9
2034	0.0	0.0	0.0	0.0	2.6	0.0	0.0	0.0	0.0	0.0	0.7	3.9	7.2
2035	0.0	0.0	0.0	0.0	4.6	0.0	0.0	0.0	0.0	0.0	0.0	35.1	39.7
2036	0.0	0.0	0.0	0.0	15.3	0.0	0.0	0.0	0.0	0.0	0.0	0.0	15.3
2037	0.0	0.0	0.0	0.0	20.9	0.0	0.0	0.0	0.0	0.0	0.0	0.0	20.9
2038	0.0	0.0	0.0	0.0	11.0	0.0	0.0	0.0	0.0	0.0	0.4	0.0	11.4
2039	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	6.1	0.0	6.1
2040	0.0	0.0	0.0	0.0	0.0	0.0	17.1	0.0	0.0	0.0	0.0	0.0	17.1
2041	0.0	0.0	0.0	0.0	0.0	0.0	4.5	0.0	0.0	0.0	0.0	0.0	4.5
2042	0.0	0.0	0.0	0.0	0.0	0.0	3.5	0.0	0.0	0.0	0.0	0.0	3.5
2043	0.0	0.0	0.0	0.0	0.0	0.0	3.6	0.0	0.0	0.0	1.6	0.0	5.2
2044	0.0	0.0	0.0	0.0	0.0	0.0	3.2	0.0	0.0	0.0	0.8	0.0	3.9
2045	0.0	0.0	0.0	0.0	0.0	0.0	3.3	0.0	0.0	0.0	2.0	0.0	5.3
2046	0.0	0.0	0.0	0.0	0.0	0.0	3.6	0.0	0.0	0.0	2.3	0.0	5.8
2047	0.0	0.0	0.0	0.0	0.0	0.0	3.6	0.0	0.0	0.0	2.2	0.0	5.8
2048	0.0	0.0	0.0	0.0	0.0	0.0	3.7	0.0	0.0	0.0	5.0	0.0	8.7
2049	0.0	0.0	0.0	0.0	0.0	0.0	4.3	0.0	0.0	0.0	3.4	0.7	8.4
2050	0.0	0.0	0.0	0.0	0.0	0.0	4.9	0.0	0.0	0.0	3.8	3.7	12.4
2051	0.0	0.0	0.0	0.0	0.0	0.0	4.4	0.0	0.0	0.0	2.0	3.7	10.2
2052	0.0	0.0	0.0	0.0	0.0	0.0	4.4	0.0	0.0	0.0	2.7	3.7	10.9
2053	0.0	0.0	0.0	0.0	0.0	0.0	4.4	0.0	0.0	0.0	2.0	3.7	10.2
2054	0.0	0.0	0.0	0.0	0.0	0.0	5.1	0.0	0.0	0.0	1.9	3.7	10.8
2055	0.0	0.0	0.0	0.0	0.0	0.0	5.2	0.0	0.0	0.0	3.0	3.7	11.9
2056	0.0	0.0	0.0	0.0	0.0	0.0	5.5	0.0	0.0	0.0	4.0	3.7	13.3
2057	0.0	0.0	0.0	0.0	0.0	0.0	5.5	0.0	0.0	0.0	4.5	3.7	13.8
2058	0.0	0.0	0.0	0.0	0.0	0.0	5.7	0.0	0.0	0.0	2.3	3.7	11.7
2059	0.0	0.0	0.0	0.0	0.0	0.0	5.9	0.0	0.0	0.0	0.2	3.7	9.9
2060	0.0	0.0	0.0	0.0	0.0	0.0	5.8	0.0	0.0	0.0	1.7	35.1	42.7
2061	0.0	0.0	0.0	0.0	0.0	0.0	5.9	0.0	0.0	0.0	1.8	0.0	7.7
2062	0.0	0.0	0.0	0.0	0.0	0.0	5.9	0.0	0.0	0.0	2.2	0.0	8.1
2063	0.0	0.0	0.0	0.0	0.0	0.0	5.9	0.0	0.0	0.0	7.7	0.0	13.6
2064	0.0	0.0	0.0	0.0	0.0	0.0	5.9	0.0	0.0	0.0	1.8	0.0	7.7
2065	0.0	0.0	0.0	0.0	0.0	0.0	6.5	0.0	0.0	0.0	1.8	0.0	8.3
2066	0.0	0.0	0.0	0.0	0.0	0.0	9.8	0.0	0.0	0.0	2.3	0.0	12.1
2067	0.0	0.0	0.0	0.0	0.0	0.0	10.8	0.0	0.0	0.0	4.3	0.0	15.0
2068	0.0	0.0	0.0	0.0	0.0	0.0	8.1	0.0	0.0	0.0	3.2	0.0	11.3
2069	0.0	0.0	0.0	0.0	0.0	0.0	5.3	0.0	0.0	0.0	3.8	0.0	9.1
2070	0.0	0.0	0.0	0.0	0.0	0.0	5.3	0.0	0.0	0.0	4.1	0.0	9.4

Table A.2.5: BAL Grid Annual Variable Operating Costs by Technology Table in 10⁶ USD (MUSD)

Year	Imports – CIV	Imports – GHA	Imports – MLI	Imports – NER	Biomass	Diesel	Natural Gas	HFO	Hydro	Nuclear	Solar	Wind	Total
2021	26.1	0.0	0.0	0.0	0.0	0.0	0.0	219.3	0.0	0.0	0.0	0.0	245.4
2022	16.5	27.9	62.5	0.0	0.0	0.0	0.0	221.3	0.0	0.0	0.0	0.0	328.2
2023	22.2	32.6	53.9	0.0	0.0	15.8	0.0	226.6	0.0	0.0	0.0	0.0	351.1
2024	27.8	37.3	42.0	0.0	2.3	0.0	0.0	228.7	0.0	0.0	0.0	0.0	338.1
2025	33.5	42.0	30.2	0.0	4.6	0.0	0.0	216.1	0.0	0.0	0.0	0.0	326.4
2026	33.4	51.2	24.3	0.0	6.9	0.0	0.0	213.6	0.0	0.0	0.0	0.0	329.4
2027	33.2	60.3	22.0	0.0	9.2	0.0	0.0	219.0	0.0	0.0	0.0	0.0	343.8
2028	33.1	69.5	25.5	0.0	11.5	0.0	0.0	208.7	0.0	0.0	0.0	0.0	348.2
2029	32.9	78.7	41.8	0.0	13.8	0.0	0.0	197.4	0.0	0.0	0.0	0.0	364.6
2030	32.8	87.8	54.1	0.0	16.1	0.0	0.0	190.1	0.0	0.0	0.0	0.0	381.0
2031	32.7	97.0	74.0	24.3	18.2	0.0	0.0	181.7	0.0	0.0	0.0	0.0	427.7
2032	32.5	106.1	81.2	40.6	20.0	0.0	0.0	179.4	0.0	0.0	0.0	0.0	459.9
2033	32.4	115.3	88.5	22.9	22.3	0.0	0.0	182.5	0.0	0.0	0.0	0.0	463.8
2034	32.4	115.3	88.5	67.7	24.6	0.2	0.0	189.8	0.0	0.0	0.0	0.0	518.4
2035	32.4	115.3	59.6	0.0	27.2	0.0	0.0	164.2	0.0	0.0	0.0	0.0	398.7
2036	32.4	115.3	88.5	0.0	39.8	0.0	0.0	133.1	0.0	0.0	0.0	0.0	409.1
2037	32.4	115.3	88.5	0.0	56.8	0.0	0.0	99.8	0.0	0.0	0.0	0.0	392.8
2038	32.4	115.3	88.5	15.0	66.8	0.0	0.0	101.3	0.0	0.0	0.0	0.0	419.2
2039	32.4	115.3	88.5	67.7	62.8	11.9	0.0	82.9	0.0	0.0	0.0	0.0	461.5
2040	32.4	115.3	0.0	0.0	65.0	0.0	190.0	13.9	0.0	0.0	0.0	0.0	416.7
2041	32.4	115.3	12.4	0.0	67.4	0.0	249.9	0.0	0.0	0.0	0.0	0.0	477.4
2042	32.4	115.3	25.0	0.0	69.4	0.0	303.5	0.0	0.0	0.0	0.0	0.0	545.6
2043	32.4	115.3	30.4	0.0	69.4	0.0	351.3	0.0	0.0	0.0	0.0	0.0	598.7
2044	32.4	115.3	44.8	0.0	70.2	0.0	400.2	0.0	0.0	0.0	0.0	0.0	662.8
2045	32.4	115.3	54.4	0.0	69.8	0.0	441.8	0.0	0.0	0.0	0.0	0.0	713.7
2046	32.4	115.3	64.9	0.0	69.6	0.0	487.9	0.0	0.0	0.0	0.0	0.0	770.0
2047	32.4	115.3	75.4	0.7	69.8	0.0	538.8	0.1	0.0	0.0	0.0	0.0	832.5
2048	32.4	115.3	86.4	2.1	70.4	0.0	594.4	0.0	0.0	0.0	0.0	0.0	900.9
2049	32.4	115.3	88.5	2.8	70.5	0.0	649.3	0.0	0.0	0.0	0.0	0.0	958.8
2050	32.4	115.3	88.5	3.8	69.9	0.0	703.0	0.0	0.0	0.0	0.0	0.0	1,012.9
2051	32.4	115.3	88.5	4.4	69.9	0.0	758.7	0.0	0.0	0.0	0.0	0.0	1,069.2
2052	32.4	115.3	88.5	5.0	69.9	0.0	815.2	0.0	0.0	0.0	0.0	0.0	1,126.3
2053	32.4	115.3	88.5	5.7	69.9	0.0	871.7	0.0	0.0	0.0	0.0	0.0	1,183.4
2054	32.4	115.3	88.5	6.3	67.6	0.0	939.5	0.0	0.0	0.0	0.0	0.0	1,249.6
2055	32.4	115.3	88.5	6.9	65.4	0.0	1007.4	0.0	0.0	0.0	0.0	0.0	1,315.8
2056	32.4	115.3	88.5	7.5	63.1	0.0	1079.8	0.0	0.0	0.0	0.0	0.0	1,386.6
2057	32.4	115.3	88.5	8.2	60.8	0.0	1152.2	0.0	0.0	0.0	0.0	0.0	1,457.4
2058	32.4	115.3	86.2	8.7	58.6	0.0	1226.3	0.0	0.0	0.0	0.0	0.0	1,527.3
2059	32.4	115.3	84.3	9.1	56.9	0.0	1313.5	0.0	0.0	0.0	0.0	0.0	1,611.3
2060	32.4	115.3	82.3	9.5	54.6	0.0	1386.5	0.0	0.0	0.0	0.0	0.0	1,680.6
2061	32.4	115.3	80.4	9.9	52.3	0.0	1460.4	0.0	0.0	0.0	0.0	0.0	1,750.6
2062	32.4	115.3	78.4	10.3	50.0	0.0	1534.3	0.0	0.0	0.0	0.0	0.0	1,820.7
2063	32.4	115.3	76.5	10.8	47.7	0.0	1608.2	0.0	0.0	0.0	0.0	0.0	1,890.8
2064	32.4	115.3	74.5	11.2	45.4	0.0	1682.1	0.0	0.0	0.0	0.0	0.0	1,960.8
2065	32.4	115.3	72.1	11.6	41.3	0.0	1764.8	0.0	0.0	0.0	0.0	0.0	2,037.5
2066	32.4	115.3	69.2	12.1	27.9	0.0	1895.4	0.0	0.0	0.0	0.0	0.0	2,152.2
2067	32.4	115.3	69.7	15.2	9.6	0.0	2042.9	0.0	0.0	0.0	0.0	0.0	2,285.0
2068	32.4	115.3	70.9	18.2	0.0	0.0	2149.1	0.0	0.0	0.0	0.0	0.0	2,385.9
2069	32.4	115.3	71.3	20.6	0.0	0.0	2214.9	0.0	0.0	0.0	0.0	0.0	2,454.4
2070	32.4	115.3	71.6	22.9	0.0	0.0	2280.7	0.0	0.0	0.0	0.0	0.0	2,522.8

Table A.2.6: BAL Emissions in kilotons CO_{2eq} and Emission Intensity in grams CO_{2eq} /kWh electricity generated

Year	kTon CO_{2eq}	g CO_{2eq} / kWh electricity generated
2021	1906.2	0.679
2022	1906.2	0.545
2023	2012.5	0.529
2024	1934.8	0.424
2025	1812.7	0.361
2026	1750.0	0.321
2027	1754.1	0.298
2028	1634.2	0.259
2029	1512.4	0.224
2030	1426.2	0.199
2031	1338.3	0.168
2032	1298.0	0.148
2033	1297.1	0.136
2034	1326.4	0.129
2035	1128.0	0.103
2036	899.5	0.076
2037	663.4	0.053
2038	662.6	0.050
2039	577.5	0.041
2040	1060.3	0.071
2041	1278.2	0.081
2042	1552.3	0.093
2043	1796.6	0.102
2044	2047.0	0.111
2045	2259.8	0.117
2046	2495.5	0.123
2047	2756.3	0.130
2048	3040.1	0.137
2049	3321.1	0.144
2050	3595.7	0.149
2051	4057.6	0.162
2052	4169.4	0.161
2053	4458.4	0.166
2054	4805.4	0.173
2055	5152.4	0.179
2056	5522.9	0.185
2057	5893.4	0.191
2058	6272.1	0.197
2059	6718.3	0.204
2060	7091.9	0.210
2061	7469.8	0.214
2062	7847.7	0.218
2063	8225.5	0.222
2064	8603.4	0.226
2065	9026.3	0.232
2066	9694.7	0.241
2067	10449.1	0.253
2068	10992.2	0.260
2069	11328.7	0.260
2070	11665.2	0.262

A.3 AMB

Table A.3.1: AMB Annual Production by Technology Table in GigaWatt-hours (GWh)

Year	Imports – CIV	Imports – GHA	Imports – MLI	Imports – NER	Biomass	Diesel	Natural Gas	HFO	Hydro	Nuclear	Solar	Wind	Total
2021	241.6	0.0	0.0	0.0	0.0	0.0	0.0	2378.6	100.4	0.0	87.6	0.0	2,808.2
2022	152.6	240.8	505.4	0.0	0.0	0.0	0.0	2378.6	100.4	0.0	134.9	0.0	3,512.7
2023	205.1	281.3	420.9	0.0	0.0	135.0	0.0	2414.3	106.2	0.0	273.3	0.0	3,836.1
2024	257.7	321.8	328.3	0.0	139.7	110.5	0.0	2721.6	187.9	0.0	1145.0	165.3	5,377.8
2025	310.2	362.3	235.7	262.7	275.4	0.0	0.0	2283.1	187.9	0.0	1876.0	314.1	6,107.3
2026	308.9	441.2	292.7	204.3	413.1	0.0	0.0	2439.4	187.9	0.0	1889.8	495.9	6,673.2
2027	307.6	520.1	349.7	136.1	558.6	0.0	0.0	2595.6	187.9	0.0	1922.1	661.2	7,239.0
2028	306.3	599.1	406.7	0.0	698.3	0.0	0.0	2751.9	187.9	0.0	2028.2	826.5	7,804.9
2029	305.0	678.0	463.7	0.0	838.0	0.0	0.0	2771.2	187.9	0.0	2135.1	991.8	8,370.7
2030	303.7	757.0	520.7	72.1	963.9	0.0	0.0	2569.3	187.9	0.0	2424.6	1137.3	8,936.6
2031	302.4	835.9	577.7	487.7	1101.6	0.0	0.0	2483.7	272.2	0.0	2722.8	1277.3	10,061.3
2032	301.0	914.8	634.8	455.5	1215.5	272.2	0.0	2415.6	385.8	0.0	3103.1	1487.6	11,186.0
2033	299.7	993.8	691.8	423.2	1244.2	293.4	0.0	2323.8	385.8	0.0	4011.8	1652.9	12,320.5
2034	299.7	993.8	691.8	423.2	1033.5	525.2	0.0	1948.6	366.3	0.0	5335.4	1776.6	13,394.0
2035	1701.3	1490.5	27.4	0.0	1052.5	0.0	1453.2	835.5	318.3	0.0	4999.0	1789.0	13,666.7
2036	1725.4	1439.0	134.0	4.5	1072.7	0.0	1567.0	853.5	347.5	0.0	5211.1	2445.5	14,800.2
2037	1729.5	1404.8	330.8	29.3	1109.7	0.0	1578.9	811.4	380.6	0.0	5318.8	3239.8	15,933.7
2038	1783.2	1549.8	581.0	158.9	1151.8	0.0	1622.9	954.9	385.8	0.0	5697.8	3181.1	17,067.2
2039	1792.2	1681.6	984.3	496.8	1195.9	0.0	1694.0	794.9	378.3	0.0	6008.6	3239.8	18,266.3
2040	1792.2	1668.6	1005.5	562.7	2155.9	0.0	1683.8	647.0	362.2	0.0	6505.9	3129.3	19,513.2
2041	1792.0	1666.8	1000.7	752.6	3030.6	0.0	1689.4	490.8	385.8	0.0	6426.9	3239.8	20,475.5
2042	1796.7	1698.5	1084.3	1191.7	3508.5	0.0	1736.4	312.9	385.8	0.0	6662.2	3239.8	21,616.8
2043	1866.6	1746.4	1228.5	1908.9	3597.7	10.5	1792.5	120.2	380.0	0.0	6867.1	3239.8	22,758.1
2044	1948.4	1723.9	1183.9	1833.1	3559.9	14.3	1663.1	31.4	381.9	1295.4	7028.5	3239.8	23,903.6
2045	1998.3	1809.7	1267.7	1750.4	3540.3	14.3	1371.6	31.4	385.8	2460.9	7170.5	3239.8	25,040.8
2046	1972.5	1776.4	1284.3	1669.2	3493.1	13.6	1095.0	24.2	385.8	4008.6	7338.1	3184.8	26,245.7
2047	1973.1	1767.8	1269.0	1859.2	3491.3	13.6	803.5	24.2	385.8	5332.6	7290.8	3239.8	27,450.7
2048	2048.6	1815.2	1256.5	2073.0	3639.9	13.5	513.8	23.4	385.8	6883.8	6841.5	3160.6	28,655.6
2049	2054.1	1807.2	1225.1	2098.1	4289.6	13.5	236.0	16.2	385.8	8091.6	6431.1	3239.8	29,888.1
2050	1652.7	1346.5	767.9	726.0	4289.6	0.0	0.0	0.0	385.8	11685.4	7018.8	3239.8	31,112.6
2051	1628.9	1288.6	745.3	674.5	4289.6	0.0	0.0	0.0	385.8	12466.4	7595.8	3239.8	32,314.7
2052	1608.0	1217.3	734.8	624.6	4289.6	0.0	0.0	0.0	385.8	13266.3	8193.4	3239.8	33,559.5
2053	1567.4	1165.5	713.4	585.7	4289.6	0.0	0.0	0.0	385.8	14066.3	8790.9	3239.8	34,804.3
2054	1519.8	1116.4	691.3	551.5	4289.6	0.0	0.0	0.0	385.8	14871.9	9393.3	3239.8	36,059.5
2055	1464.0	1081.3	671.4	517.3	4289.6	0.0	0.0	0.0	385.8	15674.9	10000.7	3239.8	37,324.8
2056	1408.8	1062.3	650.2	480.8	4289.6	0.0	0.0	0.0	385.8	16518.0	10657.2	3239.8	38,692.5
2057	1358.3	1047.0	626.2	447.1	4289.6	0.0	0.0	0.0	385.8	17354.0	11314.8	3239.8	40,062.6
2058	1307.1	1032.4	600.3	415.2	4289.6	0.0	0.0	0.0	385.8	18191.3	11973.7	3239.8	41,435.2
2059	1272.1	1017.9	567.3	390.3	4289.6	0.0	0.0	0.0	385.8	19017.6	12637.4	3239.8	42,817.9
2060	1247.9	1002.6	529.6	371.0	4289.6	0.0	0.0	0.0	385.8	19833.2	13301.1	3239.8	44,200.6
2061	1223.4	982.0	494.5	353.6	4289.6	0.0	0.0	0.0	385.8	20657.1	13971.5	3239.8	45,597.3
2062	1201.3	960.1	460.6	336.3	4289.6	0.0	0.0	0.0	385.8	21478.6	14641.9	3239.8	46,994.0
2063	1182.0	940.1	429.7	318.9	4289.6	0.0	0.0	0.0	385.8	22295.5	15315.2	3239.8	48,396.6
2064	1162.9	922.9	401.3	301.5	4289.6	0.0	0.0	0.0	385.8	23112.3	15993.3	3239.8	49,809.4
2065	1140.6	906.3	375.4	284.2	4289.6	0.0	0.0	0.0	385.8	23929.1	16671.4	3239.8	51,222.2
2066	1115.4	886.8	351.1	265.6	4289.6	0.0	0.0	0.0	385.8	24804.2	17398.4	3239.8	52,736.7
2067	1090.2	861.8	330.7	248.7	4289.6	0.0	0.0	0.0	385.8	25680.7	18126.7	3239.8	54,253.9
2068	1066.9	827.5	315.1	236.2	4289.6	0.0	0.0	0.0	385.8	26555.3	18855.0	3239.8	55,771.2
2069	1046.5	784.9	302.3	229.3	4289.6	0.0	0.0	0.0	385.8	27427.0	19583.3	3239.8	57,288.5
2070	1027.3	744.4	290.6	222.5	4289.6	0.0	0.0	0.0	385.8	28294.3	20311.6	3239.8	58,805.8

Table A.3.2: AMB Total Annual Capacity by Technology Table in MegaWatts (MW)

Year	Imports – CIV	Imports – GHA	Imports – MLI	Imports – NER	Biomass	Diesel	Natural Gas	HFO	Hydro	Nuclear	Solar	Wind	Total
2021	327.4	327.0	0.0	0.0	0.0	34.1	0.0	355.1	34.6	0.0	56.7	0.0	1,134.9
2022	327.4	327.0	250.0	760.0	0.0	34.1	0.0	355.1	34.6	0.0	87.3	0.0	2,175.5
2023	327.4	327.0	250.0	760.0	0.0	34.6	0.0	360.4	36.6	0.0	176.9	0.0	2,272.9
2024	327.4	327.0	250.0	760.0	35.0	34.6	0.0	406.3	64.8	0.0	741.2	100.0	3,046.2
2025	327.4	327.0	250.0	760.0	70.0	34.6	0.0	406.3	64.8	0.0	1256.7	200.0	3,696.8
2026	327.4	327.0	250.0	760.0	105.0	34.6	0.0	419.0	64.8	0.0	1256.7	300.0	3,844.4
2027	327.4	327.0	250.0	760.0	140.0	34.6	0.0	436.6	64.8	0.0	1265.8	400.0	4,006.2
2028	327.4	327.0	250.0	760.0	175.0	33.3	0.0	465.1	64.8	0.0	1337.5	500.0	4,240.1
2029	327.4	327.0	250.0	760.0	210.0	30.5	0.0	467.9	64.8	0.0	1410.0	600.0	4,447.6
2030	327.4	327.0	250.0	760.0	245.0	28.5	0.0	446.9	64.8	0.0	1620.0	700.0	4,769.6
2031	327.4	327.0	250.0	760.0	280.0	26.6	0.0	426.9	93.8	0.0	1803.6	800.0	5,095.3
2032	327.4	327.0	250.0	760.0	315.0	51.6	0.0	426.9	133.0	0.0	2127.8	900.0	5,618.7
2033	327.4	327.0	250.0	760.0	350.0	58.1	0.0	425.9	133.0	0.0	2816.0	1000.0	6,447.5
2034	327.4	327.0	250.0	760.0	385.0	112.9	0.0	425.9	133.0	0.0	5184.1	1100.0	9,005.4
2035	327.4	327.0	250.0	760.0	385.0	112.9	312.3	425.9	133.0	0.0	5184.1	1100.0	9,317.6
2036	327.4	327.0	250.0	760.0	385.0	107.4	336.8	369.4	133.0	0.0	5183.5	1503.7	9,683.2
2037	327.4	327.0	250.0	760.0	385.0	102.5	336.8	317.4	133.0	0.0	5183.0	1960.0	10,082.0
2038	327.4	327.0	250.0	760.0	385.0	102.5	336.8	317.4	133.0	0.0	5182.5	1960.0	10,081.5
2039	327.4	327.0	250.0	760.0	385.0	99.1	336.8	282.7	133.0	0.0	5181.6	1960.0	10,042.6
2040	327.4	327.0	250.0	760.0	681.6	95.6	336.8	246.2	133.0	0.0	5181.2	1960.0	10,298.8
2041	327.4	327.0	250.0	760.0	943.9	95.6	336.8	246.2	133.0	0.0	5146.1	1960.0	10,526.0
2042	327.4	327.0	250.0	760.0	1075.0	95.6	336.8	246.2	133.0	0.0	5130.7	1960.0	10,641.7
2043	327.4	327.0	250.0	760.0	1075.0	95.6	336.8	246.2	133.0	0.0	5128.9	1960.0	10,639.9
2044	327.4	327.0	250.0	760.0	1075.0	91.2	336.8	200.6	133.0	205.2	5128.4	1960.0	10,794.6
2045	327.4	327.0	250.0	760.0	1075.0	91.2	336.8	200.6	133.0	389.2	5127.7	1960.0	10,977.8
2046	327.4	327.0	250.0	760.0	1075.0	86.9	336.8	155.0	133.0	633.4	5097.1	1960.0	11,141.4
2047	327.4	327.0	250.0	760.0	1075.0	86.9	336.8	155.0	133.0	827.6	5007.4	1960.0	11,246.0
2048	327.4	327.0	250.0	760.0	1075.0	86.4	336.8	149.6	133.0	1027.6	4443.2	1960.0	10,876.0
2049	327.4	327.0	250.0	760.0	1075.0	86.4	336.8	103.8	133.0	1267.7	4162.9	1960.0	10,789.9
2050	327.4	327.0	250.0	760.0	1075.0	86.4	336.8	103.8	133.0	1988.8	4543.4	1960.0	11,891.5
2051	327.4	327.0	250.0	760.0	1075.0	86.4	336.8	91.1	133.0	2175.2	4916.9	1960.0	12,438.7
2052	327.4	327.0	250.0	760.0	1075.0	86.4	336.8	73.4	133.0	2361.6	5303.7	1960.0	12,994.3
2053	327.4	327.0	250.0	760.0	1075.0	86.4	336.8	31.1	133.0	2548.1	5690.4	1960.0	13,525.1
2054	327.4	327.0	250.0	760.0	1075.0	86.4	336.8	0.0	133.0	2734.5	6080.4	1960.0	14,070.4
2055	327.4	327.0	250.0	760.0	1075.0	86.4	336.8	0.0	133.0	2920.9	6473.6	1960.0	14,650.0
2056	327.4	327.0	250.0	760.0	1075.0	86.4	336.8	0.0	133.0	3119.6	6898.5	1960.0	15,273.7
2057	327.4	327.0	250.0	760.0	1075.0	61.3	336.8	0.0	133.0	3318.3	7324.2	1960.0	15,873.1
2058	327.4	327.0	250.0	760.0	1075.0	54.8	336.8	0.0	133.0	3517.0	7750.7	1960.0	16,491.7
2059	327.4	327.0	250.0	760.0	1075.0	0.0	336.8	0.0	133.0	3715.7	8180.3	1960.0	17,065.2
2060	327.4	327.0	250.0	760.0	1075.0	0.0	336.8	0.0	133.0	3914.4	8609.9	1960.0	17,693.6
2061	327.4	327.0	250.0	760.0	1075.0	0.0	336.8	0.0	133.0	4115.2	9043.9	1960.0	18,328.3
2062	327.4	327.0	250.0	760.0	1075.0	0.0	336.8	0.0	133.0	4315.9	9477.9	1960.0	18,962.9
2063	327.4	327.0	250.0	760.0	1075.0	0.0	336.8	0.0	133.0	4516.6	9913.7	1960.0	19,599.5
2064	327.4	327.0	250.0	760.0	1075.0	0.0	336.8	0.0	133.0	4717.3	10352.6	1960.0	20,239.1
2065	327.4	327.0	250.0	760.0	1075.0	0.0	336.8	0.0	133.0	4918.0	10791.6	1960.0	20,878.8
2066	327.4	327.0	250.0	760.0	1075.0	0.0	336.8	0.0	133.0	5131.8	11262.2	1960.0	21,563.2
2067	327.4	327.0	250.0	760.0	1075.0	0.0	336.8	0.0	133.0	5345.7	11733.6	1960.0	22,248.4
2068	327.4	327.0	250.0	760.0	1075.0	0.0	336.8	0.0	133.0	5559.5	12205.0	1960.0	22,933.7
2069	327.4	327.0	250.0	760.0	1075.0	0.0	336.8	0.0	133.0	5773.3	12676.5	1960.0	23,618.9
2070	327.4	327.0	250.0	760.0	1075.0	0.0	336.8	0.0	133.0	5987.1	13147.9	1960.0	24,304.1

Table A.3.3: AMB Grid Capital Investment by Technology Table in 10⁶ USD (MUSD)

Year	Imports – CIV	Imports – GHA	Imports – MLI	Imports – NER	Biomass	Diesel	Natural Gas	HFO	Hydro	Nuclear	Solar	Wind	Total
2021	0	0	0	0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
2022	0	0	0	0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
2023	0	0	0	0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
2024	0	0	0	0	87.5	0.0	0.0	66.5	84.5	0.0	599.7	125.1	963.3
2025	0	0	0	0	87.5	0.0	0.0	0.0	0.0	0.0	507.3	119.1	713.9
2026	0	0	0	0	87.5	0.0	0.0	18.4	0.0	0.0	0.0	117.0	222.9
2027	0	0	0	0	87.5	0.0	0.0	25.6	0.0	0.0	8.6	114.9	236.6
2028	0	0	0	0	87.5	0.0	0.0	61.4	0.0	0.0	66.5	112.9	328.2
2029	0	0	0	0	87.5	0.0	0.0	45.1	0.0	0.0	65.6	110.8	309.0
2030	0	0	0	0	87.5	0.0	0.0	0.0	0.0	0.0	186.0	108.7	382.2
2031	0	0	0	0	87.5	0.0	0.0	0.0	87.1	0.0	159.5	105.6	439.8
2032	0	0	0	0	87.5	30.0	0.0	0.0	117.5	0.0	276.2	102.5	613.8
2033	0	0	0	0	87.5	7.9	0.0	0.0	0.0	0.0	574.2	99.5	769.1
2034	0	0	0	0	87.5	65.7	0.0	0.0	0.0	0.0	1935.3	96.4	2184.9
2035	0	0	0	0	0.0	0.0	374.8	0.0	0.0	0.0	0.0	0.0	374.8
2036	0	0	0	0	0.0	0.0	29.4	0.0	0.0	0.0	0.0	376.6	406.0
2037	0	0	0	0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	425.7	425.7
2038	0	0	0	0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
2039	0	0	0	0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
2040	0	0	0	0	741.6	0.0	0.0	0.0	0.0	0.0	0.0	0.0	741.6
2041	0	0	0	0	655.7	0.0	0.0	0.0	0.0	0.0	0.0	0.0	655.7
2042	0	0	0	0	327.8	0.0	0.0	0.0	0.0	0.0	0.0	0.0	327.8
2043	0	0	0	0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
2044	0	0	0	0	0.0	0.0	0.0	0.0	0.0	1259.4	0.0	0.0	1259.4
2045	0	0	0	0	0.0	0.0	0.0	0.0	0.0	1128.9	0.0	0.0	1128.9
2046	0	0	0	0	0.0	0.0	0.0	0.0	0.0	1498.7	0.0	0.0	1498.7
2047	0	0	0	0	0.0	0.0	0.0	0.0	0.0	1191.9	0.0	0.0	1191.9
2048	0	0	0	0	0.0	0.0	0.0	0.0	0.0	1227.7	0.0	0.0	1227.7
2049	0	0	0	0	0.0	0.0	0.0	0.0	0.0	1473.4	170.1	93.3	1736.8
2050	0	0	0	0	0.0	0.0	0.0	0.0	96.0	4425.4	275.1	93.3	4889.7
2051	0	0	0	0	0.0	0.0	0.0	0.0	0.0	1144.1	276.6	93.3	1514.0
2052	0	0	0	0	0.0	0.0	0.0	0.0	0.0	1144.1	331.6	93.3	1568.9
2053	0	0	0	0	0.0	0.0	0.0	0.0	0.0	1144.1	332.0	93.3	1569.4
2054	0	0	0	0	87.5	0.0	0.0	0.0	0.0	1144.1	433.8	93.3	1758.6
2055	0	0	0	0	87.5	0.0	0.0	0.0	0.0	1144.1	417.0	93.3	1741.9
2056	0	0	0	0	87.5	0.0	0.0	0.0	0.0	1219.5	541.7	93.3	1942.0
2057	0	0	0	0	87.5	0.0	0.0	0.0	0.0	1219.5	805.3	93.3	2205.6
2058	0	0	0	0	87.5	0.0	0.0	0.0	0.0	1219.5	2020.6	93.3	3420.8
2059	0	0	0	0	87.5	0.0	0.0	0.0	0.0	1219.5	310.6	93.3	1710.9
2060	0	0	0	0	87.5	0.0	0.0	0.0	0.0	1219.5	310.6	0.0	1617.6
2061	0	0	0	0	87.5	0.0	0.0	0.0	0.0	1231.8	313.8	376.6	2009.7
2062	0	0	0	0	87.5	0.0	0.0	0.0	0.0	1231.8	313.8	425.7	2058.8
2063	0	0	0	0	87.5	0.0	0.0	0.0	0.0	1231.8	315.1	0.0	1634.4
2064	0	0	0	0	87.5	0.0	0.0	0.0	0.0	1231.8	317.4	0.0	1636.7
2065	0	0	0	0	0.0	0.0	0.0	0.0	7.8	1231.8	317.4	0.0	1557.0
2066	0	0	0	0	0.0	0.0	0.0	0.0	0.0	1312.1	340.2	0.0	1652.3
2067	0	0	0	0	0.0	0.0	0.0	0.0	0.0	1312.1	340.8	0.0	1653.0
2068	0	0	0	0	0.0	0.0	0.0	0.0	0.0	1312.1	340.8	0.0	1653.0
2069	0	0	0	0	0.0	0.0	0.0	0.0	0.0	1312.1	340.8	0.0	1653.0
2070	0	0	0	0	741.6	0.0	0.0	0.0	0.0	1312.1	340.8	0.0	2394.5

Table A.3.4: AMB Grid Annual Fixed Operating Costs by Technology Table in 10⁶ USD (MUSD)

Year	Imports – CIV	Imports – GHA	Imports – MLI	Imports – NER	Biomass	Diesel	Natural Gas	HFO	Hydro	Nuclear	Solar	Wind	Total
2021	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
2022	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
2023	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
2024	0.0	0.0	0.0	0.0	2.6	0.0	0.0	2.1	2.5	0.0	7.8	5.0	20.0
2025	0.0	0.0	0.0	0.0	2.6	0.0	0.0	0.0	0.0	0.0	6.6	4.8	14.0
2026	0.0	0.0	0.0	0.0	2.6	0.0	0.0	0.6	0.0	0.0	0.0	4.7	7.9
2027	0.0	0.0	0.0	0.0	2.6	0.0	0.0	0.8	0.0	0.0	0.1	4.6	8.1
2028	0.0	0.0	0.0	0.0	2.6	0.0	0.0	1.9	0.0	0.0	0.9	4.5	9.9
2029	0.0	0.0	0.0	0.0	2.6	0.0	0.0	1.4	0.0	0.0	0.9	4.4	9.3
2030	0.0	0.0	0.0	0.0	2.6	0.0	0.0	0.0	0.0	0.0	2.4	4.3	9.4
2031	0.0	0.0	0.0	0.0	2.6	0.0	0.0	0.0	2.6	0.0	2.1	4.2	11.5
2032	0.0	0.0	0.0	0.0	2.6	0.9	0.0	0.0	3.5	0.0	3.6	4.1	14.7
2033	0.0	0.0	0.0	0.0	2.6	0.2	0.0	0.0	0.0	0.0	7.5	4.0	14.3
2034	0.0	0.0	0.0	0.0	2.6	1.9	0.0	0.0	0.0	0.0	25.2	3.9	33.6
2035	0.0	0.0	0.0	0.0	0.0	0.0	10.9	0.0	0.0	0.0	0.0	0.0	10.9
2036	0.0	0.0	0.0	0.0	0.0	0.0	0.9	0.0	0.0	0.0	0.0	15.1	15.9
2037	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	17.0	17.0
2038	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
2039	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
2040	0.0	0.0	0.0	0.0	22.2	0.0	0.0	0.0	0.0	0.0	0.0	0.0	22.2
2041	0.0	0.0	0.0	0.0	19.7	0.0	0.0	0.0	0.0	0.0	0.0	0.0	19.7
2042	0.0	0.0	0.0	0.0	9.8	0.0	0.0	0.0	0.0	0.0	0.0	0.0	9.8
2043	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
2044	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	37.8	0.0	0.0	37.8
2045	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	33.9	0.0	0.0	33.9
2046	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	45.0	0.0	0.0	45.0
2047	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	35.8	0.0	0.0	35.8
2048	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	36.8	0.0	0.0	36.8
2049	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	44.2	2.2	3.7	50.1
2050	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	2.9	132.8	3.6	3.7	142.9
2051	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	34.3	3.6	3.7	41.7
2052	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	34.3	4.3	3.7	42.4
2053	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	34.3	4.3	3.7	42.4
2054	0.0	0.0	0.0	0.0	2.6	0.0	0.0	0.0	0.0	34.3	5.6	3.7	46.3
2055	0.0	0.0	0.0	0.0	2.6	0.0	0.0	0.0	0.0	34.3	5.4	3.7	46.1
2056	0.0	0.0	0.0	0.0	2.6	0.0	0.0	0.0	0.0	36.6	7.0	3.7	50.0
2057	0.0	0.0	0.0	0.0	2.6	0.0	0.0	0.0	0.0	36.6	10.5	3.7	53.4
2058	0.0	0.0	0.0	0.0	2.6	0.0	0.0	0.0	0.0	36.6	26.3	3.7	69.2
2059	0.0	0.0	0.0	0.0	2.6	0.0	0.0	0.0	0.0	36.6	4.0	3.7	47.0
2060	0.0	0.0	0.0	0.0	2.6	0.0	0.0	0.0	0.0	36.6	4.0	0.0	43.2
2061	0.0	0.0	0.0	0.0	2.6	0.0	0.0	0.0	0.0	37.0	4.1	15.1	58.7
2062	0.0	0.0	0.0	0.0	2.6	0.0	0.0	0.0	0.0	37.0	4.1	17.0	60.7
2063	0.0	0.0	0.0	0.0	2.6	0.0	0.0	0.0	0.0	37.0	4.1	0.0	43.7
2064	0.0	0.0	0.0	0.0	2.6	0.0	0.0	0.0	0.0	37.0	4.1	0.0	43.7
2065	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.2	37.0	4.1	0.0	41.3
2066	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	39.4	4.4	0.0	43.8
2067	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	39.4	4.4	0.0	43.8
2068	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	39.4	4.4	0.0	43.8
2069	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	39.4	4.4	0.0	43.8
2070	0.0	0.0	0.0	0.0	22.2	0.0	0.0	0.0	0.0	39.4	4.4	0.0	66.0

Table A.3.5: AMB Grid Annual Variable Operating Costs by Technology Table in 10⁶ USD (MUSD)

Year	Imports – CIV	Imports – GHA	Imports – MLI	Imports – NER	Biomass	Diesel	Natural Gas	HFO	Hydro	Nuclear	Solar	Wind	Total
2021	26.1	0.0	0.0	0.0	0.0	0.0	0.0	219.3	0.0	0.0	0.0	0.0	245.4
2022	16.5	27.9	64.7	0.0	0.0	0.0	0.0	221.3	0.0	0.0	0.0	0.0	330.4
2023	22.2	32.6	53.9	0.0	0.0	21.1	0.0	226.6	0.0	0.0	0.0	0.0	356.4
2024	27.8	37.3	42.0	0.0	2.3	17.4	0.0	257.8	0.0	0.0	0.0	0.0	384.6
2025	33.5	42.0	30.2	42.0	4.5	0.0	0.0	218.1	0.0	0.0	0.0	0.0	370.4
2026	33.4	51.2	37.5	32.7	6.8	0.0	0.0	238.6	0.0	0.0	0.0	0.0	400.1
2027	33.2	60.3	44.8	21.8	9.2	0.0	0.0	259.7	0.0	0.0	0.0	0.0	429.0
2028	33.1	69.5	52.1	0.0	11.5	0.0	0.0	281.6	0.0	0.0	0.0	0.0	447.7
2029	32.9	78.7	59.4	0.0	13.8	0.0	0.0	289.8	0.0	0.0	0.0	0.0	474.5
2030	32.8	87.8	66.7	11.5	15.9	0.0	0.0	274.5	0.0	0.0	0.0	0.0	489.2
2031	32.7	97.0	74.0	78.0	18.1	0.0	0.0	270.2	0.0	0.0	0.0	0.0	570.0
2032	32.5	106.1	81.2	72.9	20.0	50.1	0.0	267.6	0.0	0.0	0.0	0.0	630.5
2033	32.4	115.3	88.5	67.7	20.5	55.0	0.0	262.0	0.0	0.0	0.0	0.0	641.4
2034	32.4	115.3	88.5	67.7	17.0	100.2	0.0	223.6	0.0	0.0	0.0	0.0	644.7
2035	183.7	172.9	3.5	0.0	17.3	0.0	115.9	97.4	0.0	0.0	0.0	0.0	590.8
2036	186.3	166.9	17.2	0.7	17.7	0.0	125.8	101.2	0.0	0.0	0.0	0.0	615.8
2037	186.8	163.0	42.3	4.7	18.3	0.0	127.6	97.8	0.0	0.0	0.0	0.0	640.5
2038	192.6	179.8	74.4	25.4	19.0	0.0	132.0	117.0	0.0	0.0	0.0	0.0	740.1
2039	193.6	195.1	126.0	79.5	19.7	0.0	138.6	99.0	0.0	0.0	0.0	0.0	851.4
2040	193.6	193.6	128.7	90.0	35.5	0.0	138.7	81.9	0.0	0.0	0.0	0.0	861.9
2041	193.5	193.3	128.1	120.4	49.9	0.0	139.2	63.1	0.0	0.0	0.0	0.0	887.5
2042	194.0	197.0	138.8	190.7	57.8	0.0	143.0	40.8	0.0	0.0	0.0	0.0	962.2
2043	201.6	202.6	157.3	305.4	59.3	2.3	147.6	15.9	0.0	0.0	0.0	0.0	1,092.0
2044	210.4	200.0	151.5	293.3	58.6	3.2	137.0	4.2	0.0	2.5	0.0	0.0	1,060.8
2045	215.8	209.9	162.3	280.1	58.3	3.2	113.0	4.3	0.0	4.8	0.0	0.0	1,051.7
2046	213.0	206.1	164.4	267.1	57.5	3.1	90.2	3.4	0.0	7.9	0.0	0.0	1,012.7
2047	213.1	205.1	162.4	297.5	57.5	3.2	66.2	3.4	0.0	10.7	0.0	0.0	1,019.0
2048	221.2	210.6	160.8	331.7	60.0	3.2	42.3	3.3	0.0	13.9	0.0	0.0	1,047.0
2049	221.8	209.6	156.8	335.7	70.7	3.2	19.4	2.3	0.0	16.5	0.0	0.0	1,036.2
2050	178.5	156.2	98.3	116.2	70.7	0.0	0.0	0.0	0.0	24.1	0.0	0.0	643.9
2051	175.9	149.5	95.4	107.9	70.7	0.0	0.0	0.0	0.0	26.0	0.0	0.0	625.3
2052	173.7	141.2	94.1	99.9	70.7	0.0	0.0	0.0	0.0	27.9	0.0	0.0	607.4
2053	169.3	135.2	91.3	93.7	70.7	0.0	0.0	0.0	0.0	29.9	0.0	0.0	590.0
2054	164.1	129.5	88.5	88.2	70.7	0.0	0.0	0.0	0.0	31.9	0.0	0.0	572.9
2055	158.1	125.4	85.9	82.8	70.7	0.0	0.0	0.0	0.0	34.0	0.0	0.0	556.9
2056	152.2	123.2	83.2	76.9	70.7	0.0	0.0	0.0	0.0	36.1	0.0	0.0	542.3
2057	146.7	121.5	80.1	71.5	70.7	0.0	0.0	0.0	0.0	38.4	0.0	0.0	528.9
2058	141.2	119.8	76.8	66.4	70.7	0.0	0.0	0.0	0.0	40.6	0.0	0.0	515.5
2059	137.4	118.1	72.6	62.5	70.7	0.0	0.0	0.0	0.0	42.9	0.0	0.0	504.1
2060	134.8	116.3	67.8	59.4	70.7	0.0	0.0	0.0	0.0	45.2	0.0	0.0	494.1
2061	132.1	113.9	63.3	56.6	70.7	0.0	0.0	0.0	0.0	47.5	0.0	0.0	484.1
2062	129.7	111.4	59.0	53.8	70.7	0.0	0.0	0.0	0.0	49.9	0.0	0.0	474.4
2063	127.7	109.1	55.0	51.0	70.7	0.0	0.0	0.0	0.0	52.3	0.0	0.0	465.7
2064	125.6	107.1	51.4	48.2	70.7	0.0	0.0	0.0	0.0	54.8	0.0	0.0	457.7
2065	123.2	105.1	48.1	45.5	70.7	0.0	0.0	0.0	0.0	57.3	0.0	0.0	449.8
2066	120.5	102.9	44.9	42.5	70.7	0.0	0.0	0.0	0.0	60.0	0.0	0.0	441.4
2067	117.7	100.0	42.3	39.8	70.7	0.0	0.0	0.0	0.0	62.7	0.0	0.0	433.2
2068	115.2	96.0	40.3	37.8	70.7	0.0	0.0	0.0	0.0	65.5	0.0	0.0	425.5
2069	113.0	91.0	38.7	36.7	70.7	0.0	0.0	0.0	0.0	68.3	0.0	0.0	418.4
2070	110.9	86.3	37.2	35.6	70.7	0.0	0.0	0.0	0.0	71.2	0.0	0.0	411.9

Table A.3.6: AMB Emissions in kilotons CO_{2eq} and Emission Intensity in grams CO_{2eq} /kWh electricity generated

Year	kTon CO _{2eq}	g CO _{2eq} / kWh electricity generated
2021	1906.2	0.679
2022	1906.2	0.543
2023	2038.4	0.531
2024	2265.9	0.421
2025	1829.6	0.300
2026	1954.8	0.293
2027	2080.1	0.287
2028	2205.3	0.283
2029	2220.8	0.265
2030	2059.0	0.230
2031	1990.4	0.198
2032	2144.7	0.192
2033	2087.4	0.169
2034	1964.6	0.147
2035	1281.7	0.094
2036	1344.1	0.091
2037	1315.4	0.083
2038	1448.9	0.085
2039	1350.7	0.074
2040	1227.9	0.063
2041	1105.1	0.054
2042	982.3	0.045
2043	859.5	0.038
2044	736.7	0.031
2045	613.9	0.025
2046	491.1	0.019
2047	368.4	0.013
2048	245.6	0.009
2049	122.8	0.004
2050	0	0
2051	0	0
2052	0	0
2053	0	0
2054	0	0
2055	0	0
2056	0	0
2057	0	0
2058	0	0
2059	0	0
2060	0	0
2061	0	0
2062	0	0
2063	0	0
2064	0	0
2065	0	0
2066	0	0
2067	0	0
2068	0	0
2069	0	0
2070	0	0

